

Original Gamma Volumes and Intrinsic Spectral Sums

The unchanged packet, exact endpoint refinements and proved comparison families

Split-Zero research programme

14 September 2026

This continuation retains the original packet order k , the original dimension $q = [1 + k(m-1)](k+1)^2$, the source coordinate $S = k/2 + iy$, all roots and all four endpoints. Every new proof is printed below. The complete preceding source closure, including the positive recurrence, source transport, equilibrium and mixed kernel-boundary maps, accompanies this paper in `dependencies/cut21`.

The first part concerns the unchanged construction. The second part preserves the calculations for a separately chosen Gamma order K , with their exact comparison maps and their stated scope. The exploratory choice $K = q + 1$ is not a rule supplied by the original construction and does not replace its canonical source. The intrinsic constructions in the first part are derived from the original spaces and operators.

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Part I

The original construction and its volume calculation

1 Original source and the homogeneous remainder map

Retain the original packet integers

$$k = 4l + 1, \quad l \geq 1, \quad m \geq 1, \quad e = 1 + k(m - 1), \quad q = e(k + 1)^2 = 2n, \quad c = k/2,$$

with m an integer. Define the homogeneous comparison polynomial and its literal quotient by

$$\chi_0(S) = (S - c)^q, \quad E_0 = \mathbb{C}[S]/(\chi_0), \quad J_N^0 : \mathbb{C}[S]_{\leq N} \longrightarrow E_0. \quad (\text{HBL1})$$

Here J_N^0 is polynomial remainder, and the original source line is $S = c + iy$. The measure and its full mass remain

$$d\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy, \quad \gamma_0 = \sigma(\mathbb{R}) = \sqrt{2\pi}. \quad (\text{HBL2})$$

For $N \geq q - 1$, let H_N^σ be the source Gram in the increasing S coefficient frame, and set

$$G_N^{\sigma,0} = (J_N^0(H_N^\sigma)^{-1}(J_N^0)^*)^{-1}. \quad (\text{HBL3})$$

The map is onto in this range, so this is a positive definite quotient Gram. Define the four-endpoint homogeneous baseline with its original signs by

$$F_\sigma^{\text{hom}}(q) = \log \det G_{q-1}^{\sigma,0} + \log \det G_q^{\sigma,0} - \log \det G_{2q-1}^{\sigma,0} - \log \det G_{2q}^{\sigma,0}. \quad (\text{HBL4})$$

The result below evaluates precisely HBL4. Its comparison with a displaced original polynomial χ requires the separate original source comparison; that comparison is not asserted in this calculation.

2 The exact source/relation determinant identity

Put $r = N + 1 - q$ and retain the full real-coordinate relation matrix

$$(M_{q,r})_{ij} = \int_{\mathbb{R}} y^{2q+i+j} d\sigma(y), \quad 0 \leq i, j < r, \quad \det M_{q,0} = 1.$$

The frame

$$1, S - c, \dots, (S - c)^{q-1}, \quad \chi_0, \chi_0(S - c), \dots, \chi_0(S - c)^{r-1}$$

is the increasing translated monomial frame, so its determinant in the original coefficient coordinates is one. Its first q columns represent a basis of E_0 and its last r columns form the entire relation kernel. In this frame the remainder map is $(I_q \ 0)$. The Schur complement of the positive relation block is the quotient Gram in HBL3: this follows either by multiplying the block inverse, or by minimizing the source norm in a fixed remainder fibre. The relation columns evaluate to $(iy)^{q+j}$, so their Gram is a congruence of $M_{q,r}$ by the diagonal matrix with entries i^{q+j} . Its determinant has absolute value one. Therefore

$$\det G_N^{\sigma,0} = \frac{\det H_N^\sigma}{\det M_{q,N+1-q}}. \quad (\text{HBL5})$$

This also proves HBL5 at $r = 0$ with the empty determinant convention.

The original monic real-line polynomials $p_j(y)$ have squared norms

$$\gamma_j = \sqrt{2\pi} j!(1/2)_j = \sqrt{2\pi} \frac{(2j)!}{4^j}. \quad (\text{HBL6})$$

These are the complete HCT5 norms. The polynomials $i^j p_j((S - c)/i)$ are monic in the original S coordinate and have the same norms on $S = c + iy$. Their monic triangular coefficient map has determinant one. It follows that $\det H_N^\sigma = \prod_{j=0}^N \gamma_j$ with the full mass retained.

Let $\mu_a = \int y^{2a} d\sigma(y)$ and $H_s^{(a)} = \det[\mu_{a+i+j}]_{i,j \leq s}$, with $H_0^{(a)} = 1$. The exact even/odd permutation of the original moment matrices gives

$$\det M_{q,q} = H_n^{(q)} H_n^{(q+1)}, \quad \det M_{q,q+1} = H_{n+1}^{(q)} H_n^{(q+1)}, \quad n = q/2.$$

The permutation is applied to rows and columns alike; its determinant sign is squared. Also $\det M_{q,1} = \mu_q$. Define

$$Z_q = H_n^{(q)} H_{n+1}^{(q)} (H_n^{(q+1)})^2.$$

Substitution into HBL4–6 proves the exact finite equality

$$\boxed{F_\sigma^{\text{hom}}(q) = S_q + \log Z_q - \log \mu_q}, \quad (\text{HBL7})$$

where all endpoints in the source contribution are

$$S_q = - \sum_{j=q}^{2q-1} \log \gamma_j - \sum_{j=q+1}^{2q} \log \gamma_j = - \log \gamma_q - 2 \sum_{j=q+1}^{2q-1} \log \gamma_j - \log \gamma_{2q}. \quad (\text{HBL8})$$

3 The source products at their exact degree scale

For every integer $d \geq 1$, monotonicity of $\log x$ on each unit interval gives

$$d \log d - d + 1 \leq \log(d!) \leq d \log d - d + 1 + \log d.$$

At $d = 2j$, HBL6 consequently gives, uniformly for $q \leq j \leq 2q$,

$$\log \gamma_j = 2j \log j - 2j + O(\log q).$$

The constant $\log \sqrt{2\pi}$ is included in this finite error. For $x = j/q$, the displayed main expression is $2qx(\log q + \log x - 1)$. The left and right sums in HBL8 have q terms each. Each differs from its integral times q by $O(q \log q)$, because the total variation of this increasing function on $[1, 2]$ has that order. The $2q$ individual norm errors also sum to $O(q \log q)$. The literal integrals

$$\int_1^2 x dx = \frac{3}{2}, \quad \int_1^2 x \log x dx = 2 \log 2 - \frac{3}{4}$$

therefore prove

$$\boxed{S_q = -6q^2 \log q + (9 - 8 \log 2)q^2 + O(q \log q)}. \quad (\text{HBL9})$$

In particular both distinct endpoints in HBL8 were used before the error was estimated.

4 The complete high relation determinant at scale q^2

We use the already proved original-source free-energy theorem GEL3, GEL7–10 and the EIQ equilibrium calculation, with their original measure and coordinate factors. Their definitions are repeated here to specify every parameter entering the limit. For $\alpha, \beta > 0$,

$$\mathcal{F}(\alpha, \beta) = \sup_{\mu \in \mathcal{P}((0, \infty))} \left\{ \iint \log |x - y| d\mu(x) d\mu(y) - \int (\beta \sqrt{x} - \alpha \log x) d\mu(x) \right\}. \quad (\text{HBL10})$$

The rigorous extended definition is the single bounded-above kernel $\log |x - y| - (V(x) + V(y))/2$, so infinite terms are never subtracted. Its unique equilibrium probability has compact support bounded away from zero, and $\mathcal{F}$ is continuous on the positive quadrant.

For the exact scale $t = q^2x$, define

$$d\omega(x) = \frac{1}{2}x^{-1/2}e^{-\sqrt{x}}dx, \quad B_q(x) = 2qe^{\sqrt{x}}\sigma(q\sqrt{x}),$$

$$J_{s,q}(a) = \frac{1}{s!} \int \prod_{i < j} (x_i - x_j)^2 \prod_i x_i^a B_q(x_i) d\omega(x_i).$$

Here $d\nu(q^2x) = B_q(x)d\omega(x)$ is the exact density identity including the whole original mass and Jacobian. GEL3 and GEL10 state

$$H_s^{(a)} = q^{2as+2s(s-1)} J_{s,q}(a), \quad \frac{\log J_{s,q}(s\alpha)}{s^2} \rightarrow \mathcal{F}(\alpha, \pi\theta/2) \quad (q/s \rightarrow \theta > 0). \quad (\text{HBL11})$$

This is a previously proved theorem about these literal Gamma integrals; the present calculation applies it, without replacing its source by an unproved asymptotic density.

Its fixed- α limit also applies when $\alpha_s \rightarrow 2$. Here are the needed details. The functions $f_s(\alpha) = s^{-2} \log J_{s,q}(s\alpha)$ are convex, since their second derivatives are variances of the sum of the original logarithmic coordinates, multiplied by a positive constant. For fixed $\eta > 0$, convexity bounds their values on $[2-\eta, 2+\eta]$ above by the larger of the endpoint values. If $\alpha_s \geq 2$, the lower secant inequality is

$$f_s(\alpha_s) \geq f_s(2) + \frac{\alpha_s - 2}{\eta} (f_s(2) - f_s(2 - \eta)).$$

For $\alpha_s \leq 2$ the corresponding lower bound is

$$f_s(\alpha_s) \geq f_s(2) + \frac{\alpha_s - 2}{\eta} (f_s(2 + \eta) - f_s(2)).$$

The fixed-point convergences in HBL11 make their differences bounded, so both lower bounds have $\liminf$ at least $\mathcal{F}(2, \pi)$ when $q/s \rightarrow 2$. The upper bound and continuity let η tend to zero. This proves the moving-argument limit required below.

For Z_q , use $(s, a) = (n, q), (n+1, q), (n, q+1), (n+1, q+1)$ exactly. Each has $q/s \rightarrow 2$ and $a/s \rightarrow 2$. The four exact powers in HBL11 are

$$\frac{3}{2}q^2 - q, \quad \frac{3}{2}q^2 + 3q, \quad \frac{3}{2}q^2, \quad \frac{3}{2}q^2.$$

Their sum is $6q^2 + 2q$. The sum of the four squared sizes divided by q^2 tends to one. Thus

$$\boxed{\log Z_q = (6q^2 + 2q) \log q + q^2 \mathcal{F}(2, \pi) + o(q^2)}. \quad (\text{HBL12})$$

The scalar low moment is negligible on this scale with its full mass still present. The established original Gamma contour bound gives $\sigma(y) \leq C_U e^{-|y|}$ for a fixed $C_U > 0$, so $\mu_q \leq 2C_U(2q)!$. Positivity on $[1, 2]$ gives $\mu_q \geq 2 \int_1^2 \sigma(y) dy > 0$. Hence $\log \mu_q = O(q \log q) = o(q^2)$. Combining this with HBL7, HBL9 and HBL12 proves the exact leading value

$$\boxed{\frac{F_\sigma^{\text{hom}}(q)}{q^2} \rightarrow \mathcal{C}_B := 9 - 8 \log 2 + \mathcal{F}(2, \pi)}. \quad (\text{HBL13})$$

This holds along the original packet degrees as they tend to infinity; the homogeneous quantity depends only on q after the explicit unit-determinant translation by c . There is no error estimate of order $o(q)$ claimed in HBL12 or HBL13.

5 An exact endpoint-integral presentation of the coefficient

To evaluate the constant in the same equilibrium coordinates as GEL18, let u, v be the unique positive endpoints with $0 < u < v$ and

$$uK(\kappa) = 2, \quad vE(\kappa) = 4, \quad \kappa^2 = 1 - u^2/v^2,$$

where $K(\kappa) = \int_0^{\pi/2} (1 - \kappa^2 \sin^2 \theta)^{-1/2} d\theta$ and $E(\kappa) = \int_0^{\pi/2} (1 - \kappa^2 \sin^2 \theta)^{1/2} d\theta$. The EIQ/GEL equilibrium density is exactly

$$\rho(x) = \frac{\sqrt{(v^2 - x)(x - u^2)}}{4\pi x} \int_0^\infty \frac{\sqrt{s} ds}{(x + s)\sqrt{(u^2 + s)(v^2 + s)}}, \quad u^2 < x < v^2. \quad (\text{HBL14})$$

It has mass one. Write $\mathcal{L} = \int \log x \rho(x) dx$ and $M = \int \sqrt{x} \rho(x) dx$. Under the literal dilation $x \mapsto r^2 x$ of this probability, the value of the functional HBL10 changes by $6 \log r - \pi(r - 1)M$. The equilibrium maximizes it at $r = 1$; differentiating this explicit scalar expression there gives $\pi M = 6$. Thus $\int V_{2,\pi} \rho = 6 - 2\mathcal{L}$.

The EIQ variational identity says that $V_{2,\pi}(x) - 2 \int \log |x - y| \rho(y) dy$ is a constant ℓ on the full support, including its endpoints. Set $b = v^2$. At the right endpoint the same identity is

$$\ell = \pi v - 2 \log b - 2 \int_{u^2}^b \log(b - y) \rho(y) dy.$$

All these integrals are finite; the support is bounded away from zero and the density has square-root endpoint vanishing. Integrating the variational identity against ρ gives $\mathcal{F}(2, \pi) = -\ell/2 - 3 + \mathcal{L}$. Consequently HBL13 has the exact one-dimensional endpoint-integral presentation

$$\begin{aligned} \mathcal{C}_B &= 6 - 8 \log 2 + \mathcal{L} - \ell/2 \\ &= 6 - 8 \log 2 + \log b - \frac{\pi v}{2} + \int_{u^2}^b [\log x + \log(b - x)] \rho(x) dx. \end{aligned}$$

(HBL15)

6 A strictly positive rational bound from an explicit competitor

We prove the sign without numerical evaluation of HBL14–15. Use the arcsine probability on the literal interval $[1, 9]$,

$$d\eta(x) = \frac{1_{(1,9)}(x)}{\pi \sqrt{(x-1)(9-x)}} dx, \quad x = 5 + 4 \cos \theta, \quad 0 \leq \theta \leq \pi.$$

The substitution gives the uniform measure $d\theta/\pi$ with the orientation reversed when x increases. Its three exact integrals are

$$\int \int \log |x - y| d\eta(x) d\eta(y) = \log 2, \quad \int \log x d\eta(x) = 2 \log 2, \quad \int \sqrt{x} d\eta(x) = \frac{6}{\pi} E(\sqrt{8/9}). \quad (\text{HBL16})$$

Here are proofs retaining the constants. The double-angle identity and the substitution $\theta \mapsto \pi/2 - \theta$ give $\int_0^\pi \log \sin \theta d\theta = -\pi \log 2$; all endpoint logarithms are integrable. Factoring $\cos \theta - \cos \phi = -2 \sin((\theta + \phi)/2) \sin((\theta - \phi)/2)$, then joining the two substituted intervals, gives

$$\frac{1}{\pi} \int_0^\pi \log |\cos \theta - \cos \phi| d\phi = \log 2 + \frac{2}{\pi} \int_{(\theta-\pi)/2}^{(\theta+\pi)/2} \log |\sin t| dt = -\log 2.$$

The interval has length π and the integrand has period π . Adding $\log 4$ proves the first HBL16 identity, and also proves finite logarithmic energy of this competitor. For the second, write $5 + 4 \cos \theta = |2 +$

$e^{i\theta}|^2$ and expand $\log(1+e^{i\theta}/2)$ in its uniformly absolutely convergent power series. Every nonconstant cosine has integral zero on $[0, \pi]$, leaving $2 \log 2$. Finally $5 + 4 \cos \theta = 9(1 - (8/9) \sin^2(\theta/2))$; the substitution $t = \theta/2$ gives the third identity.

Since HBL10 is a supremum over all such probability measures, the competitor proves

$$\mathcal{F}(2, \pi) \geq 5 \log 2 - 6E(\sqrt{8/9}), \quad \mathcal{C}_B \geq 9 - 3 \log 2 - 6E(\sqrt{8/9}). \quad (\text{HBL17})$$

An elementary polynomial bound makes this strictly positive. For $0 \leq z \leq 1$ put $P(z) = 1 - z/2 - z^2/8 - z^3/16$. It is positive there, since $P(z) \geq P(1) = 5/16$, and direct multiplication gives

$$P(z)^2 - (1 - z) = \frac{5z^4}{64} + \frac{z^5}{64} + \frac{z^6}{256} \geq 0.$$

Therefore $\sqrt{1 - z} \leq P(z)$. The integrations by parts $\int_0^{\pi/2} \sin^{2r} t \, dt = (2r - 1)/(2r) \int_0^{\pi/2} \sin^{2r-2} t \, dt$ give the exact second, fourth and sixth power integrals $\pi/4, 3\pi/16, 5\pi/32$. At $z = (8/9) \sin^2 t$ the polynomial bound consequently gives

$$E(\sqrt{8/9}) \leq \frac{\pi}{2} \left(1 - \frac{2}{9} - \frac{1}{27} - \frac{10}{729} \right) = \frac{265\pi}{729}.$$

The identity $22/7 - \pi = \int_0^1 x^4(1-x)^4/(1+x^2) \, dx > 0$ follows by polynomial division and the arctangent integral. Also the positive exponential series gives $e^{7/10} > 1 + 7/10 + (7/10)^2/2 + (7/10)^3/6 = 12013/6000 > 2$, so $\log 2 < 7/10$. Thus HBL17 implies the entirely rational strict bound

$$\mathcal{C}_B \geq 9 - 3 \log 2 - \frac{530\pi}{243} > 9 - \frac{21}{10} - \frac{11660}{1701} = \frac{769}{17010} > 0. \quad (\text{HBL18})$$

This is a bound on the exact coefficient in HBL13, proved by a finite competitor calculation. In particular the homogeneous baseline has a strictly positive leading term of order q^2 . Its transfer to a displaced packet, and any subsequent combination with the arithmetic baseline or HC residual, must use the corresponding original maps and source comparisons rather than identifying those objects with HBL4.

Complete proved dependencies used. The original Gamma monic norms are HCT5 in `independent_contiguous.tex`. The exact density map, free-energy limit, and its original-source bounds are GEL1–10 in

`GAMMA_ENSEMBLE_LEADING_RETURN.tex`. The compact equilibrium, endpoint equations, positive density, and full variational identity are proved in

`EXACT_SQRT_LOG_EQUILIBRIUM.tex`. This note supplies the complete new quotient calculation, moving-argument specialization, original source expansion, coefficient identity, and its analytic sign.

7 The original packet baseline and its complete receiving maps

Retain the original parameters and all roots, counted with their orders:

$$k = 4l + 1, \quad l, m \geq 1, \quad e = 1 + k(m - 1), \quad q = e(k + 1)^2 = 2n, \quad c = k/2, \quad 0 < \delta < 1/2, \quad \gamma > 2,$$

$$\chi(S) = \prod_{a,b=0}^k [S - c - (2a - k)\delta - i(2b - k)\gamma]^e, \quad d\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy. \quad (\text{BSL1})$$

The letter e here is the positive integer order; exponential functions below are written $\exp$. Every limit in this section holds with the original m, δ, γ fixed and $l \rightarrow \infty$.

7.1 The exact source, relation, and quotient determinants

Let $\mathcal{P}_N$ be the polynomials in S of degree at most N , with the stated source inner product evaluated on $S = c + iy$. Set $E_\chi = \mathbb{C}[S]/(\chi)$ and let $J_N : \mathcal{P}_N \rightarrow E_\chi$ send a polynomial to its remainder. For $N \geq q - 1$, this is onto, and its kernel is the literal image of

$$I_{\chi,N} : \mathcal{P}_{N-q} \longrightarrow \mathcal{P}_N, \quad Q \longmapsto \chi Q. \quad (\text{BSL2})$$

The domain is the zero vector space when $N = q - 1$. Let H_N^σ be the source Gram in $(1, S, \dots, S^N)$, G_N^σ the minimum norm quotient Gram in the fixed remainder frame, and

$$A_{\chi,r} = \text{Gram}_\sigma(\chi, \chi S, \dots, \chi S^{r-1}), \quad r = N + 1 - q.$$

All these nonempty Grams are positive definite. A nonzero source polynomial is nonzero on an open real interval, and σ is strictly positive there. The quotient minimum exists and is positive because this is an onto map between finite dimensional spaces with a positive source form. The same argument applies to every relation form.

The ordered change of frame $(1, \dots, S^{q-1}, \chi, \dots, \chi S^{r-1})$ has determinant one: its column of degree j has leading coefficient one and no higher coefficient. In this frame, minimizing the quadratic form over its last r coefficients gives the Schur complement of $A_{\chi,r}$. Indeed, for blocks $\begin{pmatrix} A & C \\ C^* & D \end{pmatrix}$, the minimizing last coefficient is $-D^{-1}C^*z$, its squared norm is $z^*(A - CD^{-1}C^*)z$, and triangular block elimination has determinant one. Consequently, with the empty determinant equal to one,

$$\det G_N^\sigma = \frac{\det H_N^\sigma}{\det A_{\chi,N+1-q}}. \quad (\text{BSL3})$$

This equality retains the complete relation space before forming a quotient. It gives the same form as $G_N^\sigma = (J_N(H_N^\sigma)^{-1}J_N^*)^{-1}$: the minimizer of $P^*H_N^\sigma P$ subject to $J_N P = z$ is $(H_N^\sigma)^{-1}J_N^*G_N^\sigma z$, as multiplication by J_N and completing the square directly verify.

Let $\gamma_j = \sqrt{2\pi}(2j)!/4^j$ be the original monic Gamma norms, including $\gamma_0 = \sqrt{2\pi}$. Their complete derivation is HCT1-9. Monic orthogonalization has determinant one, so $\det H_N^\sigma = \prod_{j=0}^N \gamma_j$. Define

$$S_q = -\log \gamma_q - 2 \sum_{j=q+1}^{2q-1} \log \gamma_j - \log \gamma_{2q},$$

$$\mathcal{B}_k^\sigma = \log \frac{\det G_{q-1}^\sigma \det G_q^\sigma}{\det G_{2q-1}^\sigma \det G_{2q}^\sigma}.$$

Substitution of all four ranks $0, 1, q, q + 1$ in BSL3 proves the exact baseline formula

$$\boxed{\mathcal{B}_k^\sigma = S_q + \log \det A_{\chi,q} + \log \det A_{\chi,q+1} - \log \det A_{\chi,1}}. \quad (\text{BSL4})$$

Both high relation spaces and the one dimensional low relation space are present with their original signs.

7.2 A finite comparison retaining every original root

Define the literal power moment matrix and original coefficient map

$$(M_{q,r})_{ij} = \int_{\mathbb{R}} y^{2q+i+j} d\sigma(y), \quad T_{h,j} = \binom{j}{h} c^{j-h} i^h \quad (0 \leq h \leq j < r), \quad (\text{BSL5})$$

with other entries zero. The matrix T sends coefficients of $P(S)$ to those of $P(c + iy)$; its inverse comes from $y = (S - c)/i$. Since $|\det T| = 1$, the Gram of $(S - c)^q(1, S, \dots, S^{r-1})$ on the original source is $T^* M_{q,r} T$ and has determinant $\det M_{q,r}$. The phase i^q has squared modulus one, as used in this equality.

Put $d = \delta^2$, $g = \gamma^2$, $\epsilon = 2^{-10}$, and assume the same explicit finite threshold as CTR23 and LET4, $k \geq 2048\sqrt{d+g}$. Partition the line into $I = \{|y| < q\epsilon\}$, $B = \{q\epsilon \leq |y| \leq 64q\}$, and $F = \{|y| > 64q\}$. Retain

$$a_k = \frac{k(k+2)(d-g)}{3q} < 0, \quad t_\epsilon = \frac{k^2(d+g)}{q^2\epsilon^2} \leq \frac{1}{4}, \quad d_\chi = \frac{qt_\epsilon^2}{2(1-t_\epsilon)},$$

$$b_- = a_k\epsilon^{-2} - d_\chi, \quad b_+ = a_k64^{-2} + d_\chi. \quad (\text{BSL6})$$

Here d_χ is a nonnegative Taylor remainder bound, not a determinant or the earlier Christoffel correction. At $y = qz$ in B , the exact product gives

$$\psi_\chi(qz) := \log \frac{|\chi(c + iqz)|^2}{|qz|^{2q}} = \frac{a_k}{z^2} + E_\chi(qz), \quad |E_\chi(qz)| \leq d_\chi, \quad b_- \leq \psi_\chi(qz) \leq b_+. \quad (\text{BSL7})$$

To check every coefficient, write each factor as $iqz(1 + i\zeta_{ab}/(qz))$ with $\zeta_{ab} = (2a - k)\delta + i(2b - k)\gamma$. The logarithmic Taylor series is absolutely convergent because $|\zeta_{ab}/(qz)| \leq 1/2$. The permutation $(a, b) \mapsto (k - a, k - b)$ cancels its odd powers. The second power is $e \operatorname{Re} \sum_{a,b} \zeta_{ab}^2/(qz)^2 = a_k/z^2$, using $\sum_{a=0}^k (2a - k)^2 = (k+1)k(k+2)/3$ and $\sum_{a=0}^k (2a - k) = 0$. The even remainder has absolute value at most $q \sum_{j \geq 2} t_\epsilon^j/j \leq d_\chi$. The negative sign of a_k determines which end of the interval for z^{-2} gives each bound in BSL6.

The exact source map on $L^2(B, d\sigma)$ is multiplication by $\chi(c + iy)/(iy)^q$. It and its inverse are bounded there because the numerator and denominator have no zero on this compact set. Composing this map with $P \mapsto (iy)^q P(c + iy)$ gives precisely $P \mapsto \chi(c + iy)P(c + iy)$. Taking these Hilbert Grams and using BSL7 proves the full vector inequality

$$\exp(b_-) T^* M_{q,r}^B T \preceq A_{\chi,r}^B \preceq \exp(b_+) T^* M_{q,r}^B T. \quad (\text{BSL8})$$

No map between different remainder algebras has been presumed here; BSL8 is proved on the specified common relation coefficient space.

For either full Gram $H = H^I + H^B + H^F$, keep its exact tail value

$$t_H = \log \det \left(I_r + (H^B)^{-1/2} (H^I + H^F) (H^B)^{-1/2} \right), \quad \log \det H = \log \det H^B + t_H.$$

The complete contour and polynomial estimates CTR37–44, and the separate scalar proof LET10–16 when $r = 1$, prove

$$0 \leq t_H \leq L_r, \quad L_r = r \log(1 + \kappa_{I,r} + \kappa_F),$$

$$\kappa_{I,r} = \frac{2\epsilon r^2}{\sqrt{\cos 1}} \left(\frac{4}{3} 2^{-13} \exp(2\pi) \right)^q, \quad \kappa_F = \frac{512q}{59\sqrt{\cos 1}} \exp(-19q), \quad r \in \{1, q, q+1\}. \quad (\text{BSL9})$$

The first base is smaller than $\exp(-2)$; these are finite positive bounds for each original tail, not omitted integrals. Define the actual finite error

$$\eta_r = \log \det A_{\chi,r} - \log \det M_{q,r}.$$

Congruence of BSL8 with the positive power Gram bounds each of its r eigenvalues. Multiplying them and adding the two original tail values gives the exact expression and its enclosure

$$\eta_r = \log \frac{\det A_{\chi,r}^B}{\det M_{q,r}^B} + t_{A_{\chi,r}} - t_{M_{q,r}}, \quad (\text{BSL10})$$

$$rb_- - L_r \leq \eta_r \leq rb_+ + L_r.$$

For a further exact representation of the first term, put $A_r(t) = \text{Gram}_{1_B|y|^{2q} \exp(t\psi_\chi) d\sigma}(1, S, \dots, S^{r-1})$. Its derivative exists entrywise since ψ_χ is bounded on B , and all its matrices are positive. Jacobi's determinant derivative, proved by differentiating the multilinear determinant one column at a time, gives

$$\log \frac{\det A_{\chi,r}^B}{\det M_{q,r}^B} = \int_0^1 \text{tr}(A_r(t)^{-1} A'_r(t)) dt. \quad (\text{BSL11})$$

Thus the entire polynomial location contribution has both its original finite matrix and its exact interpolation map.

7.3 The evaluated baseline coefficient

Let $\mu_q = \int y^{2q} d\sigma(y)$, let $H_j^{(a)}$ denote the positive half-line moment determinant GEL1, and retain

$$Z_q = H_n^{(q)} H_{n+1}^{(q)} (H_n^{(q+1)})^2, \quad V_q = S_q + \log Z_q - \log \mu_q. \quad (\text{BSL12})$$

Grouping even and odd coordinate indices in $M_{q,q}$ gives $\det M_{q,q} = H_n^{(q)} H_n^{(q+1)}$. The identical permutation in $M_{q,q+1}$ gives $\det M_{q,q+1} = H_{n+1}^{(q)} H_n^{(q+1)}$. The odd cross entries vanish by evenness of σ ; the simultaneous row and column permutations have product sign one. Also $\det M_{q,1} = \mu_q$. Thus BSL4 becomes exactly

$$\boxed{\mathcal{B}_k^\sigma = V_q + \eta_q + \eta_{q+1} - \eta_1.} \quad (\text{BSL13})$$

Writing $L_* = L_q + L_{q+1} + L_1$, BSL10 retains the full finite bounds

$$\boxed{V_q + (2q+1)b_- - b_+ - L_* \leq \mathcal{B}_k^\sigma \leq V_q + (2q+1)b_+ - b_- + L_*} \quad (\text{BSL14})$$

In particular no low determinant is silently assigned the value one.

The independent homogeneous calculation HBL gives the complete result

$$\frac{V_q}{q^2} \longrightarrow C_B := 9 - 8 \log 2 + \mathcal{F}(2, \pi), \quad C_B > \frac{769}{17010} > 0, \quad (\text{BSL15})$$

where $\mathcal{F}$ is the exact variational energy GEL7 with the unique positive density EIQ11. All logarithmic source powers cancel only after the identity BSL12 is substituted: the source contribution is $-6q^2 \log q + (9 - 8 \log 2)q^2 + O(q \log q)$, while $\log Z_q = (6q^2 + 2q) \log q + q^2 \mathcal{F}(2, \pi) + o(q^2)$. The low term has size $O(q \log q)$. HBL proves the exact coefficient, its one dimensional equilibrium integral, and the strict rational lower bound by an explicit arcsine competitor. Its full proof is included with this section.

The original degree identities give

$$|a_k| \leq \frac{|d-g|}{3e}, \quad d_\chi \leq \frac{2(d+g)^2}{3e^3 k^2 \epsilon^4}.$$

Consequently $\eta_q + \eta_{q+1} - \eta_1 = O(q/e)$ by BSL10; the displayed tails decay exponentially. Since $q/e = (k+1)^2$ and $q \rightarrow \infty$, this error divided by q^2 tends to zero. The stronger signed bounds BSL14 remain in force. We have proved the original packet limit

$$\boxed{\frac{\mathcal{B}_k^\sigma}{q^2} \longrightarrow C_B > 0.} \quad (\text{BSL16})$$

7.4 Original tensor, arithmetic, kernel, and boundary receivers

The tensor-Gamma source and its mass are unchanged:

$$dm_{0,k}(y) = \beta_k |f_l(c + iy)|^2 d\sigma(y), \quad f_l(S) = \prod_{j=0}^{l-1} (S - c - 2j - \tfrac{1}{2}), \quad \beta_k = \frac{(2\pi)^{k/2}}{\sqrt{2}\Gamma(k/2)}. \quad (\text{BSL17})$$

Its full quotient baseline is the original $\mathcal{B}_k^0$. Multiplication by f_l in the original remainder fibres, including the full product Taylor jets and their invertible multiplication germ, gives TWA29–30 and the complete GCR proof. In their signs these exact identities are

$$\mathcal{B}_k^0 = \mathcal{B}_k^\sigma + \mathfrak{T}_k, \quad \mathcal{B}_k^{\text{ar}} = \mathcal{B}_k^\sigma + \delta_k^\sigma, \quad \Delta_k^\Gamma = \delta_k^\sigma - \mathfrak{T}_k. \quad (\text{BSL18})$$

All four contributions $q \log \beta_k$ and all four full resultant logarithms in that proof are retained before their signed cancellation. Here $\mathfrak{T}_k/(lq) \rightarrow C_\Gamma > 0$ is the proved GEL19 result and $\delta_k^\sigma/(kq) \rightarrow 0$ is the proved original source determinant estimate TWA15. Since $l/q \rightarrow 0$ and $k/q \rightarrow 0$, BSL16–18 give

$$\boxed{\frac{\mathcal{B}_k^0}{q^2} \rightarrow C_B, \quad \frac{\mathcal{B}_k^{\text{ar}}}{q^2} \rightarrow C_B, \quad \frac{\mathcal{R}_k}{q^2} \rightarrow C_B,} \quad \mathcal{R}_k = \mathcal{B}_k^{\text{ar}} - 4q \log(D_h k). \quad (\text{BSL19})$$

The last limit uses the original fixed $D_h > 0$ and $\log(D_h k)/q \rightarrow 0$. The full finite enclosures are obtained without any asymptotic substitution: add the original interval for $\mathfrak{T}_k$ to BSL14 for $\mathcal{B}_k^0$, add $[-a_k^-, a_k^+]$ to BSL14 for $\mathcal{B}_k^{\text{ar}}$, and subtract the original $4q \log(D_h k)$ for $\mathcal{R}_k$.

To see exactly which mixed objects receive this result, retain $\Lambda : E_\chi \rightarrow B_{\text{obs}} = \text{im } \Lambda$, $K = \ker \Lambda$, its inclusion I_K , and the original section $S_* : B_{\text{obs}} \rightarrow E_\chi$ with $\Lambda S_* = I$. For any original positive quotient Gram G_N , the square matrix $[I_K, S_*]$ is invertible: every z is uniquely $(z - S_* \Lambda z) + S_* \Lambda z$ with the first term in K . Its restricted kernel Gram is $I_K^* G_N I_K$; the boundary minimum Gram is the Schur complement after this restriction. Block elimination therefore proves, with the original frame factor,

$$\det(I_K^* G_N I_K) \det Q_N^{B_{\text{obs}}} = |\det[I_K, S_*]|^2 \det G_N. \quad (\text{BSL20})$$

Both determinant factors are positive, with the empty one equal to one at boundary rank zero or full rank. In the same four endpoint signed sum the frame contribution has coefficient $1 + 1 - 1 - 1 = 0$. The complete BRU row update construction gives these two sums as $F_K(1)$ and $F_B(1)$ and hence

$$\frac{F_K(1) + F_B(1)}{q^2} \rightarrow C_B. \quad (\text{BSL21})$$

This proves the exact combined mixed-support consequence. It keeps the individual row contractions, the full Taylor unit in Λ , and their off-diagonal terms through BSL20; no individual contribution is assigned the combined limit.

The same original HC identity TWA10 is $\mathcal{B}_k^{\text{ar}} = 4q \log \mathcal{L}_k - W_k^\sigma + S_k - \tau_k^{\text{win}}$, where all nonnegative contraction, phase, and relative-control terms occur separately in TWA9. Here $\mathcal{L}_k = 2\delta e(k+1)\lfloor(k+1)^2/4\rfloor$, $W_k^\sigma = O(q \log q)$ by its literal factorial formula TWA5, and $\tau_k^{\text{win}} = O_h(k + \log q)$ by TWA6. The explicit polynomial formula for q gives $\log \mathcal{L}_k = O_h(\log k)$ and $\log q = O_m(\log k)$. Dividing this exact HC identity by q^2 now proves

$$\boxed{\frac{S_k}{q^2} \rightarrow C_B > 0.} \quad (\text{BSL22})$$

Thus the growing source volume is connected all the way to the original contraction and phase costs. The finer Gamma law $(\mathcal{B}_k^0 - \mathcal{B}_k^{\text{ar}})/(kq) \rightarrow C_\Gamma/4$ continues to hold simultaneously. Its complete finite remainders and all packet locations remain in BSL10–14 and WGR; none is removed by the leading q^2 limit.

8 An explicit remainder for the original Gamma logarithmic derivative

Retain the original density and its logarithmic derivative:

$$\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi}, \quad L(y) = -\frac{\sigma'(y)}{\sigma(y)}, \quad y \in \mathbb{R}. \quad (\text{GDR1})$$

For every real $y \neq 0$, define

$$\epsilon(y) = yL(y) - \frac{\pi}{2}|y| - \frac{1}{2}. \quad (\text{GDR2})$$

The following two estimates hold on this entire punctured real line:

$$|\epsilon(y)| \leq \left(\frac{1}{8} + \frac{\sqrt{3}}{9} \right) \frac{1}{y^2} < \frac{1}{3y^2}, \quad (\text{GDR3})$$

$$\left| \epsilon(y) - \frac{1}{8y^2} \right| \leq \left(\frac{5}{32} + \frac{7}{12\sqrt{3}} \right) \frac{1}{y^4} < \frac{1}{2y^4}. \quad (\text{GDR4})$$

In particular, the coefficient of the inverse-square term is positive and exactly $1/8$. Both bounds are global inequalities with explicit constants; no unspecified asymptotic remainder is used. The density and $yL(y)$ have their original values at zero, namely $\sigma(0) = |\Gamma(1/4)|^2/(2\pi)$ and $yL(y) = 0$ at $y = 0$. Equations GDR3–4 make no assertion at that excluded point.

The exact convergent series. For $\Re z > 0$, repeated integration by parts in the beta integral gives

$$G_N(z) := \frac{N!N^z}{z(z+1)\cdots(z+N)} = \int_0^N u^{z-1}(1-u/N)^N du. \quad (\text{GDR5})$$

The factor $(1-u/N)^N$ is interpreted as zero above N and is bounded by e^{-u} on $[0, N]$. On a compact subset of $\Re z > 0$, use its least real part below $u = 1$ and its greatest real part above $u = 1$. The resulting integrable majorants, also multiplied by $|\log u|$, prove locally uniform convergence of both G_N and G'_N to Γ and Γ' . For $\alpha > 0$ and $v \in \mathbb{R}$,

$$\frac{|\Gamma(\alpha + iv)|}{\Gamma(\alpha)} = \lim_{N \rightarrow \infty} \prod_{r=0}^N \left(1 + \frac{v^2}{(r+\alpha)^2} \right)^{-1/2} > 0. \quad (\text{GDR6})$$

The strict positivity follows because the sum of the logarithms of the positive factors converges absolutely, by comparison with $\sum_r v^2/(r+\alpha)^2$. Logarithmic differentiation of GDR5, taking its imaginary part and then the limit, therefore proves

$$\operatorname{Im} \frac{\Gamma'(\alpha + iv)}{\Gamma(\alpha + iv)} = \sum_{r=0}^{\infty} \frac{v}{(r+\alpha)^2 + v^2}. \quad (\text{GDR7})$$

Indeed, the real term $\log N$ contributes zero imaginary part. Differentiating the two original conjugate Gamma factors in GDR1 shows that $L(y)$ equals the left side of GDR7 with $\alpha = 1/4$ and $v = y/2$. Consequently, for $y = 2v > 0$,

$$yL(y) = 2 \sum_{r=0}^{\infty} F_v(r + 1/4), \quad F_v(x) = \frac{v^2}{x^2 + v^2}, \quad x \geq 0. \quad (\text{GDR8})$$

In particular the original quarter-shift has been retained exactly.

Shifted summation with its signed remainder. We prove the needed summation identity instead of invoking an asymptotic Gamma expansion. Define the following polynomials on $[0, 1]$:

$$\begin{aligned} B_1(t) &= t - \frac{1}{2}, & B_2(t) &= t^2 - t + \frac{1}{6}, \\ B_3(t) &= t^3 - \frac{3}{2}t^2 + \frac{1}{2}t, \\ B_4(t) &= t^4 - 2t^3 + t^2 - \frac{1}{30}, \\ B_5(t) &= t^5 - \frac{5}{2}t^4 + \frac{5}{3}t^3 - \frac{1}{6}t. \end{aligned} \tag{GDR9}$$

Direct differentiation and evaluation give $B'_j = jB_{j-1}$ for $2 \leq j \leq 5$, $B_j(1) = B_j(0)$ for $2 \leq j \leq 5$, and $B_j(1-t) = (-1)^j B_j(t)$ for $1 \leq j \leq 5$. For $0 < \alpha < 1$, let

$$P_j(x) = \frac{B_j(\{x - \alpha\})}{j!}, \quad x \geq 0, \tag{GDR10}$$

where $\{u\} = u - \lfloor u \rfloor$. The functions P_j are bounded; those with $j \geq 2$ are continuous. Between successive points $r + \alpha$, $P'_1 = 1$, and at every such point P_1 has a downward jump of size one. For a continuously differentiable function f for which the expressions converge, integration on each interval and addition of the jump contributions give

$$\sum_{r \geq 0} f(r + \alpha) = \int_0^\infty f(x) dx + f(0)P_1(0) + \int_0^\infty f'(x)P_1(x) dx. \tag{GDR11}$$

One can first make this calculation on $[0, M]$, where M is an integer. The right boundary term is $-f(M)P_1(M)$ and tends to zero when $f(M) \rightarrow 0$. The left boundary is not a jump, since $0 < \alpha < 1$, and $P_j(0) = B_j(1 - \alpha)/j! = (-1)^j B_j(\alpha)/j!$. The derivative identity for P_j holds almost everywhere and P_j is locally absolutely continuous for $j \geq 2$. Repeated ordinary integration by parts in the last integral of GDR11 therefore yields, for either $p = 3$ or $p = 5$,

$$\sum_{r \geq 0} f(r + \alpha) = \int_0^\infty f(x) dx - \sum_{j=1}^p \frac{B_j(\alpha)}{j!} f^{(j-1)}(0) + \frac{(-1)^{p-1}}{p!} \int_0^\infty B_p(\{x - \alpha\}) f^{(p)}(x) dx. \tag{GDR12}$$

For the actual function $f = F_v$, every derivative in question tends to zero at infinity and is integrable on $[0, \infty)$. Thus every boundary passage and integral in this proof is justified for the function to which GDR12 will be applied. In particular, the remainder sign is positive for both odd values of p used here.

The reciprocal-square bound. Put $H(u) = (1 + u^2)^{-1}$, so that $F_v^{(r)}(x) = v^{-r} H^{(r)}(x/v)$. The necessary derivatives are

$$H''(u) = \frac{6u^2 - 2}{(1 + u^2)^3}, \quad H'''(u) = \frac{24u(1 - u^2)}{(1 + u^2)^4}. \tag{GDR13}$$

The second derivative increases from -2 to $1/2$ on $[0, 1]$ and decreases from $1/2$ to 0 on $[1, \infty)$. It follows exactly that

$$\int_0^\infty |F_v'''(x)| dx = \frac{3}{v^2}. \tag{GDR14}$$

Writing $s = t - 1/2$ gives $B_3(t) = s(s^2 - 1/4)$. On $-1/2 \leq s \leq 1/2$, differentiating this cubic shows that its largest absolute value is attained at $s = \pm 1/(2\sqrt{3})$ and equals

$$\max_{0 \leq t \leq 1} |B_3(t)| = \frac{1}{12\sqrt{3}} = \frac{\sqrt{3}}{36}. \tag{GDR15}$$

At the original shift $\alpha = 1/4$, $B_1(1/4) = -1/4$, $B_3(1/4) = 3/64$, while $F'_v(0) = 0$ and $F''_v(0) = -2/v^2$. Since $\int_0^\infty F_v(x) dx = \pi v/2$, GDR12 with $p = 3$ becomes

$$\sum_{r \geq 0} F_v(r + 1/4) = \frac{\pi v}{2} + \frac{1}{4} + \frac{1}{64v^2} + R_3(v), \quad |R_3(v)| \leq \frac{\sqrt{3}}{72v^2}. \tag{GDR16}$$

Multiply this equality by two and put $v = y/2$ as in GDR8. This proves the non-strict inequality in GDR3 for $y > 0$. The stated strict comparison of constants is equivalent to $\sqrt{3}/9 < 5/24$, whose square is $64 < 75$ after multiplying by the positive common factors.

The inverse-fourth remainder after the exact coefficient. For the same H , direct differentiation gives

$$H^{(4)}(u) = \frac{24(1 - 10u^2 + 5u^4)}{(1 + u^2)^5}, \quad H^{(5)}(u) = \frac{-240u(3 - 10u^2 + 3u^4)}{(1 + u^2)^6}. \quad (\text{GDR17})$$

The fourth derivative decreases on $[0, 1/\sqrt{3}]$, increases on $[1/\sqrt{3}, \sqrt{3}]$, and decreases on $[\sqrt{3}, \infty)$. Its endpoint values in that order are

$$24, \quad -\frac{81}{8}, \quad \frac{3}{8}, \quad 0.$$

Its total variation is therefore $(24 + 81/8) + (81/8 + 3/8) + 3/8 = 45$, and scaling back to F_v yields

$$\int_0^\infty |F_v^{(5)}(x)| dx = \frac{45}{v^4}. \quad (\text{GDR18})$$

The exact factorization

$$B_5(t) = t(t-1)(t-1/2)(t^2 - t - 1/3) = s(s^2 - 1/4)(s^2 - 7/12), \quad s = t - 1/2, \quad (\text{GDR19})$$

implies, using the maximization already proved in GDR15,

$$|B_5(t)| = |s|(1/4 - s^2)(7/12 - s^2) \leq \frac{7}{12} \frac{1}{12\sqrt{3}} = \frac{7}{144\sqrt{3}}, \quad 0 \leq t \leq 1. \quad (\text{GDR20})$$

The remaining original-shift values are $B_5(1/4) = -25/1024$, $F_v'''(0) = 0$, and $F_v^{(4)}(0) = 24/v^4$. Thus the exact $p = 5$ equality from GDR12 is

$$\begin{aligned} \sum_{r \geq 0} F_v(r + 1/4) &= \frac{\pi v}{2} + \frac{1}{4} + \frac{1}{64v^2} + \frac{5}{1024v^4} + R_5(v), \\ R_5(v) &= \frac{1}{120} \int_0^\infty B_5(\{x - 1/4\}) F_v^{(5)}(x) dx, \quad |R_5(v)| \leq \frac{7}{384\sqrt{3}v^4}. \end{aligned} \quad (\text{GDR21})$$

Equivalently, the exact signed remainder in the original coordinate is

$$\epsilon(y) = \frac{1}{8y^2} + \frac{5}{32y^4} + \frac{1}{60} \int_0^\infty B_5(\{x - 1/4\}) F_{y/2}^{(5)}(x) dx, \quad y > 0. \quad (\text{GDR22})$$

The last integral, including its factor $1/60$, has absolute value at most $7/(12\sqrt{3}y^4)$. This proves the non-strict inequality in GDR4. For its strict comparison of constants, observe that $7/(12\sqrt{3}) < 11/32$ is equivalent to $56 < 33\sqrt{3}$; squaring both positive sides gives $3136 < 3267$.

Finally, $\Gamma(1/4 - iy/2)$ is the conjugate of $\Gamma(1/4 + iy/2)$ by the original defining integral. Hence σ is even, L is odd, and $yL(y)$ is even. This proves GDR3–4 on the full stated domain $y \in \mathbb{R} \setminus \{0\}$. Every occurrence of $y/2$, the quarter-shift, the original density constant $1/(2\pi)$, and the remainder orientation is retained.

9 The original scalar Gamma endpoint to fifth-order finite error

Retain the original density and both literal moment indices:

$$\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi}, \quad I_p = \int_0^\infty y^p \sigma(y) dy \quad (p \geq 0), \quad \mu_a = \int_{\mathbb{R}} y^{2a} \sigma(y) dy = 2I_{2a}. \quad (\text{LER1})$$

Thus $\mu_a = m_{2a}$ in LET's all-power notation, and $\mu_0 = \sqrt{2\pi}$. The factors two and the original mass are kept until cancellation in the specified ratios. Let $\alpha_\Gamma = \pi/2$ and $L(y) = -\sigma'(y)/\sigma(y)$. The complete accompanying derivation DERIVATIVE_REMAINDER.tex, based on the original Gamma beta-limit series, proves

$$yL(y) = \alpha_\Gamma y + \frac{1}{2} + \frac{1}{8y^2} + E(y), \quad |E(y)| \leq \frac{1}{2y^4} \quad (y > 0). \quad (\text{LER2})$$

No asymptotic notation substitutes for this global finite estimate.

Exact moment recursion and its positive comparison data. For every integer $p \geq 4$, integrate the derivative of $y^{p+1}\sigma(y)$ on $(0, \infty)$. Its boundary values are zero because the density is bounded at zero and has the exponential bound proved in HCT1. The differentiated integrand is integrable by RWB2. Substitution of LER2 gives the exact identity

$$\alpha_\Gamma I_{p+1} = (p + 1/2)I_p - \frac{1}{8}I_{p-2} - e_p, \quad e_p = \int_0^\infty y^p E(y)\sigma(y) dy, \quad |e_p| \leq \frac{1}{2}I_{p-4}. \quad (\text{LER3})$$

The lower powers on the right are integrable at zero for $p \geq 4$. We retain the sign of e_p ; the bound on its absolute value is the only replacement made in an inequality.

RWB11 already gives, with its complete proof from the original density,

$$\frac{2v-1}{\pi} \leq \frac{I_{v+1}}{I_v} \leq \frac{2v+3}{\pi} \quad (v \geq 0). \quad (\text{LER4})$$

The lower bound is used only when it is positive. For $p \geq 10$ define

$$U_{p,2} = \frac{\pi^2}{(2p-5)(2p-3)}, \quad U_{p,4} = \frac{\pi^4}{(2p-9)(2p-7)(2p-5)(2p-3)},$$

$$b_p = \frac{U_{p,2}/8 + U_{p,4}/2}{p + 1/2}. \quad (\text{LER5})$$

Telescoping the respective two or four positive lower bounds in LER4 proves $I_{p-2}/I_p \leq U_{p,2}$ and $I_{p-4}/I_p \leq U_{p,4}$. Put

$$A_p = \frac{p + 1/2}{\alpha_\Gamma}, \quad u_p = \frac{I_{p-2}/(8I_p) + e_p/I_p}{p + 1/2}.$$

Then LER3 gives the exact positive ratio

$$\frac{I_{p+1}}{I_p} = A_p(1 - u_p), \quad |u_p| \leq b_p \leq \frac{7}{4p^3} < \frac{1}{100} \quad (p \geq 10). \quad (\text{LER6})$$

For the numerical estimate, $\pi < 22/7$ implies $\pi^2 < 10$ and $\pi^4 < 100$. Every factor $2p-9$, $2p-7$, $2p-5$, $2p-3$ is at least p when $p \geq 10$. Hence $b_p \leq 5/(4p^3) + 50/p^5 \leq 7/(4p^3)$, proving every stated constant.

The logarithmic step with a fifth-order remainder. For $p \geq 12$ define

$$V_p = \frac{\alpha_\Gamma^2}{(p-3/2)(p-1/2)}, \quad z_p = \frac{V_p}{8(p+1/2)} = \frac{\pi^2}{32(p-3/2)(p-1/2)(p+1/2)}. \quad (\text{LER7})$$

There is an exact real η_p satisfying

$$\boxed{\log \frac{I_{p+1}}{I_p} = \log \frac{p+1/2}{\alpha_\Gamma} - z_p + \eta_p, \quad |\eta_p| \leq \frac{51}{p^5}.} \quad (\text{LER8})$$

To prove it without assuming a moment expansion, multiply the two identities LER6 at $p-2, p-1$ and invert their positive product:

$$\frac{I_{p-2}}{I_p} = \frac{V_p}{(1-u_{p-2})(1-u_{p-1})}. \quad (\text{LER9})$$

For $|u| \leq b \leq 1/100, |v| \leq c \leq 1/100$, direct subtraction gives

$$\left| \frac{1}{(1-u)(1-v)} - 1 \right| \leq \frac{b+c+bc}{(1-b)(1-c)} \leq 2(b+c).$$

Indeed $bc \leq (b+c)/200$ and $(1-b)(1-c) \geq 9801/10000$; $10050/9801 < 2$ proves the last bound. LER6 and LER9 thus imply

$$\left| \frac{I_{p-2}}{I_p} - V_p \right| \leq 2V_p(b_{p-2} + b_{p-1}) \leq \frac{7V_p}{(p-2)^3}. \quad (\text{LER10})$$

Consequently the exact u_p satisfies

$$|u_p - z_p| \leq \frac{7V_p/[8(p-2)^3] + U_{p,4}/2}{p+1/2}. \quad (\text{LER11})$$

For $|u| < 1$, integration of the derivative of $\log(1-u) + u$ gives

$$|\log(1-u) + u| \leq \frac{u^2}{2(1-|u|)}.$$

Combining this with LER6 and LER11 bounds $|\eta_p|$ by

$$\frac{7V_p/[8(p-2)^3] + U_{p,4}/2}{p+1/2} + \frac{b_p^2}{2(1-b_p)}. \quad (\text{LER12})$$

Here are explicit estimates for the advertised simpler constant. For $p \geq 12, p-3/2 \geq 7p/8, p-1/2 \geq 23p/24$, and $\pi^2 < 10$ give $V_p < 3/p^2$. Also $p-2 \geq 5p/6$. The first term in LER12 is therefore bounded by

$$\frac{21}{8} \left(\frac{6}{5} \right)^3 p^{-6} + 50p^{-5} < 5p^{-6} + 50p^{-5}.$$

Using LER6, the second term is at most $[49/(16p^6)](50/99) < 2p^{-6}$. Thus LER12 is less than $50p^{-5} + 7p^{-6} < 51p^{-5}$, which proves LER8.

The original scalar endpoint and an exact telescoping correction. For integers $a \geq 6$ and $l \geq 1$ define

$$B(x) = \frac{1}{(x-3/2)(x-1/2)} \quad (x > 3/2), \quad K_{a,l} = \frac{\pi^2}{64} [B(2a) - B(2a+2l)] > 0. \quad (\text{LER13})$$

Summing LER8 at every original integer $p = 2a, \dots, 2a + 2l - 1$ gives

$$\log \frac{\mu_{a+l}}{\mu_a} = \sum_{j=0}^{2l-1} \log \frac{2a+j+1/2}{\pi/2} - K_{a,l} + \varepsilon_{a,l}, \quad |\varepsilon_{a,l}| \leq \frac{51l}{16a^5}. \quad (\text{LER14})$$

In particular no square-pushforward index is substituted for an all-power moment index: $\mu_{a+l}/\mu_a = I_{2a+2l}/I_{2a}$ is precisely the telescoped ratio. The rational term telescopes because its exact partial fraction is

$$z_p = \frac{\pi^2}{64} [B(p) - B(p+1)]. \quad (\text{LER15})$$

The sum of the error bounds in LER8 is at most $51(2l)/(2a)^5 = 51l/(16a^5)$. All constants in LER14 are therefore uniform over $a = q, \dots, q+l$, as well as over every other integer $a \geq 6$.

At $l = 1$, this gives the requested individual Christoffel endpoint:

$$\log c_0^{(a)} = \log \frac{(4a+1)(4a+3)}{\pi^2} - \frac{\pi^2}{64} [B(2a) - B(2a+2)] + \varepsilon_{a,1}, \quad |\varepsilon_{a,1}| \leq \frac{51}{16a^5}. \quad (\text{LER16})$$

The full original low moment remains the product of these coefficients; the corrections in LER16 add to the single term in LER14, with no loss from bounding the exact telescoping contribution.

Explicit finite expansion along the original packet. Retain $k = 4l + 1$, $e = 1 + k(m-1)$ and $q = e(k+1)^2 = e(4l+2)^2$, with $l, m \geq 1$. Thus $q \geq 36$. Set

$$\begin{aligned} F_{q,l} &= 2l \log \frac{4q}{\pi} + \frac{l^2}{q} - \frac{l(16l^2-1)}{48q^2} + \frac{l^2(8l^2-1)}{48q^3}, \\ T_{q,l} &= \frac{l(768l^4 - 160l^2 + 7)}{7680q^4} > 0, \quad E_{q,l} = \frac{51l}{16q^5}. \end{aligned} \quad (\text{LER17})$$

There is an exact $r_{q,l}$ with $-T_{q,l} \leq r_{q,l} \leq 0$ such that

$$\log \frac{\mu_{q+l}}{\mu_q} = F_{q,l} - K_{q,l} + r_{q,l} + \varepsilon_{q,l}, \quad |\varepsilon_{q,l}| \leq E_{q,l}. \quad (\text{LER18})$$

Indeed for every $x \geq 0$,

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \int_0^x \frac{t^3}{1+t} dt,$$

and the last term belongs to $[-x^4/4, 0]$. Apply this exact formula to $x_j = (j+1/2)/(2q)$, $0 \leq j < 2l$, in LER14. The required literal power sums, for $M = 2l$, are

$$\begin{aligned} \sum_{j=0}^{M-1} (j+1/2) &= \frac{M^2}{2}, \quad \sum_{j=0}^{M-1} (j+1/2)^2 = \frac{M(4M^2-1)}{12}, \\ \sum_{j=0}^{M-1} (j+1/2)^3 &= \frac{M^2(2M^2-1)}{8}, \quad \sum_{j=0}^{M-1} (j+1/2)^4 = \frac{M(48M^4-40M^2+7)}{240}. \end{aligned} \quad (\text{LER19})$$

Each identity follows by expanding the left-hand summand and summing integer powers; alternatively its right-hand difference at $M+1$, M equals $(M+1/2)^r$ and its value at $M=0$ is zero, proving it by induction. Substitution proves every coefficient and the sign in LER18. The additional rational correction has the elementary bound

$$0 < K_{q,l} = \sum_{p=2q}^{2q+2l-1} z_p \leq \frac{3l}{32q^3}, \quad (\text{LER20})$$

since the established $V_p < 3/p^2$ gives $z_p < 3/(8p^3)$.

Exact insertion into the entire Gamma functional. Keep the original paired high determinant $Z_a = H_n^{(a)} H_{n+1}^{(a)} (H_n^{(a+1)})^2$ with $n = q/2$, and the original product term P_k from LET19. Write

$$\mathcal{V}_k^{\text{high}} = P_k + \log \frac{Z_{q+l}}{Z_q}, \quad W_k = \mathcal{V}_k^{\text{high}} - \log \frac{\mu_{q+l}}{\mu_q}. \quad (\text{LER21})$$

These are the literal original determinant terms, with all four high blocks retained. LER18 proves the finite receiving interval

$$\boxed{\mathcal{V}_k^{\text{high}} - F_{q,l} + K_{q,l} - E_{q,l} \leq W_k \leq \mathcal{V}_k^{\text{high}} - F_{q,l} + K_{q,l} + T_{q,l} + E_{q,l}.} \quad (\text{LER22})$$

In particular the Gamma rational correction is positive in W_k , while the l^2/q term is negative there. The one-sided Taylor error is also positive after the original low-endpoint minus sign. This formula updates the scalar contribution only; it leaves the original high determinants, quotient maps, arithmetic contribution and their stated bounds intact for their own calculations.

Finally LER17–20 prove, along the actual packet, the more precise low-endpoint limit

$$\log \frac{\mu_{q+l}}{\mu_q} - 2l \log \frac{4q}{\pi} - \frac{l^2}{q} \longrightarrow 0. \quad (\text{LER23})$$

To check this directly, $q \geq (4l + 2)^2$ bounds each remaining term by a constant times respectively $l^{-1}, l^{-2}, l^{-3}, l^{-5}, l^{-9}$ as $l \rightarrow \infty$, with the displayed exact finite bounds available in place of these orders. For $m = 1, l^2/q \rightarrow 1/16$; for every fixed $m \geq 2, l^2/q \rightarrow 0$ since $q = [1 + (4l + 1)(m - 1)](4l + 2)^2$ grows cubically in l . These conclusions concern the exact original scalar moment ratio; the complete W_k continues to include both original high ranks.

10 A uniform finite partition estimate from the original equilibrium

QLG1. Exact measures and parameter domain. Fix the compact parameter rectangle

$$\mathcal{K} = [3/2, 5/2] \times [\pi/2, 3\pi/2], \quad V_{\alpha,\beta}(x) = \beta\sqrt{x} - \alpha \log x \quad (x > 0). \quad (\text{QLG1})$$

The equilibrium proved in EIQ1–33 has density

$$\rho(x) = \frac{\beta\sqrt{(b-x)(x-a)}}{4\pi^2 x} \int_0^\infty \frac{\sqrt{s} ds}{(x+s)\sqrt{(a+s)(b+s)}} \quad (a < x < b), \quad \rho(x) = 0 \quad (x \notin (a, b)), \quad (\text{QLG2})$$

where the exact endpoints are those of EIQ3–5. In particular $0 < a < b < \infty$, the density is bounded and continuous, positive in the interior, and has integral one. Write

$$d\mu(x) = \rho(x) dx, \quad P(z) = \int_a^b \log|z-t| d\mu(t) \quad (z \in \mathbb{C}), \quad I(\mu) = \iint \log|x-y| d\mu(x) d\mu(y),$$

$$F(\alpha, \beta) = I(\mu) - \int V_{\alpha,\beta} d\mu. \quad (\text{QLG3})$$

The original Euler equality and full exterior inequality in EIQ17–19 give a real number ℓ for which

$$D(x) = V_{\alpha,\beta}(x) - 2P(x) - \ell \geq 0 \quad (x > 0), \quad D(x) = 0 \quad (a \leq x \leq b). \quad (\text{QLG4})$$

Integrating the equality on the support gives the exact sign identities

$$\ell = \int V_{\alpha,\beta} d\mu - 2I(\mu), \quad F(\alpha, \beta) = -\ell - I(\mu). \quad (\text{QLG5})$$

The two integration measures treated below are, separately,

$$d\lambda_0(x) = dx, \quad d\lambda_\omega(x) = d\omega(x) = \frac{1}{2}x^{-1/2}e^{-\sqrt{x}} dx. \quad (\text{QLG6})$$

The map $x = r^2$ gives $\int d\omega = \int_0^\infty e^{-r} dr = 1$. For either of these measures, with its stated density w_λ , define the literal finite integral

$$J_n^\lambda(\alpha, \beta) = \frac{1}{n!} \int_{(0,\infty)^n} \exp\left(2 \sum_{i<j} \log|x_i - x_j| - n \sum_{i=1}^n V_{\alpha,\beta}(x_i)\right) \prod_{i=1}^n d\lambda(x_i), \quad n \geq 1. \quad (\text{QLG7})$$

No source measure, scale, factorial, or exponent has been absorbed into F . The unweighted integral without the Heine factor is exactly $n!J_n^\lambda$.

QLG2. Constants proved finite on the whole rectangle. Let

$$a_* = \min_{(\alpha,\beta) \in \mathcal{K}} a(\alpha, \beta) > 0, \quad B_* = \max_{(\alpha,\beta) \in \mathcal{K}} b(\alpha, \beta) < \infty, \quad \beta_- = \pi/2, \quad \beta_+ = 3\pi/2. \quad (\text{QLG8})$$

These are explicit compact-parameter extrema of the uniquely specified endpoint integrals EIQ3–5. Their asserted strict inequalities follow from the continuous positive endpoints proved in EIQ6 and compactness of $\mathcal{K}$. Thus their definition requires no existence or regularity assumption about an unspecified minimizing measure. For $x \in [a, b]$, the integral in QLG2 satisfies

$$\int_0^\infty \frac{\sqrt{s} ds}{(x+s)\sqrt{(a+s)(b+s)}} \leq \int_0^\infty \frac{\sqrt{s} ds}{(a_*+s)^2} = \frac{\pi}{2\sqrt{a_*}}.$$

For the last equality substitute $s = a_*r^2$, followed by $r = \tan \theta$; the resulting integral is

$$2a_*^{-1/2} \int_0^{\pi/2} \sin^2 \theta d\theta.$$

Since $\sqrt{(b-x)(x-a)} \leq (b-a)/2 \leq B_*/2$, it follows that

$$0 \leq \rho(x) \leq M_* := \frac{\beta_+ B_*}{16\pi a_*^{3/2}} \quad \text{for all real } x \text{ and all parameters in } \mathcal{K}. \quad (\text{QLG9})$$

In particular $B_* M_* \geq 1$, by the mass-one identity and support length at most B_* .

The entropy relative to either stated measure is

$$H_\lambda(\mu) = \int_a^b \rho(x) \log \frac{\rho(x)}{w_\lambda(x)} dx, \quad 0 \log 0 = 0. \quad (\text{QLG10})$$

It is absolutely integrable: the function $u |\log u|$ is bounded on $[0, M_*]$, and $\log w_\lambda$ is bounded on the fixed compact interval $[a_*, B_*]$. In the two cases its upper bounds are

$$H_{\lambda_0}(\mu) \leq \log M_*, \quad H_{\lambda_\omega}(\mu) \leq \log M_* + \log 2 + \frac{1}{2} \log B_* + \sqrt{B_*}. \quad (\text{QLG11})$$

All pair logarithmic energies on this support are absolutely integrable. For the negative part, integration against a density bounded by M_* is bounded above by $M_* \int_{-1}^1 -\log |s| ds = 2M_*$, uniformly in the centre. For the positive part the bound $\log^+ B_*$ suffices.

Put

$$T_\lambda = \int_0^\infty e^{-D(x)} d\lambda(x). \quad (\text{QLG12})$$

For $\lambda = \lambda_\omega$, QLG4 and the mass of ω give $0 < T_{\lambda_\omega} \leq 1$. For the Lebesgue measure a finite uniform bound can also be displayed entirely. For every $x > 0$,

$$P(x) \leq \log(1+x) + \log(1+B_*), \quad P(x) \geq -2M_*.$$

The lower bound uses only the negative part estimate just proved. Evaluating QLG4 at any point of $[a, b]$ therefore yields

$$\ell \leq L_* := \beta_+ \sqrt{B_*} + \frac{5}{2} \max\{|\log a_*|, |\log B_*|\} + 4M_*.$$

Consequently

$$T_{\lambda_0} \leq e^{L_*} (1+B_*)^2 \int_0^\infty (1+x)^2 (1+x^{5/2}) e^{-\beta-\sqrt{x}} dx =: T_* < \infty. \quad (\text{QLG13})$$

Here $x^\alpha \leq 1+x^{5/2}$ for all parameters in $\mathcal{K}$. The integral in QLG13 equals, by $x = r^2$ and elementary repeated integration by parts,

$$2 \left(\frac{1!}{\beta_-^2} + \frac{2 \cdot 3!}{\beta_-^4} + \frac{5!}{\beta_-^6} + \frac{6!}{\beta_-^7} + \frac{2 \cdot 8!}{\beta_-^9} + \frac{10!}{\beta_-^{11}} \right). \quad (\text{QLG14})$$

QLG3. The planar logarithmic comparison, with its proof. We use the plane only to construct an exact comparison of logarithmic kernels; the original particles and potential remain on $(0, \infty)$. If η is a finite signed measure of compact support on $\mathbb{R}^2$, has total mass zero, and $\iint |\log |z-w|| d|\eta|(z) d|\eta|(w) < \infty$, then

$$I(\eta) := \iint \log |z-w| d\eta(z) d\eta(w) \leq 0. \quad (\text{QLG15})$$

To prove this, for $d > 0$ integrate the elementary identity

$$-\log d = \frac{1}{2} \int_0^\infty \frac{e^{-td^2} - e^{-t}}{t} dt.$$

It follows by differentiation in d and evaluation at $d = 1$. On any finite subinterval of $(0, \infty)$, the integral on the right has the same sign as $-\log d$ and absolute value at most $|\log d|$. The stipulated logarithmic integrability therefore permits dominated passage to the full integral against $d\eta^2$. The constant term vanishes by the zero mass. The elementary Gaussian Fourier integral in each of the two real coordinates gives

$$-I(\eta) = \frac{1}{2} \int_0^\infty \frac{dt}{t} \iint e^{-t|z-w|^2} d\eta(z) d\eta(w),$$

$$\iint e^{-t|z-w|^2} d\eta(z) d\eta(w) = \frac{1}{4\pi t} \int_{\mathbb{R}^2} e^{-|\xi|^2/(4t)} \left| \int e^{i\xi \cdot z} d\eta(z) \right|^2 d\xi \geq 0. \quad (\text{QLG16})$$

For completeness the scalar Fourier identity is obtained by differentiating $\int e^{-u^2/(4t)} e^{isu} du$ with respect to s , integrating by parts in u , and using its value $\sqrt{4\pi t}$ at zero. The product of the two scalar identities has the displayed factor $(4\pi t)^{-1}$. All finite Gaussian integrations against the measures are absolutely convergent. This proves QLG15 without requiring an equilibrium theorem on the plane.

Let $c_{x,\varepsilon}$ be probability arc measure on the circle $z = x + \varepsilon e^{i\theta}$, where the real centre x is arbitrary and $\varepsilon > 0$. Its exact logarithmic potential is

$$\int \log |z - w| dc_{x,\varepsilon}(z) = \log \max\{\varepsilon, |x - w|\}. \quad (\text{QLG17})$$

Indeed, if $|x - w| > \varepsilon$, expand $\log |1 - re^{i\theta}| = -\sum_{j \geq 1} r^j \cos(j\theta)/j$ with $r = \varepsilon/|x - w| < 1$; all nonconstant terms average to zero. If $|x - w| < \varepsilon$, use the same expansion with $r = |x - w|/\varepsilon$. At equality the assertion is the elementary integral $\int_0^{2\pi} \log |1 - e^{i\theta}| d\theta = 0$. To verify this last integral directly, write $|1 - e^{i\theta}| = 2 \sin(\theta/2)$, and put $A = \int_0^{\pi/2} \log(\sin u) du$. Reflection gives the same integral for $\log \cos u$; the double-angle identity then gives $2A = -(\pi/2) \log 2 + A$, whence $A = -(\pi/2) \log 2$. The endpoint logarithms are integrable, for example from $2u/\pi \leq \sin u \leq u$ on $[0, \pi/2]$.

For finite centres x, y , the logarithmic kernel is absolutely integrable against $c_{x,\varepsilon} \otimes c_{y,\varepsilon}$. Its positive part is bounded by a finite constant depending on the centres and ε . Applying QLG17 first to this bounded positive part minus the negative part, Tonelli's theorem and the finite value $\log \max\{\varepsilon, |x - w|\}$ show that the negative part also has finite double integral. This argument includes identical and tangent circles. The same argument proves absolute integrability of the circle- μ cross kernel, using compactness of the support of μ ; the μ - μ term was proved above. In particular QLG15 applies to a finite mixture of these circles minus μ .

Two consequences of QLG17, retaining both self and cross terms, are

$$I(c_{x,\varepsilon}) = \log \varepsilon, \quad I(c_{x,\varepsilon}, c_{y,\varepsilon}) \geq \log \max\{\varepsilon, |x - y|\} \geq \log |x - y|. \quad (\text{QLG18})$$

The first equality applies QLG17 at w on its own circle. For the second inequality, the first circle average is $\log \max\{\varepsilon, |x - w|\} \geq \log |x - w|$; averaging the right side on the second circle and applying QLG17 again gives its asserted value. When $x = y$ the last lower bound is understood in the extended sense.

QLG4. A finite upper estimate on the original particle domain. For fixed particles $x_1, \dots, x_n > 0$, put $\nu_\varepsilon = n^{-1} \sum_i c_{x_i, \varepsilon}$. By QLG18 and QLG15, respectively,

$$2 \sum_{i < j} \log |x_i - x_j| \leq n^2 I(\nu_\varepsilon) - n \log \varepsilon, \quad I(\nu_\varepsilon) \leq 2I(\nu_\varepsilon, \mu) - I(\mu). \quad (\text{QLG19})$$

There is no potential evaluated at a complex particle in these identities. The exact comparison back to the original real potential is

$$0 \leq \int P(z) dc_{x,\varepsilon}(z) - P(x) = \int_{|t-x| < \varepsilon} \log \frac{\varepsilon}{|t-x|} \rho(t) dt \leq 2M_* \varepsilon. \quad (\text{QLG20})$$

The last constant is the explicit integral $M_* \int_{-\varepsilon}^{\varepsilon} \log(\varepsilon/|s|) ds = 2M_*\varepsilon$. Combining QLG19–20 and then substituting QLG4–5 proves, for every original particle configuration,

$$2 \sum_{i < j} \log |x_i - x_j| - n \sum_i V_{\alpha, \beta}(x_i) \leq n^2 F - n \sum_i D(x_i) - n \log \varepsilon + 4M_* n^2 \varepsilon. \quad (\text{QLG21})$$

The coefficient $4M_*$ comes from twice the logarithmic potential and the bound $2M_*\varepsilon$; the coefficient of the self energy is $-n \log \varepsilon$. Take $\varepsilon = 1/n$. Since $D \geq 0$ and $n \geq 1$, $\int e^{-nD} d\lambda \leq T_\lambda$. Integrating QLG21 gives the finite positive upper bound

$$\log J_n^\lambda \leq n^2 F + 4M_* n + n \log T_\lambda + n \log n - \log(n!). \quad (\text{QLG22})$$

In particular the original integrals in QLG7 are finite. The elementary inequality $\log(n!) \geq \int_1^n \log t dt = n \log n - n + 1$ also proves

$$\log J_n^\lambda \leq n^2 F + n(1 + 4M_* + \log T_\lambda) - 1. \quad (\text{QLG23})$$

The inequality for $n!$ follows by monotonicity of $\log t$ on each interval $[j-1, j]$, for $j = 2, \dots, n$, and is exact at $n = 1$.

QLG5. Quantile lower estimate and the exact factorial cancellation. Because ρ is continuous and positive on (a, b) , there are unique endpoints

$$a = t_0 < t_1 < \dots < t_n = b, \quad \int_{t_{j-1}}^{t_j} \rho(x) dx = \frac{1}{n}, \quad \Delta_j = t_j - t_{j-1} > 0, \quad p_j(x) = n\rho(x) \mathbf{1}_{(t_{j-1}, t_j)}(x). \quad (\text{QLG24})$$

The endpoints have measure zero. Each $p_j(x)dx$ is a probability measure with bounded density, on a compact interval of positive length. Every pair logarithmic integral below is therefore absolutely integrable, including self integrals and integrals over touching intervals, by the same bounded-density estimate used after QLG11. Also $\int p_j |\log(p_j/w_\lambda)| dx < \infty$, since $p_j \leq nM_*$ and $\log w_\lambda$ is bounded on the support.

The product cell $\prod_{j=1}^n (t_{j-1}, t_j)$ and its $n!$ coordinate permutations are disjoint. Symmetry of the integrand and the product integration measure in QLG7 implies that the Heine factor $1/n!$ cancels these $n!$ equal integrals exactly. Restricting to their union therefore gives

$$J_n^\lambda \geq \int_{\prod_j (t_{j-1}, t_j)} \exp \left(2 \sum_{i < j} \log |x_i - x_j| - n \sum_i V(x_i) \right) \prod_j w_\lambda(x_j) dx_j. \quad (\text{QLG25})$$

On this cell use the probability density $\prod_j p_j(x_j)$. The logarithm of the integrand relative to this probability is integrable in absolute value, by the preceding estimates and boundedness of V on $[a, b]$. The elementary convex inequality $e^u \geq e^v(1+u-v)$, integrated with v equal to the expectation of this logarithm, proves Jensen's lower bound directly. Since $\sum_j p_j = n\rho$, its pair and potential contributions are exactly

$$\sum_{i \neq j} I(p_i, p_j) - n \sum_j \int V p_j dx = n^2 F - \sum_j I(p_j).$$

Here $I(p_i, p_j) = \iint \log |x-y| p_i(x) p_j(y) dx dy$ and $I(p_j) = I(p_j, p_j)$. Its entropy contribution is exactly

$$\sum_j \int p_j \log \frac{p_j}{w_\lambda} dx = n \log n + n H_\lambda(\mu). \quad (\text{QLG26})$$

Moreover $|x-y| \leq \Delta_j$ on the j -th cell, whence $I(p_j) \leq \log \Delta_j$. Concavity of the logarithm, proved for example by its negative second derivative, and $\sum_j \Delta_j = b-a$ now give

$$\sum_j I(p_j) \leq \sum_j \log \Delta_j \leq n \log \frac{b-a}{n}. \quad (\text{QLG27})$$

Thus the $n \log n$ in QLG26 cancels that in QLG27, and the complete lower bound is

$$\boxed{\log J_n^\lambda(\alpha, \beta) \geq n^2 F(\alpha, \beta) - n\{\log(b-a) + H_\lambda(\mu)\}.} \quad (\text{QLG28})$$

This also proves strict positivity of QLG7. No lower bound on the density at an endpoint or on a quantile interval length has been used.

QLG6. The uniform order- n theorem. For every $(\alpha, \beta) \in \mathcal{K}$, every integer $n \geq 1$, and each of the two measures QLG6, QLG22 and QLG28 are the explicit two-sided finite estimate

$$\boxed{-n\{\log(b-a) + H_\lambda(\mu)\} \leq \log J_n^\lambda - n^2 F \leq 4M_* n + n \log T_\lambda + n \log n - \log(n!).} \quad (\text{QLG29})$$

For entirely parameter-uniform constants set

$$\begin{aligned} L_0 &= \log(B_* M_*), & L_\omega &= \log(B_* M_*) + \log 2 + \frac{1}{2} \log B_* + \sqrt{B_*}, \\ C_0 &= \max\{0, L_0, 1 + 4M_* + \log^+ T_*\}, & C_\omega &= \max\{0, L_\omega, 1 + 4M_*\}. \end{aligned} \quad (\text{QLG30})$$

These constants are finite by QLG8–14. Since $b-a \leq B_*$, QLG11 bounds the lower side of QLG29 by $-nL_\lambda$; QLG12–14 and QLG23 bound its upper side by nC_λ . Consequently

$$\boxed{\left| \log J_n^{\lambda_0}(\alpha, \beta) - n^2 F(\alpha, \beta) \right| \leq C_0 n, \quad \left| \log J_n^{\lambda_\omega}(\alpha, \beta) - n^2 F(\alpha, \beta) \right| \leq C_\omega n.} \quad (\text{QLG31})$$

For the integral Z_n^λ without the Heine factor, the exact relation $\log Z_n^\lambda = \log J_n^\lambda + \log(n!)$ gives

$$\left| \log Z_n^\lambda - n^2 F - \log(n!) \right| \leq C_\lambda n, \quad \left| \log Z_n^\lambda - n^2 F \right| \leq n \log n + C_\lambda n. \quad (\text{QLG32})$$

The theorem therefore retains the distinction between the original Heine determinant integral and the integral with labelled particles. The gain to order n in QLG31 follows from the proved circle comparison and the quantile-cell cancellation, with the same exact potential, positive-half-line particle domain, and equilibrium of EI1–33 throughout.

11 The original Gamma Hankel return with a finite moving-shift error

Original objects and exact square pushforward. Retain the original packet integers and the original density:

$$k = 4l+1, \quad e = 1+k(m-1), \quad q = e(k+1)^2 = 2n, \quad l, m \geq 1, \quad \sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi}. \quad (\text{QGT1})$$

In particular $n \geq 18$ and $n \geq 18l$: indeed $n/l \geq 8l + 8 + 2/l \geq 18$, the last expression increasing for $l \geq 1$. For every real $a > -1/2$ define the actual even moments and their Hankel determinant by

$$\mu_a = \int_{\mathbb{R}} |y|^{2a} \sigma(y) dy, \quad H_s^{(a)} = \det[\mu_{a+i+j}]_{0 \leq i, j < s}, \quad s \geq 1. \quad (\text{QGT2})$$

For the integer values used in the programme this is precisely $\mu_a = m_{2a}$, with $\mu_0 = \sqrt{2\pi}$ retained. The extension to real a is only the displayed integral on the same original measure. Evenness of σ and $t = y^2$ give, including both real half-lines,

$$\mu_a = \int_0^\infty t^{a-1/2} \sigma(\sqrt{t}) dt, \quad H_s^{(a)} = \frac{1}{s!} \int_{(0, \infty)^s} \prod_{i < j} (t_i - t_j)^2 \prod_{i=1}^s t_i^{a-1/2} \sigma(\sqrt{t_i}) dt_i. \quad (\text{QGT3})$$

To verify the second identity, expand the two determinants $\det[t_i^{j-1}]_{i,j=1}^s$ as finite permutation sums, multiply them, and integrate each monomial against the displayed positive product measure. Every integral is finite: the density is bounded at zero, $a > -1/2$, and the global Gamma upper bound below gives exponential decay in $\sqrt{t}$ at infinity. Relabeling the integration variables in each term gives $s!$ copies of the defining moment determinant. The Vandermonde identity follows by its alternating zeros and its leading coefficient one. This proves QGT3 with its literal factorial. It is strictly positive because its integrand is positive on each open cell of distinct positive coordinates.

The actual Gamma density, including its prefactor. Write $\alpha_\Gamma = \pi/2$, $D = 1/16 + \sqrt{3}/18$, and set

$$C_+ = \max\{\sigma(0)e^{\alpha_\Gamma}, \sigma(1)e^{\alpha_\Gamma+D}\}, \quad r(y) = y^{1/2}e^{\alpha_\Gamma y} \sigma(y) \quad (y > 0). \quad (\text{QGT4})$$

Integrating the proved original logarithmic derivative GDR3 establishes, by the complete deduction below, on the full stated domains

$$\sigma(y) \geq \sigma(1)(1+y)^{-1/2}e^{-\alpha_\Gamma y} \quad (y \geq 0), \quad \sigma(y) \leq C_+ y^{-1/2}e^{-\alpha_\Gamma y} \quad (y > 0). \quad (\text{QGT5})$$

Here is also the short full deduction needed for this use. GDR3 says $L(y) = \alpha_\Gamma + 1/(2y) + \epsilon(y)/y$ and $|\epsilon(y)| \leq (1/8 + \sqrt{3}/9)y^{-2}$. Integrating from 1 to $y \geq 1$ shows

$$\log r(y) = \log \sigma(1) + \alpha_\Gamma - \int_1^y \epsilon(u) du/u, \quad \left| \int_1^y \epsilon(u) du/u \right| \leq D.$$

Since $\alpha_\Gamma > D$, this gives $r(y) \geq \sigma(1)$ for $y \geq 1$ and the upper in QGT4 there. GDR7 implies $L(y) > 0$ for $y > 0$, so for $0 \leq y \leq 1$ one has $\sigma(1) \leq \sigma(y) \leq \sigma(0)$. On that interval $(1+y)^{-1/2}e^{-\alpha_\Gamma y} \leq 1$ and $y^{-1/2}e^{-\alpha_\Gamma y} \geq e^{-\alpha_\Gamma}$ when $y > 0$. These facts prove both inequalities in QGT5 and keep the density's finite value at zero. The elementary inequality $\alpha_\Gamma > D$ follows, for example, from $\pi > 3$ and $\sqrt{3} < 2$.

Use exactly the dilation $t_i = q^2 x_i$ in QGT3 and put

$$\alpha = \frac{a-3/4}{s}, \quad \beta = \frac{\pi q}{2s}, \quad \mathcal{J}_{s,q}(\alpha, \beta) = \frac{1}{s!} \int_{(0, \infty)^s} \prod_{i < j} (x_i - x_j)^2 \prod_i e^{-s(\beta\sqrt{x_i} - \alpha \log x_i)} r(q\sqrt{x_i}) dx_i. \quad (\text{QGT6})$$

Then the exact original determinant is

$$H_s^{(a)} = q^{2as+2s(s-1)+s/2} \mathcal{J}_{s,q}(\alpha, \beta). \quad (\text{QGT7})$$

Indeed the Vandermonde contributes $q^{2s(s-1)}$, and each density factor transforms as

$$(q^2x)^{a-1/2}\sigma(q\sqrt{x})q^2dx = q^{2a+1/2}x^{a-3/4}e^{-\alpha\Gamma q\sqrt{x}}r(q\sqrt{x})dx.$$

This accounts separately for the original Jacobian, the quarter shift, and the factor $q^{s/2}$. None of them is incorporated in the equilibrium free energy.

A uniform finite partition theorem for this exact source. Let $\mathcal{K} = [3/2, 5/2] \times [\pi/2, 3\pi/2]$. Use the uniquely proved equilibrium $\mu_{\alpha,\beta}$ of EIQ on the positive half-line, its actual endpoints $a(\alpha, \beta)$, $b(\alpha, \beta)$, and its energy

$$F(\alpha, \beta) = \iint \log|x-y| d\mu_{\alpha,\beta}(x)d\mu_{\alpha,\beta}(y) - \int (\beta\sqrt{x} - \alpha \log x) d\mu_{\alpha,\beta}(x). \quad (\text{QGT8})$$

In this paragraph only, the endpoint $a(\alpha, \beta)$ carries arguments to distinguish it from the moment exponent a . Let a_* , B_* , M_* , C_0 be the finite constants explicitly defined and proved in QLG8–14 and QLG30. Set

$$L_0 = \log(B_*M_*), \quad C_* = \sigma(1) \left(\frac{\sqrt{a_*}}{1 + \sqrt{a_*}} \right)^{1/2} > 0, \\ C = \max\{0, C_0 + \log C_+, L_0 - \log C_*\}. \quad (\text{QGT9})$$

For every $q \geq 1$, integer $s \geq 1$ and $(\alpha, \beta) \in \mathcal{K}$,

$$\boxed{|\log \mathcal{J}_{s,q}(\alpha, \beta) - s^2 F(\alpha, \beta)| \leq Cs.} \quad (\text{QGT10})$$

For the upper estimate QGT5 gives $r(q\sqrt{x}) \leq C_+$ everywhere on the original positive half-line. QLG31 for the Lebesgue reference measure therefore proves $\log \mathcal{J} \leq s^2 F + (C_0 + \log C_+)s$. For the lower estimate retain the precise quantile cells QLG24–28 for this same equilibrium and its same s . Every coordinate in these cells lies in $[a_*, B_*]$, where QGT5 gives

$$r(q\sqrt{x}) \geq \sigma(1) \left(\frac{q\sqrt{x}}{1 + q\sqrt{x}} \right)^{1/2} \geq C_*.$$

The last inequality uses $q \geq 1$, $x \geq a_*$ and monotonicity of $u/(1+u)$. QLG25–28 bound the integral over precisely these cells, with their $s!$ permutations cancelling the original Heine factor. Multiplying its integrand by the lower factor C_*^s yields $\log \mathcal{J} \geq s^2 F - (L_0 - \log C_*)s$. This proves QGT10. A global lower bound by $x^{-3/4}$ at zero was not used; the original density and its exact positive-domain factor $r(q\sqrt{x})$ remain in QGT6.

Combining QGT7 and QGT10 gives the exact finite decomposition

$$\boxed{\log H_s^{(a)} = [2as + 2s(s-1) + s/2] \log q + s^2 F \left(\frac{a-3/4}{s}, \frac{\pi q}{2s} \right) + E_{s,q}(a), \quad |E_{s,q}(a)| \leq Cs.} \quad (\text{QGT11})$$

The bound holds whenever the displayed parameter pair belongs to $\mathcal{K}$. The quantity $E_{s,q}(a)$ denotes the exact difference in this equality; no derivative estimate for it has been assumed.

A quantitative moving secant, proved from convexity. Write $\mathcal{L}(\alpha, \beta) = \partial_\alpha F(\alpha, \beta)$, the smooth original logarithmic moment proved in EIQ26 and EIQ29–30. The fixed-interval integral EIQ29 and its proved differentiated majorants give a finite constant

$$M = \max_{(\alpha,\beta) \in \mathcal{K}} |\partial_\alpha \mathcal{L}(\alpha, \beta)|, \quad A = M/2 + 2C. \quad (\text{QGT12})$$

For the two original ranks $s = n, n + 1$, every real initial exponent $a \in [q, q + l + 1]$, and the original positive integer shift l , one has

$$\left| \log \frac{H_s^{(a+l)}}{H_s^{(a)}} - 2sl \log q - s^2 \left[F\left(\frac{a+l-3/4}{s}, \frac{\pi q}{2s}\right) - F\left(\frac{a-3/4}{s}, \frac{\pi q}{2s}\right) \right] \right| \leq Al\sqrt{s}. \quad (\text{QGT13})$$

All parameter neighbourhoods used to prove this bound lie in the stated rectangle. Indeed put $u = (a - 3/4)/s, t = l/s$ and $\eta = s^{-1/2} < 1/4$. For either rank,

$$u \geq 2 - \frac{11}{4s} \geq 2 - \frac{11}{72}, \quad u + t \leq 2 + \frac{2l + 1/4}{n} \leq 2 + \frac{1}{8}.$$

Here $n \geq 18l$ and $n \geq 18$ give $(2l + 1/4)/n \leq 1/9 + 1/72 = 1/8$. Thus $[u - \eta, u + t + \eta] \subset (3/2, 5/2)$. The second coordinate is π for $s = n$ and $\pi n/(n + 1)$ for $s = n + 1$, both lying in $[\pi/2, 3\pi/2]$.

For fixed q, s, β , define $g(v) = s^{-2} \log \mathcal{J}_{s,q}(v, \beta)$ on this neighbourhood. This function is twice continuously differentiable and convex. To justify differentiation under its integral, on every compact subinterval of $v > 0$ the exponent at zero is a positive power of x , and the Gamma upper in QGT5 bounds the tails by a polynomial times $\exp(-s\beta\sqrt{x})$. The finite Vandermonde polynomial, multiplied by any of the finitely many $\log x_i$ or their products arising in the first two derivatives, has an integrable majorant. Division by the strictly positive integral is legitimate. Under its exact probability density,

$$(\log \mathcal{J}_{s,q})''(v) = s^2 \text{Var}_v \left(\sum_i \log x_i \right), \quad g''(v) = \text{Var}_v \left(\sum_i \log x_i \right) \geq 0.$$

This proves convexity without a property of the remainder.

For each $v \in [u, u + t]$, convexity gives

$$\frac{g(v) - g(v - \eta)}{\eta} \leq g'(v) \leq \frac{g(v + \eta) - g(v)}{\eta}.$$

QGT10 implies $|g - F| \leq C/s$. The integral formula for a difference quotient of F and the bound in QGT12 give, for both signs,

$$|g'(v) - \mathcal{L}(v, \beta)| \leq \frac{M\eta}{2} + \frac{2C}{s\eta}. \quad (\text{QGT14})$$

In detail, the upper quotient of F is $\eta^{-1} \int_v^{v+\eta} \mathcal{L}(w, \beta) dw$ and differs from $\mathcal{L}(v, \beta)$ by at most $\eta^{-1} \int_0^\eta Mr dr = M\eta/2$; the lower quotient has the identical bound. Integrating QGT14 over $[u, u + t]$ and multiplying by s^2 gives $Al\sqrt{s}$ because $\eta = s^{-1/2}$ and $t = l/s$. In QGT7 the terms $2s(s - 1) \log q$ and $(s/2) \log q$ cancel exactly between the two values of a , while $2as \log q$ contributes $2sl \log q$. This proves QGT13, retaining the rank $n + 1$ and the source prefactor.

All four original high blocks in one exact centre. Define the same positive high product as in LET and LER,

$$Z_a = H_n^{(a)} H_{n+1}^{(a)} (H_n^{(a+1)})^2.$$

For the original pair of endpoints $a = q, q + l$ put

$$\begin{aligned} c_{q,l} = & n^2 \left[F\left(\frac{q+l-3/4}{n}, \pi\right) - F\left(\frac{q-3/4}{n}, \pi\right) \right] \\ & + (n+1)^2 \left[F\left(\frac{q+l-3/4}{n+1}, \frac{\pi n}{n+1}\right) - F\left(\frac{q-3/4}{n+1}, \frac{\pi n}{n+1}\right) \right] \\ & + 2n^2 \left[F\left(\frac{q+l+1/4}{n}, \pi\right) - F\left(\frac{q+1/4}{n}, \pi\right) \right], \end{aligned} \quad (\text{QGT15})$$

$$\mathcal{U}_{q,l} = Al(3\sqrt{n} + \sqrt{n+1}). \quad (\text{QGT16})$$

Adding QGT13 with the displayed coefficients 1, 1, 2 yields the finite exact decomposition

$$\log \frac{Z_{q+l}}{Z_q} = (4q+2)l \log q + C_{q,l} + \xi_{q,l}, \quad |\xi_{q,l}| \leq \mathcal{U}_{q,l}. \quad (\text{QGT17})$$

The coefficient of the logarithm is exactly $2l[n + (n+1) + 2n] = 2l(2q+1)$. The two odd blocks, the different high ranks, and the original quarter shifts are all explicit in QGT15.

Finite return with the low endpoint and every product correction. Use the original P_k of WGP1 and the original scalar moment of LER1, so that

$$W_k = P_k + \log(Z_{q+l}/Z_q) - \log(\mu_{q+l}/\mu_q). \quad (\text{QGT18})$$

Put $\tilde{P}_{q,l} = P_k + 4lq \log q$, the exact finite sum WGP3, and retain $F_{q,l}, K_{q,l}, T_{q,l}, E_{q,l}$ with exactly the definitions LER13 and LER17. Define the following centre in terms of the already proved original objects:

$$\mathcal{V}_{q,l} = \tilde{P}_{q,l} + C_{q,l} + 2l \log q - F_{q,l} + K_{q,l}. \quad (\text{QGT19})$$

QGT17 and the signed remainder LER18 give

$$\mathcal{V}_{q,l} - \mathcal{U}_{q,l} - E_{q,l} \leq W_k \leq \mathcal{V}_{q,l} + \mathcal{U}_{q,l} + T_{q,l} + E_{q,l}. \quad (\text{QGT20})$$

To check the signs, LER18 is $\log(\mu_{q+l}/\mu_q) = F_{q,l} - K_{q,l} + r_{q,l} + \varepsilon_{q,l}$ with $-T_{q,l} \leq r_{q,l} \leq 0$ and $|\varepsilon_{q,l}| \leq E_{q,l}$. Substitution in QGT18 gives $W_k = \mathcal{V}_{q,l} + \xi_{q,l} - r_{q,l} - \varepsilon_{q,l}$, which proves QGT20. The original logarithmic powers cancel exactly: $-4lq + (4q+2)l - 2l = 0$, since the logarithmic part of $F_{q,l}$ is $2l \log(4q/\pi)$.

For an entirely integral-and-rational centre instead of the finite source sum, set

$$\begin{aligned} p_{q,l}^c &= -4lq(2 \log 2 - 1) - 4l^2 \log 2 + \frac{8l^3 + l}{6q} - \frac{l^4 + l^2/4}{q^2}, \\ p_{q,l}^- &= \frac{l}{6q} + \frac{8l^3 + l}{12q^3}, \quad p_{q,l}^+ = \frac{l^2}{3q^2} + \frac{8l^4 + 2l^2}{6q^4} + \frac{7(192l^5 + 80l^3 - 2l)}{1440q^3}, \\ \mathcal{V}_{q,l}^c &= p_{q,l}^c + C_{q,l} + 2l \log q - F_{q,l} + K_{q,l}. \end{aligned} \quad (\text{QGT21})$$

WGP13 is the proved finite statement $-p_{q,l}^- \leq \tilde{P}_{q,l} - p_{q,l}^c \leq p_{q,l}^+$. Consequently the full original Gamma functional satisfies

$$\mathcal{V}_{q,l}^c - p_{q,l}^- - \mathcal{U}_{q,l} - E_{q,l} \leq W_k \leq \mathcal{V}_{q,l}^c + p_{q,l}^+ + \mathcal{U}_{q,l} + T_{q,l} + E_{q,l}. \quad (\text{QGT22})$$

In particular this bound uses the exact Gamma source and both original low and high endpoint contributions; its finite constants were obtained from their integrals and the proved equilibrium.

An explicit bound around the previously proved leading coefficient. Write $\mathcal{L}_0 = \mathcal{L}(2, \pi)$ and $\mathcal{C}_\Gamma = 2\mathcal{L}_0 - 4(2 \log 2 - 1)$, exactly the positive coefficient proved in GEL. The actual dilation proved in EIQ32 gives $\mathcal{L}(v, \pi n/(n+1)) = \mathcal{L}(v, \pi) + 2 \log(1+1/n)$. For $d \in \{-3/4, -11/4, 1/4\}$, the fundamental theorem of calculus and QGT12 give

$$\left| s^2 \int_{2+d/s}^{2+(d+l)/s} \mathcal{L}(v, \pi) dv - s\mathcal{L}_0 \right| \leq M \left(|d|l + \frac{l^2}{2} \right). \quad (\text{QGT23})$$

Indeed set $v = 2 + (d+r)/s$, $0 \leq r \leq l$; the integrand's difference from $\mathcal{L}_0$ is at most $M|d+r|/s$, and $|d+r| \leq |d| + r$. Apply this respectively at ranks $n, n+1, n$ with coefficients 1, 1, 2 in QGT15. Since $3/4 + 11/4 + 2(1/4) = 4$, the result is

$$|\mathcal{C}_{q,l} - (2q+1)l\mathcal{L}_0 - 2(n+1)l \log(1+1/n)| \leq M(2l^2 + 4l). \quad (\text{QGT24})$$

Every v in these integrals is in the rectangle already verified above. Define the fully explicit finite quantity

$$\begin{aligned} \mathcal{D}_{q,l} = & M(2l^2 + 4l) + l|\mathcal{L}_0| + 2(n+1)l \log(1 + 1/n) + 4l^2 \log 2 \\ & + \frac{8l^3 + l}{6q} + \frac{l^4 + l^2/4}{q^2} + \max\{p_{q,l}^-, p_{q,l}^+\} + 2l|\log(4/\pi)| + \frac{l^2}{q} \\ & + \frac{l(16l^2 - 1)}{48q^2} + \frac{l^2(8l^2 - 1)}{48q^3} + K_{q,l} + T_{q,l} + E_{q,l}. \end{aligned} \quad (\text{QGT25})$$

All displayed terms are nonnegative for $l \geq 1$. Combining QGT18, QGT17, WGP13, LER17–18 and QGT24, with the triangle inequality applied only after their signs have been recorded, proves

$$|W_k - lq\mathcal{C}_\Gamma| \leq \mathcal{U}_{q,l} + \mathcal{D}_{q,l}. \quad (\text{QGT26})$$

The sign-sensitive sharper statement remains QGT22; QGT26 is its explicit estimate about the original leading coefficient.

Along the actual packet, $\mathcal{D}_{q,l} = O(l^2 + l)$ with a constant independent of m . This follows term by term from $q \geq (4l + 2)^2$, the displayed rational quantities, LER20, and $\log(1 + 1/n) \leq 1/n$ (integrate $1/(1+t) \leq 1$ on $[0, 1/n]$). Furthermore $\mathcal{U}_{q,l} \leq 4Al\sqrt{n+1}$. Thus the quantitative theorem proves, for the same original integers,

$$W_k = lq\mathcal{C}_\Gamma + O(l\sqrt{q} + l^2). \quad (\text{QGT27})$$

The finite bounds QGT22 and QGT26 specify this error whenever an order symbol is insufficient. For each fixed $m \geq 2$, the exact formula $q = [1 + (4l + 1)(m - 1)](4l + 2)^2$ gives $l/\sqrt{q} \rightarrow 0$ and $l^2/q \rightarrow 0$. Dividing QGT26 by q therefore proves the stronger original-return limit

$$\frac{W_k - lq\mathcal{C}_\Gamma}{q} \rightarrow 0 \quad (l \rightarrow \infty, m \geq 2 \text{ fixed}). \quad (\text{QGT28})$$

For $m = 1$, the same exact calculation gives a finite bound on $(W_k - lq\mathcal{C}_\Gamma)/q$, since $q = (4l + 2)^2$ and $l/\sqrt{q} \rightarrow 1/4$. The bound does not identify the limit of that quotient: the retained high-block uncertainty $\mathcal{U}_{q,l}/q$ need not tend to zero in this case. This identifies precisely what this finite estimate resolves at the original q scale, with every packet relation unchanged.

12 The finite original $m = 1$ return bound at the q scale

Retain the same original measure, determinants, source products, and constants A, M, C_Γ of QGT1–28. At the original value $m = 1$, the packet is exactly

$$k = 4l + 1, \quad q = (4l + 2)^2, \quad n = q/2, \quad l \geq 1. \quad (\text{QGQ1})$$

The explicit two finite nonnegative bounds in QGT16 and QGT25 have the following limits on this packet:

$$\frac{\mathcal{U}_{q,l}}{q} \longrightarrow \frac{A}{\sqrt{2}}, \quad \frac{\mathcal{D}_{q,l}}{q} \longrightarrow \frac{M}{8} + \frac{\log 2}{4}. \quad (\text{QGQ2})$$

For the first limit use the exact identity

$$\frac{\mathcal{U}_{q,l}}{q} = A \frac{l}{\sqrt{q}} \left(\frac{3}{\sqrt{2}} + \sqrt{\frac{1}{2} + \frac{1}{q}} \right).$$

Here $l/\sqrt{q} = l/(4l + 2) \rightarrow 1/4$, so the limit is $A(1/4)(4/\sqrt{2})$, as displayed.

For the second limit, extract from the actual QGT25 sum the two terms $2Ml^2$ and $4l^2 \log 2$. Their quotients by q tend to $M/8$ and $(\log 2)/4$, since $l^2/q \rightarrow 1/16$. Every other summand tends to zero after division by q . Here is the term-by-term justification using the original summands. The terms $4Ml$, $l|\mathcal{L}_0|$, and $2l|\log(4/\pi)|$ are fixed constants times l ; their quotients by q tend to zero. The actual rank correction satisfies

$$0 \leq 2(n + 1)l \log(1 + 1/n) \leq 2(n + 1)l/n \leq 4l,$$

because $\log(1 + u) = \int_0^u (1 + t)^{-1} dt \leq u$ and $n \geq 1$. The two next rational terms of QGT25 have sizes $O(l)$ and $O(1)$, respectively:

$$\frac{8l^3 + l}{6q}, \quad \frac{l^4 + l^2/4}{q^2}.$$

To verify these orders directly, substitute $q = l^2(4 + 2/l)^2$; the remaining denominators are bounded below by positive constants for $l \geq 1$. The two nonnegative product bounds have orders $p_{q,l}^- = O(l^{-1})$ and $p_{q,l}^+ = O(l^{-1})$. For p^- its two terms have orders l^{-1}, l^{-3} . For p^+ its three displayed terms have orders l^{-2}, l^{-4}, l^{-1} , respectively. Each follows by the same literal substitution in QGT21; the maximum therefore also has order l^{-1} . The remaining three displayed polynomial low terms

$$\frac{l^2}{q}, \quad \frac{l(16l^2 - 1)}{48q^2}, \quad \frac{l^2(8l^2 - 1)}{48q^3}$$

have orders $1, l^{-1}, l^{-2}$, respectively. Finally the actual LER bounds retained in QGT25 give $0 < K_{q,l} \leq 3l/(32q^3) = O(l^{-5})$, $T_{q,l} = O(l^{-3})$, and $E_{q,l} = O(l^{-9})$. These are all the terms of QGT25, and each of these remaining terms divided by $q \geq 16l^2$ tends to zero. This proves the second limit in QGQ2.

Apply the finite original inequality QGT26 for every integer l , divide it by the same q , and use both proved limits in QGQ2. The result is the quantitative statement

$$\limsup_{l \rightarrow \infty} \left| \frac{W_k - lqC_\Gamma}{q} \right| \leq \frac{A}{\sqrt{2}} + \frac{M}{8} + \frac{\log 2}{4} \quad (m = 1). \quad (\text{QGQ3})$$

All constants in this inequality are the finite constants of the proved original-source QGT calculation. The two limits in QGQ2 evaluate its explicit error bounds. Inequality QGQ3 alone does not determine a limit for the actual expression inside its absolute value; the exact centre $C_{q,l}$ in QGT15 and the signed finite interval QGT22 remain available for that continuing calculation.

The exact q -scale limit of the proved finite centre. The centre in QGT22 can itself be evaluated one scale further. Use the smooth original equilibrium logarithmic moment of EIQ29–30, with its same parameter domain, and put

$$\mathcal{L}_1 = \partial_\alpha \mathcal{L}(2, \pi), \quad M_2 = \max_{(\alpha, \beta) \in \mathcal{K}} |\partial_\alpha^2 \mathcal{L}(\alpha, \beta)| < \infty, \quad \mathcal{C}_\Gamma^{[q,1]} = \frac{\mathcal{L}_1}{8} - \frac{\log 2}{4}. \quad (\text{QGQ4})$$

The maximum is finite by the smoothness proved from the actual fixed-interval density integral in EIQ29 and compactness of the same rectangle $\mathcal{K}$ used in QGT12. Thus $\mathcal{L}_1$ and M_2 are derivatives of that explicitly proved integral.

For the original triples $(s, d) = (n, -3/4), (n+1, -11/4), (n, 1/4)$, all intermediate parameter points below belong to $\mathcal{K}$, by the domain calculation in QGT13. Integrating the derivative of $\mathcal{L}$ twice gives

$$|\mathcal{L}(2+v, \pi) - \mathcal{L}_0 - v\mathcal{L}_1| \leq \frac{M_2 v^2}{2}$$

whenever the interval joining 2 and $2+v$ is in that rectangle. Indeed the exact remainder is $\int_0^v \int_0^u \partial_\alpha^2 \mathcal{L}(2+w, \pi) dw du$; using oriented integrals gives the same absolute bound for $v < 0$. Now use exactly $v = (d+r)/s, 0 \leq r \leq l$, in the integral of $\mathcal{L}$ defining the difference of F . The result is

$$s^2 \left[F \left(2 + \frac{d+l}{s}, \pi \right) - F \left(2 + \frac{d}{s}, \pi \right) \right] = s\mathcal{L}_0 + (dl + l^2/2)\mathcal{L}_1 + \zeta_{s,d,l}, \quad (\text{QGQ5})$$

$$|\zeta_{s,d,l}| \leq \frac{M_2}{6s} [(|d|+l)^3 - |d|^3].$$

For the bound, the exact integral of the remainder is bounded by $M_2(2s)^{-1} \int_0^l (|d|+r)^2 dr$, which is the quantity displayed.

Apply QGQ5 to the literal three terms of QGT15 with their weights 1, 1, 2. The rank- $n+1$ beta change contributes exactly $2(n+1)l \log(1+1/n)$ by EIQ32. Therefore

$$C_{q,l} = (2q+1)l\mathcal{L}_0 + (2l^2 - 3l)\mathcal{L}_1 + 2(n+1)l \log(1+1/n) + \zeta_{q,l},$$

$$|\zeta_{q,l}| \leq \frac{M_2}{6} \left\{ \frac{(l+3/4)^3 - (3/4)^3}{n} + \frac{(l+11/4)^3 - (11/4)^3}{n+1} + 2 \frac{(l+1/4)^3 - (1/4)^3}{n} \right\}. \quad (\text{QGQ6})$$

The weighted rank sum is $n + (n+1) + 2n = 2q+1$. The weighted starting shift is exactly $-3/4 - 11/4 + 2(1/4) = -3$, and the weights have sum four; these give the coefficient $2l^2 - 3l$ in this formula.

Substitute QGQ6 into the exact centre $\mathcal{V}_{q,l}^c$ of QGT21 and use the full expression for $F_{q,l}$ of LER17. After displaying all original terms, the resulting identity is

$$\begin{aligned} \mathcal{V}_{q,l}^c - lq\mathcal{C}_\Gamma &= (2\mathcal{L}_1 - 4\log 2)l^2 + l\mathcal{L}_0 - 3l\mathcal{L}_1 + 2(n+1)l \log(1+1/n) \\ &\quad + \frac{8l^3 + l}{6q} - \frac{l^4 + l^2/4}{q^2} - 2l \log(4/\pi) - \frac{l^2}{q} \\ &\quad + \frac{l(16l^2 - 1)}{48q^2} - \frac{l^2(8l^2 - 1)}{48q^3} + K_{q,l} + \zeta_{q,l}. \end{aligned} \quad (\text{QGQ7})$$

The equality retains the original source, rank correction, low endpoint, and signed cubic terms. Its first coefficient comes from the actual four-block high contribution $2l^2\mathcal{L}_1$ and the original source contribution $-4l^2 \log 2$.

On the original packet QGQ1, the finite bound in QGQ6 gives $\zeta_{q,l} = O(l)$, because $n = q/2 \geq 8l^2$ and each numerator there is a cubic polynomial in l with zero constant term. Every term of QGQ7 after its first term is consequently $O(l)$ or smaller: the rank correction was bounded by $4l$ above, the rational terms have the orders individually proved before QGQ3, and $K_{q,l} \leq 3l/(32q^3)$. Dividing QGQ7 by the literal $q = (4l+2)^2$ proves

$$\boxed{\frac{\mathcal{V}_{q,l}^c - lq\mathcal{C}_\Gamma}{q} \longrightarrow \mathcal{C}_\Gamma^{[q,1]} = \frac{\partial_\alpha \mathcal{L}(2, \pi)}{8} - \frac{\log 2}{4}.} \quad (\text{QGQ8})$$

Finally QGT22 gives, before taking any limit, the two signed errors

$$-p_{q,l}^- - \mathcal{U}_{q,l} - E_{q,l} \leq W_k - \mathcal{V}_{q,l}^c \leq p_{q,l}^+ + \mathcal{U}_{q,l} + T_{q,l} + E_{q,l}.$$

The two product bounds and both low errors divided by q tend to zero by the estimates already proved above. The remaining bound $\mathcal{U}_{q,l}/q$ tends to $A/\sqrt{2}$ by QGQ2. Combining this finite interval with QGQ8 proves the sharper conclusion

$$\boxed{\limsup_{l \rightarrow \infty} \left| \frac{W_k - lq\mathcal{C}_\Gamma}{q} - \mathcal{C}_\Gamma^{[q,1]} \right| \leq \frac{A}{\sqrt{2}} \quad (m = 1).} \quad \text{(QGQ9)}$$

Here the centre is forced by the original finite formula. The remaining width is the proved error of its high-block transport. The estimate does not replace that width by zero or assign an unproved limit to the actual original Gamma functional.

13 The Gamma orders produced by the existing exterior and sum maps

The source order in this section is one of the two orders already present in the construction. The determinant size is the actual exterior degree. Every relation coefficient remains in the resulting density.

13.1 Original coordinates and the exact source measure

Retain

$$k = 4l + 1, \quad l, m \geq 1, \quad e = 1 + k(m - 1), \quad q = e(k + 1)^2, \quad c = k/2, \quad 0 < \delta < 1/2, \quad \gamma > 2,$$

$$\chi(S) = \prod_{a,d=0}^k [S - c - (2a - k)\delta - i(2d - k)\gamma]^e, \quad s \in \{1, k\}, \quad b = s/2, \quad M_s = (2\pi)^{s/2}. \quad (\text{IG01})$$

Here $s = 1$ is the fixed source σ and $s = k$ is the original tensor Gamma source. The original line throughout is $S = c + iy$. The change of letter from the second root index to d leaves every root and its order unchanged.

For each real $v > 0$ define a density on the real coordinate by

$$\rho_v(y) = \frac{2^v}{4\pi\Gamma(v)} |\Gamma(v/2 + iy/2)|^2, \quad dm_{0,s}(y) = M_s \rho_b(y) dy. \quad (\text{IG02})$$

Thus the full mass M_s is retained. In particular $dm_{0,1} = |\Gamma(1/4 + iy/2)|^2 dy / (2\pi) = d\sigma$. For the other existing source, the recurrence of the Gamma function at $k/4 = l + 1/4$ gives

$$dm_{0,k}(y) = \beta_k \prod_{j=0}^{l-1} [y^2 + (2j + \frac{1}{2})^2] d\sigma(y), \quad \beta_k = \frac{(2\pi)^{k/2}}{\sqrt{2}\Gamma(k/2)}.$$

Indeed the squared Gamma recurrence contributes 4^{-l} times the displayed product. The coefficient in IG02, multiplied by 4^{-l} and divided by the coefficient $1/(2\pi)$ of σ , is $M_k 2^{k/2-1-2l} / \Gamma(k/2) = \beta_k$. This proves agreement with the original tensor source including its full polynomial factor and mass. For real $|t| < \pi/2$ and real ξ , the exact transforms are

$$\begin{aligned} \int_{\mathbb{R}} e^{ty} \rho_v(y) dy &= (\cos t)^{-v}, \\ \int_{\mathbb{R}} e^{i\xi y} \rho_v(y) dy &= (\cosh \xi)^{-v}, \\ \phi_s(t) &:= \int e^{ty} dm_{0,s} = M_s (\cos t)^{-b}. \end{aligned} \quad (\text{IG03})$$

Here is a direct justification of the source identities and the integrability used below. The beta integral with the substitution $u = \exp(2x)$ gives

$$\frac{\Gamma(v/2 + iy/2)\Gamma(v/2 - iy/2)}{\Gamma(v)} = 2 \int_{\mathbb{R}} e^{iyx} (2 \cosh x)^{-v} dx.$$

Consequently $\rho_v(y) = (2\pi)^{-1} \int e^{iyx} (\cosh x)^{-v} dx$. Fourier inversion applies: the function and its first two derivatives are integrable, and two integrations by parts make its transform integrable. This proves the characteristic identity and mass one in IG03, with the displayed Fourier convention. More strongly, for every $0 < \theta < \pi/2$, move the x contour to $x + i\theta \operatorname{sgn}(y)$. The analytic branch of $(\cosh x)^{-v}$ is fixed by its positive values on the real axis; it exists in this strip because $\cosh x$ has no zero there. Its modulus decays as $\exp(-v|\operatorname{Re} x|)$ on each closed smaller strip. The vertical sides of the rectangle therefore tend to zero and $\rho_v(y) \leq C_{v,\theta} \exp(-\theta|y|)$. This proves absolute convergence of every

polynomial moment with an exponential factor e^{ty} when $|t| < \pi/2$. Holomorphic continuation of the characteristic identity through this strip then gives the first identity in IG03. Strict positivity follows from the zero-free Euler Gamma product in $\text{Re } z > 0$:

$$\frac{1}{\Gamma(z)} = z \exp(\gamma_E z) \prod_{n=1}^{\infty} (1 + z/n) \exp(-z/n).$$

Indeed the logarithms of the factors after their linear terms are removed converge locally uniformly, and none of the factors vanishes there. This product follows by taking the beta-integral limit $\Gamma(z) = \lim_{n \rightarrow \infty} n! n^z / [z(z+1) \cdots (z+n)]$; the linear sum is absorbed by $\gamma_E = \lim_n (\sum_{j=1}^n 1/j - \log n)$.

The natural line of the source of order s is $U = b + iy$. Its presentation on the original line is the exact translation

$$U = S + (b - c), \quad F(S) \mapsto F(U - (b - c)). \quad (\text{IG04})$$

This is an algebra isomorphism, with inverse given by the opposite translation, and its coefficient matrix on each degree filtration is triangular with diagonal one. In particular, using $s = 1$ in IG02 retains the original line $c + i\mathbb{R}$ through IG04. The original tensor identification is exact in the real coordinate:

$$dm_{0,k} = (dm_{0,1})^{*k}.$$

Indeed the characteristic function of the convolution on the right is

$$[M_1(\cosh \xi)^{-1/2}]^k = M_k(\cosh \xi)^{-k/2},$$

which is the characteristic function of the left side by IG03. The finite-measure Fourier uniqueness argument given after IG017 proves equality. The natural individual coordinates are $1/2 + iy_j$, whose sum is $c + i \sum_j y_j$. If the individual sources are first written on the fixed original line as $S_j = c + iy_j$, the same sum is obtained by the exact map $\sum_j S_j \mapsto \sum_j S_j + k(1/2 - c)$, applying IG04 to all k factors. This retains the full tensor mass $M_1^k = M_k$ and its original centre c .

13.2 The actual relation exterior vector and its sum measure

The relation multiplier is the full polynomial

$$W_\chi(y) = |\chi(c + iy)|^2 = \prod_{a,d=0}^k ([(2a - k)\delta]^2 + [y - (2d - k)\gamma]^2)^e = \sum_{j=0}^{2q} w_j y^j. \quad (\text{IG05})$$

All its original parameters occur in the displayed product. It is real, even, strictly positive on $\mathbb{R}$, and monic of degree $2q$. Evenness follows by the permutation $d \mapsto k - d$. Strict positivity follows because k is odd, so $(2a - k)\delta \neq 0$ for every a . Thus $w_{2q} = 1$ and $w_j = 0$ for odd j ; no other coefficient is removed.

For $r \geq 1$ put

$$\begin{aligned} \Delta_r(y) &= \prod_{1 \leq i < j \leq r} (y_j - y_i), \\ d\lambda_{\chi,s,r}(y_1, \dots, y_r) &= \frac{1}{r!} \Delta_r(y)^2 \prod_{i=1}^r W_\chi(y_i) dm_{0,s}(y_i), \\ \Sigma_r(y) &= \sum_{i=1}^r y_i, \\ \nu_{\chi,s,r} &= (\Sigma_r)_* \lambda_{\chi,s,r}. \end{aligned} \quad (\text{IG06})$$

The notation with subscript $\emptyset$ instead of χ means that each factor W_χ is replaced by the literal factor 1. For $r = 0$, the product space is the one-point space with mass one, the sum is 0, and both pushforward measures are δ_0 .

This is precisely the existing exterior measure. To check all phases and factorials, write $S_i = c + iy_i$ and define the exterior vector in the product source Hilbert space by

$$\psi_{\chi,r}(y) = \frac{1}{\sqrt{r!}} \det[\chi(S_i) S_i^{j-1}]_{i,j=1}^r = \frac{i^{r(r-1)/2}}{\sqrt{r!}} \Delta_r(y) \prod_{i=1}^r \chi(c + iy_i). \quad (\text{IG07})$$

Its squared modulus times $\prod dm_{0,s}(y_i)$ is exactly IG06. The phase in IG07 is retained in the exterior map below. Expanding both determinants and relabelling integration variables shows that $\|\psi_{\chi,r}\|^2$ equals the Gram determinant of $\chi, \chi S, \dots, \chi S^{r-1}$ in the original source. The $r!$ identical permutation contributions cancel the displayed $1/r!$. This also proves the following full Laplace identity:

$$\begin{aligned} g_{\chi,s}(t) &= \int e^{ty} W_{\chi}(y) dm_{0,s}(y) = W_{\chi}(\partial_t) \phi_s(t), \\ H_{\chi,s,r}(t) &:= \int e^{tY} d\nu_{\chi,s,r}(Y) = \det[g_{\chi,s}^{(i+j)}(t)]_{i,j=0}^{r-1}. \end{aligned} \quad (\text{IG08})$$

All expansions and differentiations here are absolutely convergent by the exponential estimate after IG03, since the remaining factors are fixed polynomials. For real $|t| < \pi/2$ the matrix on the right is the Gram matrix of $1, y, \dots, y^{r-1}$ for the strictly positive measure $e^{ty} W_{\chi}(y) dm_{0,s}(y)$. It is positive definite: a nonzero polynomial is nonzero on a real open interval. Thus $H_{\chi,s,r}(t) > 0$.

13.3 The exact differential polynomial and its degree

Put $z = \tan t$ and define polynomials recursively by

$$\begin{aligned} P_0(z) &= 1, \\ P_{j+1}(z) &= bzP_j(z) + (1+z^2)P_j'(z), \\ Q_0(z) &= \sum_{j=0}^{2q} w_j P_j(z), \\ Q_{j+1}(z) &= bzQ_j(z) + (1+z^2)Q_j'(z). \end{aligned} \quad (\text{IG09})$$

The product rule proves $\phi_s^{(j)}(t) = \phi_s(t)P_j(\tan t)$ and $g_{\chi,s}^{(j)}(t) = \phi_s(t)Q_j(\tan t)$. These are exact polynomial recursions in all the w_j , with no limit or discarded term.

For polynomials $F_0, \dots, F_{r-1}$ write $\text{Wr}_z(F_0, \dots, F_{r-1}) = \det[\partial_z^i F_j]_{i,j=0}^{r-1}$. Then

$$B = br + r(r-1), \quad D_{\chi,s,r}(z) = \text{Wr}_z(Q_0, \dots, Q_{r-1}), \quad H_{\chi,s,r}(t) = M_s^r (\cos t)^{-B} D_{\chi,s,r}(\tan t). \quad (\text{IG010})$$

To prove this without losing a determinant factor, the matrix in IG08 is the t -Wronskian of $g, g', \dots, g^{(r-1)}$. In a Wronskian, multiplying every column function by a common function h multiplies the determinant by h^r : the product-rule matrix on its rows is lower triangular with diagonal h . Changing the independent variable from t to z multiplies its rows by a lower triangular matrix with diagonal $1, z', \dots, (z')^{r-1}$. Its determinant is $(z')^{r(r-1)/2}$. Here $h = \phi_s$ and $z' = 1 + z^2 = (\cos t)^{-2}$. Multiplication of these two factors gives exactly $M_s^r (\cos t)^{-br-r(r-1)}$ in IG010.

The polynomial P_j has parity j , degree j , and leading coefficient $(b)_j = \prod_{h=0}^{j-1} (b+h)$; this follows directly from IG09. The polynomial Q_j has parity j , degree $2q+j$, and leading coefficient $(b)_{2q}(b+2q)_j$. Therefore

$$D_{\chi,s,r}(z) = \sum_{j=0}^{qr} d_j z^{2j}, \quad d_{qr} = ((b)_{2q})^r \prod_{j=0}^{r-1} j!(b+2q)_j > 0, \quad D_{\chi,s,r}(z) > 0 \quad (z \in \mathbb{R}). \quad (\text{IG011})$$

For the degree and leading coefficient, factor the leading coefficient of each Q_j from the Wronskian. If its degrees are $n_j = 2q+j$, the coefficient of its largest possible power of z is $\det[(n_j)_i]_{i,j=0}^{r-1}$. The falling

factorial in row i is a monic polynomial of degree i in n_j ; subtracting lower rows therefore changes it to the ordinary Vandermonde determinant. Its value is $\prod_{i < j} (n_j - n_i) = \prod_{j=0}^{r-1} j!$. The power is $\sum_j (2q + j) - \sum_i i = 2qr$, so the coefficient cannot cancel. The parity of entry (i, j) is $i + j$ modulo two; each determinant term has even total parity $\sum_i i + \sum_j j = r(r-1)$. Finally positivity of D for every real z follows from the positive Gram in IG08 by the bijection $z = \tan t$ between $(-\pi/2, \pi/2)$ and $\mathbb{R}$.

13.4 The Gamma order actually produced without relation factors

For the literal exterior of the source alone, use $Q_0 = 1$ and $Q_j = P_j$ in the same Wronskian calculation. Its degrees are j , so its Wronskian is the positive constant

$$C_{b,r} = \prod_{j=0}^{r-1} j! (b)_j, \quad H_{\emptyset,s,r}(t) = M_s^r C_{b,r} (\cos t)^{-B}, \quad d\nu_{\emptyset,s,r}(Y) = M_s^r C_{b,r} \rho_B(Y) dY. \quad (\text{IG012})$$

The transform identity and Fourier uniqueness prove the density formula, including its whole mass $M_s^r C_{b,r}$. The source order in this sum law is forced by the actual exterior degree:

$$2B = sr + 2r(r-1), \quad d\nu_{\emptyset,s,r} = \frac{M_s^r C_{b,r}}{(2\pi)^B} dm_{0,2B}. \quad (\text{IG013})$$

Here $dm_{0,a}(Y) := (2\pi)^{a/2} \rho_{a/2}(Y) dY$ names the Gamma measure at the produced positive order $a = 2B$; the same notation below is used for the produced orders $2B + 4h$. The input order remains the original $s \in \{1, k\}$. In the second expression $dm_{0,2B}$ is written in the real coordinate Y . Its natural complex line is $U = B + iY$, whereas the original sum coordinate is $T = rc + iY$. The exact translation of these lines and their polynomial algebras is

$$U = T + (B - rc), \quad F(T) \mapsto F(U - (B - rc)). \quad (\text{IG014})$$

It has inverse translation by $rc - B$, preserves every degree, and has coefficient determinant one on every finite degree filtration. If IG04 is first applied to all r factors, their sum is $rb + iY$. The further displacement in IG014 is exactly $r(r-1)$, since $B - rb = r(r-1)$. Thus both the original sum centre and the naturally produced Gamma centre have explicit maps. Equations IG012–IG014 concern $r \geq 1$; the unit case $r = 0$ remains the point mass specified after IG06.

13.5 All relation terms: finite signed Gamma decomposition

For the actual polynomial W_χ , define every coefficient

$$a_h = \sum_{j=h}^{qr} (-1)^{j+h} \binom{j}{h} d_j, \quad 0 \leq h \leq qr. \quad (\text{IG015})$$

Analytic continuation of IG010 to $t = i\xi$, or the characteristic integral in IG08 directly, gives

$$\widehat{\nu}_{\chi,s,r}(\xi) = M_s^r (\cosh \xi)^{-B} D_{\chi,s,r}(i \tanh \xi) = M_s^r \sum_{h=0}^{qr} a_h (\cosh \xi)^{-(B+2h)}. \quad (\text{IG016})$$

Indeed each term $d_j(i \tanh \xi)^{2j}$ is $d_j(-1)^j [1 - (\cosh \xi)^{-2}]^j$; expanding the finite binomial power gives IG015 with its stated signs. Thus

$$d\nu_{\chi,s,r}(Y) = M_s^r \sum_{h=0}^{qr} a_h \rho_{B+2h}(Y) dY = \sum_{h=0}^{qr} \frac{M_s^r a_h}{(2\pi)^{B+2h}} dm_{0,2B+4h}(Y). \quad (\text{IG017})$$

Fourier uniqueness applies to these finite signed measures. For completeness, a finite signed measure with zero Fourier transform has zero convolution with every Gaussian, by Fourier inversion and

Fubini. Letting the Gaussian widths decrease to zero and testing against compactly supported continuous functions gives zero measure. This proves the asserted equality of measures, without a sign assumption on any a_h .

For each term in the second expression of IG017 the original sum line is still $T = rc + iY$. Its natural source line is $U_h = B + 2h + iY$, and its pullback is the explicit translation $U_h = T + (B + 2h - rc)$, with the polynomial map and inverse as in IG014. Thus IG017 adds measures on the same original line; the different natural centres have not been identified implicitly.

The top coefficient satisfies $a_{qr} = d_{qr} > 0$. The largest order forced by the full relation operation is consequently

$$2B + 4qr = sr + 2r(r - 1) + 4qr. \quad (\text{IG018})$$

It equals $sq + 6q^2 - 2q$ at $r = q$, and $s(q + 1) + 6q(q + 1)$ at $r = q + 1$. These orders result from IG06 and the leading coefficient of its original polynomial. All the lower signed constituents in IG017 remain part of the same exact measure.

13.6 A positive polynomial density with all masses retained

The recurrence $\Gamma(z + 1) = z\Gamma(z)$ gives, with no residual power of two,

$$\frac{\rho_{B+2h}(Y)}{\rho_B(Y)} = \frac{\Gamma(B)}{\Gamma(B + 2h)} \prod_{j=0}^{h-1} [Y^2 + (B + 2j)^2]. \quad (\text{IG019})$$

In detail, the ratio of the constants in IG02 is $2^{2h}\Gamma(B)/\Gamma(B + 2h)$; the squared Gamma ratio is $\prod_{j < h} [(B/2 + j)^2 + Y^2/4]$. Each of its h factors contributes $1/4$, cancelling 2^{2h} exactly. Define

$$R_{\chi,s,r}(X) = \sum_{h=0}^{qr} a_h \frac{\Gamma(B)}{\Gamma(B + 2h)} \prod_{j=0}^{h-1} [X + (B + 2j)^2]. \quad (\text{IG020})$$

Then the exact positive measure, in both conventions for the underlying Gamma mass, is

$$\begin{aligned} d\nu_{\chi,s,r}(Y) &= M_s^r \rho_B(Y) R_{\chi,s,r}(Y^2) dY \\ &= \frac{M_s^r}{(2\pi)^B} R_{\chi,s,r}(Y^2) dm_{0,2B}(Y), \end{aligned} \quad (\text{IG021})$$

$$\deg R_{\chi,s,r} = qr, \quad [X^{qr}] R_{\chi,s,r} = d_{qr} \frac{\Gamma(B)}{\Gamma(B + 2qr)} > 0.$$

The source on the second line has the centre map IG014. In its complex coordinate the multiplier is exactly $R_{\chi,s,r}(-(U - B)^2)$; on the original sum line it is $R_{\chi,s,r}(-(T - rc)^2)$.

To prove its strict positivity at every real Y , as opposed to positivity only almost everywhere from the measure identity, write the density of IG06 relative to $dy_1 \cdots dy_r$ as

$$w_{\chi,s,r}(y) = \frac{M_s^r}{r!} \Delta_r(y)^2 \prod_{i=1}^r W_\chi(y_i) \rho_b(y_i).$$

For $r \geq 2$ the coordinate change $y_j = u_j$ for $j < r$ and $y_r = Y - \sum_{j < r} u_j$ has absolute Jacobian one. It gives the actual fibre density

$$p_{\chi,s,r}(Y) = \int_{\mathbb{R}^{r-1}} w_{\chi,s,r}(u_1, \dots, u_{r-1}, Y - \sum_{j < r} u_j) du_1 \cdots du_{r-1}. \quad (\text{IG022})$$

It is finite and continuous. On a compact set of Y , the exponential estimate following IG03 bounds the integrand by an integrable polynomial in $|u|$ times $\exp(-\theta \sum_{j < r} |u_j|)$, with one fixed $0 < \theta < \pi/2$. This also justifies dominated convergence. For each Y , choose any $t_0 \neq 0$ and distinct coordinates $y_i = Y/r + t_0(i - (r + 1)/2)$. Their sum is Y . The Vandermonde is nonzero there, and all the other

factors are strictly positive. They remain positive in an open set of the u variables. Hence $p_{\chi,s,r}(Y) > 0$. For $r = 1$ the same conclusion follows from $p_{\chi,s,1}(Y) = M_s W_\chi(Y) \rho_b(Y) > 0$. The density in IG021 is continuous and equals this density almost everywhere by IG017. Their equality therefore holds everywhere. In particular

$$R_{\chi,s,r}(X) > 0 \quad (X \geq 0), \quad \nu_{\chi,s,r}(\mathbb{R}) = M_s^r D_{\chi,s,r}(0) = M_s^r \sum_{h=0}^{qr} a_h = M_s^r \int \rho_B(Y) R_{\chi,s,r}(Y^2) dY. \quad (\text{IG023})$$

The coefficient sum equals $D(0)$ by the binomial identity in IG016 at $\xi = 0$. Thus the full original relation determinant is present as this mass. The Gamma order in IG013, the full polynomial in IG020, and the signed constituents in IG017 are related by the proved identities IG019–IG023. In particular, no choice of a larger source parameter is needed to form this operation, and the polynomial coefficients are required to recover its actual mass.

13.7 The complete sum map, its adjoint, and the retained kernel

Let $\mathcal{H}_\lambda = L^2(\mathbb{R}^r, \lambda_{\chi,s,r})$ and $\mathcal{H}_\nu = L^2(\mathbb{R}, \nu_{\chi,s,r})$, using the full measures IG06. The sum pullback

$$U_r : \mathcal{H}_\nu \longrightarrow \mathcal{H}_\lambda, \quad (U_r F)(y) = F(\Sigma_r y) \quad (\text{IG024})$$

is an isometry, since its squared norm is exactly the defining pushforward integral. Its range is closed. We use complex inner products linear in the second variable. For $r \geq 2$ its adjoint is the exact fibre integral

$$(U_r^* h)(Y) = \frac{1}{p_{\chi,s,r}(Y)} \int_{\mathbb{R}^{r-1}} h(u, Y - \sum u_j) w_{\chi,s,r}(u, Y - \sum u_j) du. \quad (\text{IG025})$$

Every factor in this formula is specified in IG022. The denominator is the actual positive fibre mass; the source measure and its total mass in IG023 are unchanged. In the Euclidean surface measure on $\Sigma_r^{-1}(Y)$, the same numerator is $r^{-1/2} \int_{\Sigma_r^{-1}(Y)} h w d\mathcal{H}^{r-1}$. Indeed $|\nabla \Sigma_r| = \sqrt{r}$, and the parametrization in IG022 has surface Jacobian $\sqrt{r}$.

For a proof of the adjoint formula and its domain, Cauchy–Schwarz on each fibre gives

$$|\int h w du|^2 \leq p(Y) \int |h|^2 w du.$$

After division by $p(Y)$ and integration in Y , this gives $\|U_r^* h\|_\nu \leq \|h\|_\lambda$, so the formula is well defined for every equivalence class in $\mathcal{H}_\lambda$. Fubini applied first to bounded integrable functions proves $\langle U_r F, h \rangle_\lambda = \langle F, U_r^* h \rangle_\nu$; the bound extends it to all L^2 functions. Taking $h = U_r F$ in IG025 gives $U_r^* U_r F = F$. Consequently the exact orthogonal projection and complete kernel are

$$\begin{aligned} \Pi_r &= U_r U_r^*, \quad \mathcal{H}_\lambda = U_r \mathcal{H}_\nu \oplus \ker U_r^*, \\ \ker U_r^* &= \left\{ h \in \mathcal{H}_\lambda : \right. \\ &\quad \left. \int_{\mathbb{R}^{r-1}} h(u, Y - \sum u_j) w_{\chi,s,r}(u, Y - \sum u_j) du = 0 \right. \\ &\quad \left. \text{for almost every } Y \right\}, \\ \|h\|_\lambda^2 &= \|U_r^* h\|_\nu^2 + \|h - U_r U_r^* h\|_\lambda^2. \end{aligned} \quad (\text{IG026})$$

This identifies both the map and every discarded direction in a passage to the sum. For $r = 1$, Σ_1 is the identity, U_1 and U_1^* are the identity between the identical spaces, and the kernel is zero. For $r = 0$ the two Hilbert spaces are the one-dimensional spaces at their unit points, the maps are the identity under that identification, and the kernel is again zero.

For $r \geq 2$ the kernel remains nonzero even in the symmetric subspace. To see this with its exact fibre dependence, let $A(y) = \sum_i y_i^2$ and let

$$\bar{A}(Y) = \frac{1}{p(Y)} \int_{\mathbb{R}^{r-1}} A(u, Y - \sum u_j) w(u, Y - \sum u_j) du.$$

All its moments here exist by the same exponential bound. The symmetric function $A - U_r \bar{A}$ lies in $\ker U_r^*$ by IGO25. It is nonzero: for each fixed Y , A is a nonconstant quadratic function on the sum fibre, and the weight is positive on the open set of distinct coordinates. It therefore has strictly positive fibre variance. This also records a concrete direction that the sum coordinate alone does not reconstruct.

Finally, for $r \geq 1$, the exterior Hilbert space has a complete typed map from the same sum space. Realize $\bigwedge^r L^2(dm_{0,s})$ as the antisymmetric subspace of the product L^2 space with the determinant convention in IGO7. Multiplication by $\psi_{\chi,r}$ gives a unitary map from the symmetric subspace of $\mathcal{H}_\chi$ to that antisymmetric subspace: its norm identity is IGO7, and its inverse is division by $\psi_{\chi,r}$ away from the diagonal. The diagonal has product source measure zero, and χ has no zero on the original line, so this inverse exists almost everywhere. Division takes an antisymmetric function to a symmetric function and the norm identity proves its L^2 membership. Thus

$$\mathcal{J}_r : \mathcal{H}_\nu \longrightarrow \bigwedge^r L^2(dm_{0,s}), \quad \mathcal{J}_r F = \psi_{\chi,r} F(\Sigma_r y) \quad (\text{IGO27})$$

is an isometry with the original determinant phase retained. For an antisymmetric vector v , its exact adjoint is

$$(\mathcal{J}_r^* v)(Y) = \frac{1}{p(Y)} \int_{\mathbb{R}^{r-1}} \overline{\psi_{\chi,r}(u, Y - \sum u_j)} v(u, Y - \sum u_j) \prod_{i=1}^r [M_s \rho_b(y_i)] du, \quad y = (u, Y - \sum u_j). \quad (\text{IGO28})$$

Its kernel is precisely the vectors for which the numerator in IGO28 vanishes for almost every Y ; equivalently it is $\psi_{\chi,r}$ times the symmetric part of the kernel in IGO26. The proof is multiplication of IGO25 by the exact identity $w = |\psi_{\chi,r}|^2 \prod_i [M_s \rho_b(y_i)]$. It also gives the full orthogonal decomposition with projection $\mathcal{J}_r \mathcal{J}_r^*$. The nonzero symmetric kernel vector just constructed supplies a nonzero exterior kernel vector through this unitary multiplication. At $r = 0$, the empty determinant is 1 and the zeroth exterior space is $\mathbb{C}$; the exterior map and its adjoint are the identity between the unit spaces, as for IGO24–IGO26.

For a bounded measurable function f , multiplication by $f(Y)$ on $\mathcal{H}_\nu$ intertwines through IGO24 and IGO27 with multiplication by $f(\Sigma_r y)$ on their target spaces. For the unbounded original sum coordinate, the domain on $\mathcal{H}_\nu$ is $\{F : \int |Y F(Y)|^2 d\nu(Y) < \infty\}$; the identical integral on the target proves the same intertwining on exactly its transported domain. Adding the original constant centre rc and the factor i gives the intertwining for $T = rc + iY$ as well. Together with the translations in IGO14 and after IGO17, these formulas specify the spaces, measures, actions, adjoints, and kernels through which the intrinsically produced Gamma density enters the original spectral-sum construction.

14 The intrinsic exterior mass and the original four-endpoint relation graph

The source order in this section is $s \in \{1, k\}$, with $k = 4l + 1$, $l, m \geq 1$, $e = 1 + k(m - 1)$, $q = e(k + 1)^2$, and $c = k/2$. Retain $0 < \delta < 1/2$, $\gamma > 2$ and the complete original polynomial

$$\chi(S) = \prod_{a,b=0}^k [S - c - (2a - k)\delta - i(2b - k)\gamma]^e. \quad (\text{IFB1})$$

The repeated index b in this product is local to the product; below $b_s = s/2$ denotes the source parameter. We write

$$M_s = (2\pi)^{s/2}, \quad b_s = s/2, \quad dm_{0,s}(y) = \frac{M_s 2^{b_s-2}}{\pi \Gamma(b_s)} \left| \Gamma\left(\frac{b_s}{2} + \frac{iy}{2}\right) \right|^2 dy. \quad (\text{IFB2})$$

For $s = 1$ this is exactly $d\sigma$ in BSL1. For $s = k$ it is exactly the original tensor-Gamma measure in BSL17: repeated use of $\Gamma(z + 1) = z\Gamma(z)$ gives

$$\left| \Gamma\left(l + \frac{1}{4} + \frac{iy}{2}\right) \right|^2 = 4^{-l} \prod_{j=0}^{l-1} (y^2 + (2j + \frac{1}{2})^2) |\Gamma(\frac{1}{4} + \frac{iy}{2})|^2,$$

and the coefficient in IFB2 times 4^{-l} equals $M_k/(2\pi\sqrt{2}\Gamma(k/2))$. Thus no source has been selected by matching its size to q . Both sources are the measures already present in the original calculation. The original evaluation line in both cases is $S = c + iy$.

14.1 The original orthogonal coordinates and the complete quotient map

The original source transform, with its full mass, is

$$\int_{\mathbb{R}} e^{ty} dm_{0,s}(y) = M_s (\cos t)^{-b_s}, \quad |t| < \pi/2. \quad (\text{IFB3})$$

Here is a direct check of its scale. With $\lambda = b_s/2$ and $g(x) = \exp(\lambda x - e^x)$, Euler's integral gives $\widehat{g}(\xi) = \Gamma(\lambda + i\xi)$ for $\widehat{g}(\xi) = \int g(x) e^{i\xi x} dx$. Autocorrelation gives

$$\int e^{iu\xi} |\Gamma(\lambda + i\xi)|^2 d\xi = 2\pi \int g(x + u/2) g(x - u/2) dx = \frac{2\pi \Gamma(2\lambda)}{(2 \cosh(u/2))^{2\lambda}}.$$

The first equality follows by Fourier inversion of the integrable autocorrelation. The function g and every derivative decay exponentially at $-\infty$ and faster than exponentially at $+\infty$, so these Fourier operations and the displayed integrals converge. The last equality follows from $v = e^x$. Substituting $y = 2\xi$, $u = 2w$, and the coefficient in IFB2 gives $M_s (\cosh w)^{-b_s}$. Rotation of Euler's integral through $\theta \operatorname{sgn}(y)$, $0 < \theta < \pi/2$, proves an upper bound $C_{s,\theta} \exp(-\theta|y|)$ for the squared Gamma factor. The small sector arc tends to zero because $\lambda > 0$; the large arc tends to zero because $\cos \theta > 0$. This proves analytic continuation of the characteristic transform to $|\operatorname{Im} w| < \pi/2$ and hence IFB3. In particular all polynomial moments below exist and $\int dm_{0,s} = M_s$.

Define real monic polynomials $p_n^{(s)}(y)$ by

$$\sum_{n=0}^{\infty} p_n^{(s)}(y) \frac{z^n}{n!} = (1 + z^2)^{-b_s/2} \exp(y \arctan z). \quad (\text{IFB4})$$

Branches are analytic at zero. For sufficiently small real z, w , IFB3 and the cosine addition formula show that the integral of the product of these two generating functions is $M_s (1 - zw)^{-b_s}$. The exponential bound used for IFB3 permits each derivative in a neighborhood of zero under the integral. Coefficient comparison gives the exact norms and recurrence

$$h_n^{(s)} = M_s n! (b_s)_n, \quad \int p_n^{(s)} p_j^{(s)} dm_{0,s} = \mathbf{1}_{n=j} h_n^{(s)},$$

$$yp_n^{(s)} = p_{n+1}^{(s)} + n(b_s + n - 1)p_{n-1}^{(s)}, \quad p_0^{(s)} = 1, \quad p_{-1}^{(s)} = 0. \quad (\text{IFB5})$$

The recurrence also follows by differentiating IFB4 and multiplying by $1 + z^2$. Each $p_n^{(s)}$ has parity n . We henceforth retain the superscript s on h_n where needed and suppress it on p_n .

Let $E_\chi = \mathbb{C}[S]/(\chi)$ with its fixed remainder frame $(1, S, \dots, S^{q-1})$, and define actual source polynomials

$$u_n(S) = \frac{p_n((S - c)/i)}{\sqrt{h_n^{(s)}}}, \quad C_s : \mathbb{C}^q \longrightarrow E_\chi, \quad C_s x = \sum_{n=0}^{q-1} x_n [u_n]. \quad (\text{IFB6})$$

The positive square root is used. On the original line $u_n(c + iy)$ equals $p_n(y)/\sqrt{h_n^{(s)}}$. The source polynomials u_n are therefore an orthonormal frame, with all source factors explicitly present. The leading coefficient of u_n in S is $i^{-n}/\sqrt{h_n^{(s)}}$. Thus C_s is triangular and invertible, and

$$|\det C_s|^2 = \prod_{n=0}^{q-1} (h_n^{(s)})^{-1}. \quad (\text{IFB7})$$

This is an exact change of coordinates on the original remainder space, not a replacement of that space or of its frame.

For $j \geq 0$ put

$$v_j = C_s^{-1}[u_{q+j}] \in \mathbb{C}^q, \quad V_r = [v_0 \cdots v_{r-1}], \quad K_{s,r} = I_q + V_r V_r^*, \quad \Omega_{s,r} = \det K_{s,r}. \quad (\text{IFB8})$$

The definitions for $r = 0$ are V_0 the empty q by zero matrix, $K_{s,0} = I_q$, and $\Omega_{s,0} = 1$. These vectors use the same C_s and the same remainder map for every endpoint. Fix $r \geq 0$ and $N = q + r - 1$. The orthonormal source-coordinate map

$$U_{s,N} : \mathcal{P}_N \longrightarrow \mathbb{C}^q \oplus \mathbb{C}^r, \quad \sum_{n < q} x_n u_n + \sum_{j < r} z_j u_{q+j} \longmapsto (x, z)$$

is a linear isometry. Polynomial division and IFB8 give the exact commuting relation

$$J_N U_{s,N}^{-1}(x, z) = C_s(x + V_r z), \quad U_{s,N}(\chi \mathcal{P}_{r-1}) = \{(-V_r z, z) : z \in \mathbb{C}^r\}. \quad (\text{IFB9})$$

To verify equality on the right, a vector in the displayed graph has zero remainder by the first formula and hence is divisible by the original monic χ . Its degree is at most $q + r - 1$, so its quotient has degree at most $r - 1$. Conversely every such multiple has zero remainder, which forces its low coordinate to be $-V_r z$. The map $z \mapsto U_{s,N}^{-1}(-V_r z, z)$ is consequently an exact bijection from the high source coordinates to the original relation space. No relation direction is projected out. When $r = 0$, this means the zero space on both sides.

For a fixed original remainder $a \in E_\chi$, write $w = C_s^{-1}a$. The unique minimum source-norm lift has coordinates

$$(x, z) = (K_{s,r}^{-1}w, V_r^* K_{s,r}^{-1}w), \quad \|P_{\min}\|^2 = w^* K_{s,r}^{-1}w. \quad (\text{IFB10})$$

Indeed $x + V_r z = K_{s,r} K_{s,r}^{-1}w = w$. The displayed vector is orthogonal to every $(-V_r z', z')$, since its inner product with that vector is zero. Every other lift differs by such a relation vector, so the Pythagorean identity proves uniqueness and the minimum formula. It follows in the original fixed remainder frame that

$$G_{q+r-1}^{(s)} = C_s^{-*} K_{s,r}^{-1} C_s^{-1}, \quad \det G_{q+r-1}^{(s)} = \frac{\prod_{n=0}^{q-1} h_n^{(s)}}{\Omega_{s,r}}. \quad (\text{IFB11})$$

This gives the entire quotient Gram, including every off-diagonal entry, before its determinant is taken.

14.2 The relation determinant as a positive finite sum

The integer q is divisible by four. Pairing both original signed root coordinates in IFB1 gives the exact real polynomial

$$\vartheta(y) := \chi(c + iy) = \prod_{A \in \{\delta, 3\delta, \dots, k\delta\}} \prod_{B \in \{\gamma, 3\gamma, \dots, k\gamma\}} [((y - B)^2 + A^2)((y + B)^2 + A^2)]^e. \quad (\text{IFB12})$$

Each quartet contributes two negative factors from its two opposite real-root pairs, so their product has the positive sign displayed. Each factor is strictly positive on $\mathbb{R}$ because $A > 0$. Thus ϑ is real, even, strictly positive on the real line, and monic of degree q ; its squared modulus is exactly $W_\chi(y) = \vartheta(y)^2$. Write

$$\vartheta(y) = \sum_{a=0}^q c_a y^a, \quad W_\chi(y) = \sum_{d=0}^{2q} w_d y^d, \quad w_d = \sum_{a+a'=d} c_a c_{a'}. \quad (\text{IFB13})$$

These coefficients retain all roots and all repetitions. For an explicit finite rule producing them, each factor of IFB12 is $(y^4 + 2(A^2 - B^2)y^2 + (A^2 + B^2)^2)^e$. Expand it as

$$\sum_{u+v+w=e} \frac{e!}{u!v!w!} [2(A^2 - B^2)]^v (A^2 + B^2)^{2w} y^{4u+2v}, \quad (\text{IFB14})$$

and multiply these finite coefficient lists. In particular this rule does not discard their signed middle coefficients.

Let $T_{n,h}$ be the coefficient of p_h in y^n . The exact finite recurrence following from IFB5 is

$$T_{0,0} = 1, \quad T_{0,h} = 0 \ (h \neq 0), \quad T_{n+1,h} = T_{n,h-1} + (h+1)(b_s + h)T_{n,h+1}, \quad (\text{IFB15})$$

with indices outside $0 \leq h \leq n$ interpreted as zero. Induction using $yp_h = p_{h+1} + h(b_s + h - 1)p_{h-1}$ proves the formula and $y^n = \sum_h T_{n,h} p_h$. In particular $T_{n,n} = 1$, and only $h \equiv n \pmod{2}$ can occur. Define the full rectangular matrix

$$A_{h,j} = \sum_{a=0}^q c_a T_{a+j,h}, \quad 0 \leq h \leq q+r-1, \quad 0 \leq j < r, \quad L_{h,j} = \sqrt{h_h^{(s)}} A_{h,j}. \quad (\text{IFB16})$$

Here $h_h^{(s)}$ means the source norm with index h ; the matrix letter L denotes the full weighted coefficient matrix only in this section. The definition says exactly that $\vartheta(y)y^j = \sum_h A_{h,j} p_h(y)$. Thus the Gram of these r polynomials is L^*L .

In the original S -coefficient frame the relation vectors are $(\chi, \chi S, \dots, \chi S^{r-1})$. Its coefficient change to the vectors $(\vartheta, \vartheta y, \dots, \vartheta y^{r-1})$ is the exact triangular matrix

$$(T_c)_{h,j} = \binom{j}{h} c^{j-h} i^h \quad (0 \leq h \leq j < r). \quad (\text{IFB17})$$

Its inverse is substitution $y = (S - c)/i$, and $|\det T_c| = 1$. Consequently the original relation determinant $Z_{s,r} := \det \text{Gram}_{m_{0,s}}(\chi, \dots, \chi S^{r-1})$ equals $\det(L^*L)$. We set $Z_{s,0} = 1$.

Let L_{low} comprise rows $0, \dots, q-1$ of L and R comprise rows $q, \dots, q+r-1$. Since ϑy^j is monic of degree $q+j$, the matrix R is upper triangular with diagonal $\sqrt{h_{q+j}^{(s)}}$. Hence it is invertible and

$$|\det R|^2 = \prod_{j=0}^{r-1} h_{q+j}^{(s)}, \quad L_{\text{low}} R^{-1} = -V_r. \quad (\text{IFB18})$$

For the second identity, the column space of L is the relation space in IFB9. Its high coordinates are precisely R times its coefficient vector; expressing a relation by its high coordinate therefore gives

the low graph matrix $L_{\text{low}} R^{-1}$. Uniqueness of the graph in IFB9 proves the asserted sign and every matrix entry. Therefore

$$Z_{s,r} = \left(\prod_{j=0}^{r-1} h_{q+j}^{(s)} \right) \det(I_r + V_r^* V_r) = \left(\prod_{j=0}^{r-1} h_{q+j}^{(s)} \right) \Omega_{s,r}. \quad (\text{IFB19})$$

To justify the second equality without discarding zero singular directions, eliminate either diagonal identity block in $\begin{pmatrix} I_q & V_r \\ -V_r^* & I_r \end{pmatrix}$. The two block eliminations have determinant one and give $\det(I_q + V_r V_r^*) = \det(I_r + V_r^* V_r)$, for all dimensions.

The full finite expansion of the same determinant is

$$\Omega_{s,r} = \sum_{\substack{I \subset \{0, \dots, q+r-1\} \\ |I|=r}} |\det A_{I, \{0, \dots, r-1\}}|^2 \frac{\prod_{h \in I} h_h^{(s)}}{\prod_{j=0}^{r-1} h_{q+j}^{(s)}}. \quad (\text{IFB20})$$

For completeness, expand each column of L in the standard orthonormal coordinates of $\mathbb{C}^{q+r}$. Their exterior product has, in the exterior frame indexed by r -element subsets I , coefficient $\det L_I$. The exterior frame is orthonormal, so its squared norm is $\sum_I |\det L_I|^2$. Direct expansion of that same squared norm gives $\det(L^* L)$ by the permutation formula for a determinant. Substituting IFB16 and dividing by the explicit positive factor in IFB19 proves IFB20. In particular the subset $I = \{q, \dots, q+r-1\}$ contributes exactly one. All other terms are nonnegative; none of their signed coefficient calculations has been replaced by a degree estimate. When $r = 0$, the single empty minor contributes one.

There is also a direct moment evaluation of exactly the same mass. Define polynomials $P_n(z)$ by

$$P_0(z) = 1, \quad P_{n+1}(z) = b_s z P_n(z) + (1 + z^2) P_n'(z).$$

Induction by differentiation gives $\frac{d^n}{dt^n} (\cos t)^{-b_s} = (\cos t)^{-b_s} P_n(\tan t)$. Differentiating IFB3 at zero is justified by its exponential bound. It follows that

$$Z_{s,r} = M_s^r \det \left[\sum_{d=0}^{2q} w_d P_{d+i+j}(0) \right]_{i,j=0}^{r-1}. \quad (\text{IFB21})$$

The moment degree required at the largest original endpoint $r = q + 1$ is exactly $2q + 2r - 2 = 4q$. The finite recurrences IFB14–16 and IFB21 therefore calculate every contribution from the original parameters in finitely many additions, multiplications and divisions by the displayed nonzero positive constants. IFB20 certifies the positivity of the result even when coefficients in IFB21 cancel.

14.3 The actual spectral-sum mass at the four ranks

For $r \geq 1$, let $\Delta(y) = \prod_{1 \leq i < j \leq r} (y_j - y_i)$ and define the actual exterior measure and actual sum map

$$d\mathbf{m}_{s,\chi,r}(y_1, \dots, y_r) = \frac{1}{r!} \Delta(y)^2 \prod_{j=1}^r W_\chi(y_j) dm_{0,s}(y_j), \quad \Sigma_r(y_1, \dots, y_r) = \sum_{j=1}^r y_j. \quad (\text{IFB22})$$

Set $\nu_{s,\chi,r} = (\Sigma_r)_* \mathbf{m}_{s,\chi,r}$. At $r = 0$ the source space is a one-point space of mass one, $\Sigma_0 = 0$, and $\nu_{s,\chi,0} = \delta_0$. Expanding both Vandermonde determinants in IFB22 gives

$$\nu_{s,\chi,r}(\mathbb{R}) = Z_{s,r}. \quad (\text{IFB23})$$

To check the prefactor, write $\Delta = \det[y_i^{j-1}]$ and expand it twice. Integrating a product gives a product of the moments of $W_\chi dm_{0,s}$. Holding the first permutation fixed and reindexing the second gives the

determinant of its r by r moment matrix; the first permutation has $r!$ possible values. Those $r!$ values cancel precisely the $1/r!$ in IFB22. Absolute integrability follows from the polynomial degree and the exponential source bound. The moment determinant agrees with the original S -relation determinant by IFB17. This proves IFB23 with its entire original mass.

Here is the precise retained fibre map. Write ρ_s for the density in IFB2 and, for $r \geq 2$, put

$$y_r = Y - \sum_{a=1}^{r-1} y_a, \quad F_r(Y; y_1, \dots, y_{r-1}) = \frac{1}{r!} \Delta(y)^2 \prod_{a=1}^r W_\chi(y_a) \rho_s(y_a),$$

$$g_r(Y) = \int_{\mathbb{R}^{r-1}} F_r(Y; y') dy'. \quad (\text{IFB24})$$

The change of variables $(y_1, \dots, y_r) \mapsto (y_1, \dots, y_{r-1}, Y)$ has Jacobian determinant one. Therefore $d\nu_{s,\chi,r}(Y) = g_r(Y) dY$. These integrals are finite for every Y : a polynomial in $|Y| + \sum |y_a|$ times $\exp(-\theta \sum_{a < r} |y_a|)$ is an integrable dominating function when Y is fixed. Each $g_r(Y) > 0$, because the source densities and W_χ are positive and the distinct-coordinate configurations form a nonempty open subset of the fibre. For $r = 1$, the formula is $g_1(Y) = W_\chi(Y) \rho_s(Y) > 0$.

The map

$$\mathcal{U}_r : L^2(\nu_{s,\chi,r}) \longrightarrow L^2(\mathfrak{m}_{s,\chi,r}), \quad (\mathcal{U}_r f)(y) = f(\Sigma_r y) \quad (\text{IFB25})$$

is an isometry, directly by the defining integral of a pushforward. Its full adjoint, for $r \geq 2$, is

$$(\mathcal{U}_r^* H)(Y) = \frac{1}{g_r(Y)} \int H(y_1, \dots, y_{r-1}, Y - \sum y_a) F_r(Y; y') dy'. \quad (\text{IFB26})$$

The integral exists for almost every Y by Cauchy-Schwarz in the fibre and Fubini. The same inequality shows its squared modulus, multiplied by $g_r(Y)$, is bounded by the fibre integral of $|H|^2 F_r$; hence the adjoint is a contraction. Multiplying by a test function $f(Y)$ and applying Fubini proves the adjoint identity and thus IFB26. Its kernel is exactly the functions with weighted fibre integral zero for almost every Y . For $r = 1$, both maps are the identity; at $r = 0$ they are the identity of the one-dimensional one-point L^2 space. Thus the fibre kernel and the full source norm have specified domains and codomains; passing to a sum does not silently identify the entire exterior space with its image.

The original complex spectral sum is $\sum_{a=1}^r (c + iy_a) = rc + iY$. Accordingly the actual target line of this pushforward is $rc + i\mathbb{R}$. If a Gamma density of order a is written on its natural line $a/2 + i\mathbb{R}$, the coordinate isomorphism for that constituent is

$$a/2 + iY \longmapsto rc + iY, \quad \text{translation by } rc - a/2. \quad (\text{IFB27})$$

It preserves dY and the complete density and mass. Different constituents use their own stated translation; none changes the original target centre rc .

Let $\mathcal{D}_{s,\chi,r}(z)$ denote the polynomial provided by the intrinsic exterior/sum calculation, so its exact Laplace transform is

$$\int e^{tY} d\nu_{s,\chi,r}(Y) = M_s^r(\cos t)^{-[b_s r + r(r-1)]} \mathcal{D}_{s,\chi,r}(\tan t). \quad (\text{IFB28})$$

This polynomial is completely determined by IFB21 as a differentiated moment determinant, or by the full Wronskian construction in the intrinsic source provider. To check this definition directly, let $Q_0(z) = \sum_d w_d P_d(z)$ and $Q_{j+1}(z) = b_s z Q_j(z) + (1 + z^2) Q_j'(z)$. The Laplace version of IFB23 has moment entries $M_s(\cos t)^{-b_s} Q_{i+j}(\tan t)$. Thus one may take

$$\mathcal{D}_{s,\chi,r}(z) = (1 + z^2)^{-r(r-1)/2} \det[Q_{i+j}(z)]_{i,j < r}.$$

This expression is a polynomial: its determinant is the t -Wronskian of the first r derivatives of $(\cos t)^{-b_s} Q_0(\tan t)$, and the following elementary two identities show the factors explicitly. For a

common factor f , successive row subtractions in $[\partial_t^i(fg_j)]$ give determinant f^r times the Wronskian of the g_j ; the subtraction coefficients come from the product rule, and the triangular derivative matrix has diagonal f . For a coordinate $z = z(t)$, the chain rule expresses ∂_t^i as $(z')^i \partial_z^i$ plus lower derivatives; the determinant multiplier is $(z')^{r(r-1)/2}$. Apply these identities with $f = (\cos t)^{-b_s}$, $g_j = Q_j(z)$ and $z' = 1 + z^2$ to obtain $\mathcal{D}_{s,\chi,r}(z) = \det[\partial_z^i Q_j(z)]_{i,j < r}$. This also proves IFB28 without any assumed statement about the degree or signs of this polynomial. Its full degree and Gamma constituents are evaluated in the intrinsic source provider.

At the original four ranks its constant term is now calculated by the positive original relation graph, not by size counting:

$$\mathcal{D}_{s,\chi,r}(0) = \left(\prod_{j=0}^{r-1} (q+j)!(b_s)_{q+j} \right) \Omega_{s,r}, \quad r = 0, 1, q, q+1. \quad (\text{IFB29})$$

Indeed IFB28 at $t = 0$ gives $M_s^T \mathcal{D}(0) = Z_{s,r}$. Substituting IFB19 and every factor $h_{q+j}^{(s)} = M_s(q+j)!(b_s)_{q+j}$ proves IFB29. In particular the full exterior-source mass appears to power r on both sides before its algebraic cancellation. Equation IFB29 is the exact factor relating the intrinsic Gamma expression to the original quotient determinant.

14.4 The original four-endpoint identity and its positive update chain

Write $\mathcal{B}_k^{(1)} = \mathcal{B}_k^\sigma$ and $\mathcal{B}_k^{(k)} = \mathcal{B}_k^0$. The monic source norms in the original S -basis are $h_n^{(s)}$: multiplication of u_n by $i^n \sqrt{h_n^{(s)}}$ makes it monic and leaves its squared norm $h_n^{(s)}$. The triangular monic source/relation frame in BSL2–3 therefore gives, at the original four degrees, the full signed formula

$$\begin{aligned} \mathcal{B}_k^{(s)} &= S_q^{(s)} + \log Z_{s,q} + \log Z_{s,q+1} - \log Z_{s,1}, \\ S_q^{(s)} &= -\log h_q^{(s)} - 2 \sum_{j=q+1}^{2q-1} \log h_j^{(s)} - \log h_{2q}^{(s)}. \end{aligned} \quad (\text{IFB30})$$

The omitted rank-zero logarithm is exactly $\log Z_{s,0} = 0$. One can also prove this directly from IFB11 by taking the prescribed $+, +, -, -$ signs at $r = 0, 1, q, q+1$. Substituting IFB19 into the displayed formula cancels the monic norm factors as follows: the two high products contain h_q^2 , each $h_{q+1}, \dots, h_{2q-1}$ twice, and h_{2q} once; division by the low factor h_q leaves precisely the factors canceled by $S_q^{(s)}$. In particular the full source mass exponent in the relation terms is $q + (q+1) - 1 = 2q$, while that in $S_q^{(s)}$ is $-1 - 2(q-1) - 1 = -2q$. The exact result is

$$\mathcal{B}_k^{(s)} = \log \Omega_{s,q} + \log \Omega_{s,q+1} - \log \Omega_{s,1}. \quad (\text{IFB31})$$

Every original sign and all low and high polynomial factors have now been retained and evaluated before this cancellation.

Define the finite, strictly positive update factors

$$d_{s,j} = 1 + v_j^* K_{s,j}^{-1} v_j, \quad j \geq 0. \quad (\text{IFB32})$$

Since $K_{s,j} = I_q + \sum_{a < j} v_a v_a^*$ is positive definite, its inverse exists and $d_{s,j} \geq 1$. Multiplying $K_{s,j+1} = K_{s,j} + v_j v_j^*$ by $K_{s,j}^{-1/2}$ on both sides reduces its determinant ratio to that of $I + w w^*$, $w = K_{s,j}^{-1/2} v_j$. In an orthonormal frame with first vector $w/\|w\|$ when $w \neq 0$, that operator has eigenvalues $1 + \|w\|^2, 1, \dots, 1$; the case $w = 0$ is the identity. Consequently

$$\Omega_{s,r} = \prod_{j=0}^{r-1} d_{s,j}, \quad K_{s,j+1}^{-1} = K_{s,j}^{-1} - \frac{K_{s,j}^{-1} v_j v_j^* K_{s,j}^{-1}}{d_{s,j}}. \quad (\text{IFB33})$$

The inverse identity follows by multiplication by $K_{s,j} + v_j v_j^*$; its two rank-one terms cancel exactly. Starting with $K_{s,0}^{-1} = I_q$, it is a finite formula for every inverse required in IFB32. The original four-endpoint expression therefore equals the complete nonnegative sum

$$\mathcal{B}_k^{(s)} = \log d_{s,0} + 2 \sum_{j=1}^{q-1} \log d_{s,j} + \log d_{s,q}. \quad (\text{IFB34})$$

This is proved from the signed identity IFB31: each factor with $1 \leq j < q$ occurs in both high endpoints, $j = q$ occurs once, and the rank-one low endpoint removes one of the two occurrences of $j = 0$. Formula IFB34 preserves all singular directions and all off-diagonal interactions through the full inverse $K_{s,j}^{-1}$. It is not an assertion that these interactions are independent.

14.5 A finite companion calculation of every update vector

Let $E_\vartheta = \mathbb{C}[y]/(\vartheta)$ in its monomial frame and let Y_ϑ be the matrix of multiplication by y . Thus its column $j < q - 1$ is the vector with a one in row $j + 1$, and its last column is $(-c_0, \dots, -c_{q-1})^t$. The exact algebra isomorphism is

$$E_\vartheta \longrightarrow E_\chi, \quad [P(y)] \longmapsto [P((S - c)/i)], \quad (\text{IFB35})$$

because substituting $y = (S - c)/i$ in $\vartheta(y) = \chi(c + iy)$ gives $\chi(S)$, and the inverse substitution is $S = c + iy$. Let E_s be the coefficient matrix of $p_0/\sqrt{h_0^{(s)}}, \dots, p_{q-1}/\sqrt{h_{q-1}^{(s)}}$ in the y -monomial frame; its diagonal is $1/\sqrt{h_n^{(s)}}$. Hence E_s is invertible. Define

$$Y_s = E_s^{-1} Y_\vartheta E_s, \quad a_n = \sqrt{n(b_s + n - 1)} \ (n \geq 1), \quad a_0 = 0. \quad (\text{IFB36})$$

All entries are now determined by the finite original coefficients in IFB14 and the finite original recurrence IFB5. For all $n \geq 0$ let $z_n = C_s^{-1}[u_n]$. Its initial values are $z_{-1} = 0$ and z_n the n th coordinate vector for $0 \leq n < q$. Dividing IFB5 by $\sqrt{h_n^{(s)}}$ and using $h_n^{(s)}/h_{n-1}^{(s)} = n(b_s + n - 1)$ gives

$$z_{n+1} = \frac{Y_s z_n - a_n z_{n-1}}{a_{n+1}}, \quad v_j = z_{q+j}. \quad (\text{IFB37})$$

Multiplication in the quotient is exactly Y_s by IFB35–36, so this is a recurrence for the actual remainders, including all of their original polynomial relation coefficients. Every denominator a_{n+1} is positive. It suffices to run it through $n = 2q - 1$ to obtain the last vector $v_q = z_{2q}$ used in IFB34. Combining IFB14, IFB36–37, and IFB32–33 is thus a finite exact calculation of the original four-endpoint quantity, using a fixed q -dimensional space and its full matrices.

As an explicit coefficient check, summing the coefficients of y^{q-2} in the original quartets gives

$$c_{q-2} = \frac{qk(k+2)(\delta^2 - \gamma^2)}{6}. \quad (\text{IFB38})$$

Indeed $\sum_{a \in \{1,3,\dots,k\}} a^2 = k(k+1)(k+2)/6$ and there are $(k+1)/2$ choices of the other positive coordinate. Each quartet contributes $2e(A^2 - B^2)$, which proves IFB38. The coefficient of y^{n-2} in p_n is

$$-A_n^{(s)}, \quad A_n^{(s)} = \sum_{j=1}^{n-1} j(b_s + j - 1) = \frac{n(n-1)(3b_s + 2n - 4)}{6}. \quad (\text{IFB39})$$

The first equality follows successively from the coefficient of y^{n-1} in the recurrence for p_{n+1} , starting with $p_1 = y$; the second uses the finite sums of j and j^2 . Consequently the coefficient of p_{q-2} in the full original ϑ is exactly $c_{q-2} + A_q^{(s)}$. In IFB20 the minor obtained by replacing the row q in the

highest-degree minor by $q-2$ is therefore $c_{q-2} + A_q^{(s)}$, for every $r \geq 1$. To check that assertion, its first column has this entry in the replacement row and is zero in all remaining rows, while the remaining top-degree block is upper triangular with diagonal one. It follows that

$$\Omega_{s,r} \geq 1 + \frac{(c_{q-2} + A_q^{(s)})^2 h_{q-2}^{(s)}}{h_q^{(s)}} \quad (r \geq 1). \quad (\text{IFB40})$$

For $r \geq 2$, the analogous replacement of row $q+1$ by $q-1$ has squared minor $(c_{q-2} + A_{q+1}^{(s)})^2$: its two lowest rows and first two columns have determinant $-(c_{q-2} + A_{q+1}^{(s)})$, and the remaining block is upper triangular. Making both replacements gives the product of the two coefficients, because their opposite-parity cross entries vanish. Keeping these four distinct nonnegative summands in IFB20 proves the stronger finite bound

$$\Omega_{s,r} \geq \left(1 + \frac{(c_{q-2} + A_q^{(s)})^2 h_{q-2}^{(s)}}{h_q^{(s)}}\right) \left(1 + \frac{(c_{q-2} + A_{q+1}^{(s)})^2 h_{q-1}^{(s)}}{h_{q+1}^{(s)}}\right), \quad r \geq 2. \quad (\text{IFB41})$$

These are specific calculated terms in the full original polynomial mass; IFB20 and IFB34 continue to retain every remaining term. In particular they assert no asymptotic replacement of the full quantity by these selected summands.

Finally ϑ is even and each p_n has its original parity. Division by this even monic polynomial preserves parity. Hence v_j has coordinates only in indices congruent to $q+j$, and $K_{s,r}$ has zero entries between the two source parity subspaces. This follows entrywise from IFB8, not by removing coordinates. The full matrix C_s in IFB11 carries this exact structure back to the original S -remainder frame. The downstream GMB calculation then composes that same map with the original mixed boundary map; its contractions use the entire matrix $K_{s,r}$ and all its original source phases.

15 The exact map from Gamma remainders to the original mixed rows

This calculation uses only the two sources already present in the original construction. It retains the packet order k , its centre $c = k/2$, its monic polynomial χ of degree q , and the original surjection $\Lambda : E_\chi \rightarrow B_{\text{obs}}$, where $E_\chi = \mathbb{C}[S]/(\chi)$. The source order is $s \in \{1, k\}$. Write $b_s = s/2$ and $M_s = (2\pi)^{s/2}$. For the full source measure $dm_{0,s}$, the monic real orthogonal polynomials $p_d(y)$ and their squared norms are

$$\begin{aligned} p_0 &= 1, \quad p_1 = y, \quad p_{d+1}(y) = yp_d(y) - d(b_s + d - 1)p_{d-1}(y), \\ h_d^{(s)} &= M_s d! (b_s)_d, \quad u_d(S) = \frac{p_d((S - c)/i)}{\sqrt{h_d^{(s)}}}. \end{aligned} \quad (\text{GMB1})$$

The recurrence and every source mass are those proved for the original Gamma source. On $S = c + iy$ the u_d are orthonormal; their leading coefficients in S are $i^{-d}/\sqrt{h_d^{(s)}}$. Thus no phase or mass is removed by introducing these explicit coordinates. Inner products below are conjugate linear in the first variable.

Let $C_s : \mathbb{C}^q \rightarrow E_\chi$ have columns $[u_0], \dots, [u_{q-1}]$ in the fixed original remainder frame. It is invertible, with the literal determinant

$$\det C_s = i^{-q(q-1)/2} \prod_{d=0}^{q-1} (h_d^{(s)})^{-1/2}. \quad (\text{GMB2})$$

For $j \geq 0$ define the actual remainder coordinates and matrices

$$\begin{aligned} v_j &= C_s^{-1}[u_{q+j}], \quad K_j = I_q + \sum_{a=0}^{j-1} v_a v_a^*, \quad t_j = v_j^* K_j^{-1} v_j, \quad d_j^{\text{gain}} = 1 + t_j, \\ A_j &= [I_q, v_0, \dots, v_{j-1}] : \mathbb{C}^{q+j} \rightarrow \mathbb{C}^q. \end{aligned} \quad (\text{GMB3})$$

The empty sum is zero. In particular $K_0 = I_q$, $K_j = A_j A_j^*$ is positive definite, and $d_j^{\text{gain}} \geq 1$. The original polynomial remainder map in these orthonormal source coordinates is exactly $C_s A_j$. This follows by applying it to each displayed basis vector, including all high degrees.

For $x \in \mathbb{C}^q$, the unique minimum norm source lift of $C_s x$ at degree $q + j - 1$ has coefficient vector $A_j^* K_j^{-1} x$. Indeed its remainder coordinates are x , and it is orthogonal to $\ker A_j$. Every other lift differs by an element of that kernel, so its squared norm is $x^* K_j^{-1} x$ plus the squared norm of that difference. Hence the original quotient Gram is

$$G_{q+j-1} = C_s^{-*} K_j^{-1} C_s^{-1}. \quad (\text{GMB4})$$

This proves the actual quotient map and its metric, not only its determinant. The source kernel has its full dimension j .

15.1 The monic relation, its original phase, and its row

Put $D = q + j$. In degree at most D , with the last coordinate corresponding to u_D , form

$$z_j = \begin{pmatrix} -A_j^* K_j^{-1} v_j \\ 1 \end{pmatrix}, \quad \mathfrak{d}_{D-1}(S) = i^D \sqrt{h_D^{(s)}} \left(u_D(S) - \sum_{a=0}^{D-1} (A_j^* K_j^{-1} v_j)_a u_a(S) \right). \quad (\text{GMB5})$$

The first vector has zero remainder because $-A_j A_j^* K_j^{-1} v_j + v_j = 0$. Thus $\mathfrak{d}_{D-1}$ is divisible by the original χ . Its leading coefficient in S is exactly one by GMB1. Moreover z_j is orthogonal to every old relation: its old coordinates lie in $\text{im } A_j^*$. It follows that $\mathfrak{d}_{D-1} = \chi p_j^{\chi,s}$, where $p_j^{\chi,s}$ is the original monic polynomial orthogonal to lower degrees in the measure $|\chi(c + iy)|^2 dm_{0,s}(y)$. To verify uniqueness without any assumption on coordinates, the difference of two such monic relations is an old relation,

and its inner product with that difference is zero. Positivity makes it zero. Consequently GMB5 is exactly the relation line used in BRU and HMR32, with its prescribed monic orientation.

Let $a_{D-1} = \|\mathfrak{d}_{D-1}\|^2$ and let α_{D-1} be the row with entries $\langle \mathfrak{d}_{D-1}, S^a \rangle$, $0 \leq a < q$. GMB3–5 give the complete formulas

$$\begin{aligned} a_{D-1} &= h_D^{(s)} d_j^{\text{gain}}, \\ \alpha_{D-1} &= -i^{-D} \sqrt{h_D^{(s)}} v_j^* K_j^{-1} C_s^{-1}. \end{aligned} \quad (\text{GMB6})$$

For the first equality, the squared norm of z_j is $1 + v_j^* K_j^{-1} A_j A_j^* K_j^{-1} v_j = 1 + t_j$. For the second, the degree D orthogonal polynomial has zero inner product with every S^a , $a < q$. A polynomial of degree less than q has its low orthonormal coordinates $C_s^{-1}[P]$, so the old lift term in GMB5 has inner product $v_j^* K_j^{-1} C_s^{-1}[P]$ with it. Conjugating the scalar $i^D \sqrt{h_D^{(s)}}$ proves precisely the phase and minus sign in GMB6.

Direct multiplication proves

$$(K_j + v_j v_j^*)^{-1} = K_j^{-1} - \frac{K_j^{-1} v_j v_j^* K_j^{-1}}{d_j^{\text{gain}}}.$$

Using GMB4 and GMB6 therefore gives the original BRU update, including its complete coefficient row:

$$G_{D-1} = G_D + \frac{\alpha_{D-1}^* \alpha_{D-1}}{a_{D-1}}, \quad \frac{\det G_{D-1}}{\det G_D} = d_j^{\text{gain}}. \quad (\text{GMB7})$$

The determinant equality follows either by taking determinants in GMB4, or by conjugating $K_j + v_j v_j^*$ by $K_j^{-1/2}$: the resulting rank-one update has eigenvalue $1 + t_j$ in its one possible nontrivial direction and eigenvalue one elsewhere. This also proves it when $v_j = 0$.

15.2 The unchanged observation and both mixed contractions

The observation in these explicit source coordinates is

$$R_s = \Lambda C_s : \mathbb{C}^q \rightarrow B_{\text{obs}}, \quad M_j^{\text{obs}} = R_s K_j R_s^*, \quad w_j = R_s v_j, \quad \xi_j = w_j^* (M_j^{\text{obs}})^{-1} w_j. \quad (\text{GMB8})$$

The notation R_s here names a matrix of the fixed observation, not a new observation. Every Taylor unit and primary coefficient already in Λ is still in this product. For zero boundary rank the matrices are empty and $\xi_j = 0$. Otherwise M_j^{obs} is positive definite by surjectivity. Define

$$r_j^{\text{ker}} = v_j - K_j R_s^* (M_j^{\text{obs}})^{-1} w_j. \quad (\text{GMB9})$$

Then $R_s r_j^{\text{ker}} = 0$. Its two summands give the exact K_j^{-1} -orthogonal decomposition

$$t_j = \xi_j + (r_j^{\text{ker}})^* K_j^{-1} r_j^{\text{ker}}, \quad 0 \leq \xi_j \leq t_j. \quad (\text{GMB10})$$

Indeed the cross inner product is $(r_j^{\text{ker}})^* R_s^* (M_j^{\text{obs}})^{-1} w_j = 0$, and the squared norm of the second summand in GMB9 is ξ_j . The map C_s restricts to a bijection $\ker R_s \rightarrow \ker \Lambda$, since $\Lambda C_s x = R_s x$. This supplies the exact map of the retained kernel, including its original coefficient embedding.

Let $I : \ker \Lambda \hookrightarrow E_\chi$ be that original embedding. At the new degree D , retain the original forms and minimum section

$$\begin{aligned} H_D^{\text{ker}} &= I^* G_D I, \quad Q_D = (\Lambda G_D^{-1} \Lambda^*)^{-1}, \quad S_D = G_D^{-1} \Lambda^* Q_D, \\ \beta_{D-1} &= (\alpha_{D-1} I) (H_D^{\text{ker}})^{-1} (\alpha_{D-1} I)^*, \quad \nu_{D-1} = (\alpha_{D-1} S_D) Q_D^{-1} (\alpha_{D-1} S_D)^*. \end{aligned} \quad (\text{GMB11})$$

All inverses on a zero dimensional space have their unique empty meaning. The section obeys $\Lambda S_D = I$ and $I^* G_D S_D = 0$. The map $[I, S_D]$ is a bijection: the inverse sends z to its kernel component

$z - S_D \Lambda z$ and its boundary component Λz . In this frame the metric has diagonal blocks $H_D^{\ker}$ and Q_D . Inverting the congruence proves

$$G_D^{-1} = I(H_D^{\ker})^{-1}I^* + S_D Q_D^{-1} S_D^*. \quad (\text{GMB12})$$

This proves all decomposition identities used below while retaining the original source, kernel, boundary and section maps.

Set $d = 1 + t_j$, $h = h_D^{(s)}$, and $N = M_j^{\text{obs}} + w_j w_j^*$. Equations GMB4 and GMB8 imply $Q_D = N^{-1}$ and $S_D = C_s(K_j + v_j v_j^*) R_s^* N^{-1}$. Direct multiplication yields $N^{-1} w_j = (M_j^{\text{obs}})^{-1} w_j / (1 + \xi_j)$. GMB6 now gives

$$\alpha_{D-1} S_D = -i^{-D} \sqrt{h} d w_j^* N^{-1}, \quad \nu_{D-1} = h d^2 \frac{\xi_j}{1 + \xi_j}.$$

On the other hand GMB6 and GMB12 give

$$\beta_{D-1} + \nu_{D-1} = \alpha_{D-1} G_D^{-1} \alpha_{D-1}^* = h(t_j + t_j^2) = h d t_j.$$

Subtracting the displayed expression for ν_{D-1} , with no sign assumption, proves the exact full mixed row formula

$$\boxed{\begin{aligned} \alpha_{D-1} &= h d, & \beta_{D-1} &= h d \frac{t_j - \xi_j}{1 + \xi_j}, & \nu_{D-1} &= h d^2 \frac{\xi_j}{1 + \xi_j}, \\ 1 + \frac{\beta_{D-1}}{\alpha_{D-1}} &= \frac{1 + t_j}{1 + \xi_j}, & 1 + \frac{\nu_{D-1}}{\alpha_{D-1} + \beta_{D-1}} &= 1 + \xi_j. \end{aligned}} \quad (\text{GMB13})$$

Both factors are at least one by GMB10, and their product is $1 + t_j$, the original complete increment in GMB7. These are the individual BRU kernel and boundary factors, in their original order. In particular the kernel factor has the denominator $1 + \xi_j$; it is not replaced by $1 + t_j - \xi_j$.

15.3 The four original endpoints and their finite calculation

For a sequence f_j , put

$$\mathcal{W}_q(f) = f_0 + 2 \sum_{j=1}^{q-1} f_j + f_q. \quad (\text{GMB14})$$

The original signed quotient volume is $\log \det G_{q-1} + \log \det G_q - \log \det G_{2q-1} - \log \det G_{2q}$. Telescoping GMB7 at these four precise degrees gives

$$\begin{aligned} \mathcal{B}_k^{(s)} &= \mathcal{W}_q(\log(1 + t_j)), \\ F_K^{(s)} &= \mathcal{W}_q\left(\log \frac{1 + t_j}{1 + \xi_j}\right), & F_B^{(s)} &= \mathcal{W}_q(\log(1 + \xi_j)), \\ F_K^{(s)} + F_B^{(s)} &= \mathcal{B}_k^{(s)}. \end{aligned} \quad (\text{GMB15})$$

To check the indices, the relation added at this step has degree $D = q + j$ and is the BRU row with index $D - 1$. Thus $j = 0, q$ are exactly the two weight-one rows $q - 1, 2q - 1$; all intervening rows have weight two. GMB13 proves the individual factors, so GMB15 is the full original mixed decomposition at $s = 1$ or $s = k$. It does not assign an arithmetic source to either Gamma order. The established arithmetic comparison retains its own exact transition scalar and its original maps.

For completeness this is a finite calculation from the original polynomial, rather than a definition by an unspecified projection. Let T_χ be multiplication by S on the fixed remainder frame and set $Y_\chi = (T_\chi - cI)/i$. Starting with $r_0 = [1]$, $r_1 = Y_\chi[1]$, compute

$$r_{d+1} = Y_\chi r_d - d(b_s + d - 1)r_{d-1}, \quad C_s = \left[\frac{r_0}{\sqrt{h_0^{(s)}}}, \dots, \frac{r_{q-1}}{\sqrt{h_{q-1}^{(s)}}} \right], \quad v_j = \frac{C_s^{-1} r_{q+j}}{\sqrt{h_{q+j}^{(s)}}}. \quad (\text{GMB16})$$

Polynomial division and GMB1 prove by induction that $r_d = [p_d(Y)]$. Every entry of T_χ is a literal coefficient of the original χ , including its primary multiplicities. The matrices K_j, M_j^{obs} and the scalars in GMB13–15 are then obtained by the displayed finite operations. Their inverses exist by the proved positivity, including the stated empty cases. This closes the exact morphism from the intrinsic exterior mass calculation, through the original polynomial remainders, to both original mixed contractions. It introduces no new source order or factor.

Part II

Exploratory independent-order comparison with exact maps

The independent order in this part is a tested comparison parameter. Its source changes are carried simultaneously through the baseline, the Gamma return and the arithmetic identity. All original parameters remain present in the comparison. No selection of a replacement canonical order is inferred from these calculations.

16 An independent Gamma order on the complete original packet

Exploratory comparison status. This section preserves the tested independent Gamma family and its complete comparison maps. The choice $K = q + 1$ was selected to compare orders of growth; the original tensor construction does not select that choice here. Following the user's instruction against ad hoc insertions, it is not substituted into the canonical source. The canonical order remains k with $q = [1 + k(m - 1)](k + 1)^2$. The exact translations, fibre maps, and arithmetic transition law below state precisely how this auxiliary family relates to that unchanged construction.

The original tensor order k and the original dimension q retain BSL1. In particular $q = [1 + k(m - 1)](k + 1)^2$, so $k/q \rightarrow 0$ at fixed m . We now introduce a second, independent order

$$K = 4L + 1, \quad L \geq 0, \quad f_L(S) = \prod_{j=0}^{L-1} (S - c - t_j), \quad t_j = 2j + \frac{1}{2}, \quad \beta_K = \frac{(2\pi)^{K/2}}{\sqrt{2}\Gamma(K/2)}. \quad (\text{IK01})$$

Every parameter and root of χ remains unchanged. Since q is a multiple of four, the exact choice

$$L = q/4, \quad K = q + 1 \quad (\text{IK02})$$

is available in this family. Thus the ratio K/q tends to one without identifying the original order k with the original dimension q .

16.1 The coordinate and source maps, with the whole mass

The natural Gamma source of order K lies on $U = K/2 + iy$. Its pullback to the original line $S = c + iy$ is the translation

$$U = S + (K - k)/2, \quad \mathcal{T}_K P(U) = P(U - (K - k)/2). \quad (\text{IK03})$$

This is a unital algebra isomorphism on polynomial rings, with inverse given by the opposite translation; it preserves every degree and has triangular coefficient matrix with diagonal one. It sends χ to $\chi_K(U) = \chi(U - (K - k)/2)$ and induces an isomorphism $\overline{\mathcal{T}}_K : E_\chi \rightarrow \mathbb{C}[U]/(\chi_K)$. Every root is translated by $(K - k)/2$, so each divided Taylor jet at its translated root is exactly the original divided Taylor jet. The observed boundary map on that presentation is $\Lambda_K = \Lambda \overline{\mathcal{T}}_K^{-1}$, whose kernel is exactly $\overline{\mathcal{T}}_K(\ker \Lambda)$.

On the original line define the pulled-back measure

$$dm_{0,K}(y) = \beta_K |f_L(c + iy)|^2 d\sigma(y). \quad (\text{IK04})$$

Repeated use of $\Gamma(z + 1) = z\Gamma(z)$ at $z = 1/4 + iy/2$ gives

$$|\Gamma(K/4 + iy/2)|^2 = 4^{-L} |\Gamma(1/4 + iy/2)|^2 \prod_{j=0}^{L-1} (y^2 + t_j^2).$$

Consequently IKO4 is precisely the original order- K Gamma density after IKO3, with its mass $(2\pi)^{K/2}$. More explicitly, the exact Gamma integral of the source gives

$$\int dm_{0,K} = (2\pi)^{K/2}, \quad \gamma_{K,j} = (2\pi)^{K/2} \frac{j! \Gamma(j + K/2)}{\Gamma(K/2)} = \beta_K \gamma_j \prod_{a=0}^{2L-1} (j + a + \frac{1}{2}). \quad (\text{IKO5})$$

Here is a direct derivation of the parameter dependence. The Euler double integral, with $u + v = w$ and $\log(u/v) = 2x$, gives the density characteristic function

$$\int e^{ity} \frac{|\Gamma(a + iy/2)|^2}{2\pi} dy = 2\Gamma(2a)(2 \cosh t)^{-2a}, \quad a > 0.$$

This is the same full calculation as CTR50: the Jacobian evaluates the positive w integral, and Fourier inversion is justified by the integrable exponentially decaying derivatives of $(2 \cosh x)^{-2a}$. At $a = K/4$ multiply by $\beta_K 4^L$; the constant becomes $\beta_K 2^{2L+1-K/2} \Gamma(K/2) = (2\pi)^{K/2}$. It follows that the characteristic function of IKO4 is $(2\pi)^{K/2} (\cosh t)^{-K/2}$, including its value at zero. Its moment generating function for real $|t| < \pi/2$ is the same formula with $\cos t$; analytic continuation is justified by the Gamma exponential bound times its fixed polynomial Φ_L . Define the monic polynomials by

$$\sum_{j \geq 0} p_{K,j}(y) \frac{z^j}{j!} = (1 + z^2)^{-K/4} \exp(y \arctan z).$$

For sufficiently small real z, w , integrate the product of these two generating functions. The identity $\cos(\arctan z + \arctan w) = (1 - zw)/\sqrt{(1 + z^2)(1 + w^2)}$ makes the result exactly $(2\pi)^{K/2} (1 - zw)^{-K/2}$. Coefficient differentiation is justified by the integrable exponential majorant. Its $z^i w^j$ coefficients are zero for $i \neq j$ and give the squared norm $(2\pi)^{K/2} j! (K/2)_j$ when $i = j$. The generating function shows leading coefficient one. Multiplication by i^j after $y = (S - c)/i$ gives monic polynomials on the original line with the same norms, proving IKO5. For its last equality insert $\gamma_j = \sqrt{2\pi} j! (1/2)_j$ and retain $\Gamma(1/2) = \sqrt{\pi}$ before cancellation. For $L = 0$ one has $K = 1$, $\beta_K = 1$, and the original fixed measure σ .

16.2 Exact multiplication on every remainder fibre

For $N \geq q - 1$, retain the same remainder map $J_N : \mathcal{P}_N \rightarrow E_\chi$. Write E_χ for the complete ordered divided Taylor map from E_χ to its q primary jets. All original roots have nonzero imaginary displacement from c because k is odd and $(2b - k)\gamma \neq 0$. The roots $c + t_j$ of f_L are distinct and real, so f_L and χ have no common root. Taylor multiplication by f_L on the complete χ jets is therefore invertible; call it U_{f_L} . Its determinant is exactly

$$\det U_{f_L} = \prod_{a,b=0}^k \prod_{j=0}^{L-1} ((2a - k)\delta - t_j + i(2b - k)\gamma)^e \neq 0. \quad (\text{IKO6})$$

Multiplication gives the bijection of specified affine spaces

$$\{P \in \mathcal{P}_N : J_N P = z\} \longrightarrow \{Q \in \mathcal{P}_{N+L} : E_{f_L} Q = 0, E_\chi Q = U_{f_L} E_\chi z\}, \quad P \longmapsto f_L P. \quad (\text{IKO7})$$

Indeed zero evaluation at every distinct f_L root implies divisibility by f_L , and division has degree at most N ; its χ jets are recovered using $U_{f_L}^{-1}$. This proves the inverse on the stated fibre, not just injectivity on polynomials. The full relation injection becomes $Q \mapsto f_L \chi Q$ on the original domain $\mathcal{P}_{N-q}$. On the product-jet image the observed map is exactly

$$(0, Y) \longmapsto \Lambda E_\chi^{-1} U_{f_L}^{-1} Y. \quad (\text{IKO8})$$

Thus the original observation, unit, kernel, and all of its nilpotent directions are carried through this independent order.

Here is its finite Gram formula. Let $p_j(S)$ be the original monic Gamma polynomials in the S coordinate with norms γ_j . The kernel for any ordered list of divided evaluations $\mathcal{E}$ at degree d is $K_{\mathcal{E},d} = \sum_{j=0}^d \mathcal{E}(p_j) \mathcal{E}(p_j)^* / \gamma_j$. At degree $N + L$ take the full product list $(\mathbf{E}_\chi, \mathbf{E}_{f_L})$. It is onto because Hermite interpolation modulo χf_L has dimension $q + L \leq N + L + 1$. Its kernel Gram is positive definite. Its bottom block is $K_{f_L, N+L}$ and its Schur complement is

$$C_{L,N} = K_{\chi, N+L} - K_{\chi, f_L; N+L} K_{f_L, N+L}^{-1} K_{f_L, \chi; N+L} > 0.$$

The minimum of the original squared norm for prescribed product jets $(U_{f_L} \mathbf{E}_\chi z, 0)$ is their quadratic form in the inverse joint kernel. This follows by the same constrained minimum calculation as BSL3; its upper-left inverse block is $C_{L,N}^{-1}$. Multiplying by the exact source factor β_K gives

$$G_N^{(K)} = \beta_K \mathbf{E}_\chi^* U_{f_L}^* C_{L,N}^{-1} U_{f_L} \mathbf{E}_\chi. \quad (\text{IK09})$$

For $L = 0$, interpret the bottom block as empty; the formula is the fixed Gamma quotient G_N^σ .

For completeness, this is also an exact determinant of size $2L$. Let A have columns

$$\mathbf{E}_\chi(p_{N+j}) / \sqrt{\gamma_{N+j}},$$

$1 \leq j \leq L$, and let

$$B = K_{\chi, f_L; N+L} K_{f_L, N+L}^{-1/2}.$$

The finite sum identity $K_{\chi, N+L} = K_{\chi, N} + AA^*$ gives

$$d_{L,N} := \frac{\det C_{L,N}}{\det K_{\chi, N}} = \det \left(I_{2L} + \text{diag}(I_L, -I_L) [A, B]^* K_{\chi, N}^{-1} [A, B] \right) > 0. \quad (\text{IK010})$$

The identity $\det(I + AB) = \det(I + BA)$ follows by eliminating either diagonal block in $\begin{pmatrix} I & A \\ -B & I \end{pmatrix}$. This proves IK010, including its mixed cross entries and its sign. Taking determinants of IK09 gives at each individual degree

$$\log \det G_N^{(K)} - \log \det G_N^\sigma = q \log \beta_K + 2 \log |\det U_{f_L}| - \log d_{L,N}. \quad (\text{IK011})$$

16.3 The exact four-endpoint return for arbitrary L

For the original signs $+, +, -, -$ at $N = q - 1, q, 2q - 1, 2q$, define $\mathcal{B}_k^{(K)}$ using $G_N^{(K)}$ and put

$$\mathfrak{T}_{k;L} = \mathcal{B}_k^{(K)} - \mathcal{B}_k^\sigma.$$

The two degree-independent terms in IK011 have signed coefficient zero only after their complete values have been retained. Hence

$$\mathfrak{T}_{k;L} = -\log d_{L,q-1} - \log d_{L,q} + \log d_{L,2q-1} + \log d_{L,2q}. \quad (\text{IK012})$$

The complementary source/relation presentation follows from the same monic frame used in BSL3. Define

$$\begin{aligned} D_{L,N} &= \prod_{j=0}^N \prod_{a=0}^{2L-1} (j + a + \tfrac{1}{2}), \\ P_{q,L} &= \log D_{L,q-1} + \log D_{L,q} - \log D_{L,2q-1} - \log D_{L,2q}, \\ R_{\chi,L,r} &= \frac{\det \text{Gram}_\sigma[f_L \chi S^j]_{j < r}}{\det \text{Gram}_\sigma[\chi S^j]_{j < r}}. \end{aligned}$$

The source norm products IK05 and the exact relation ranks $0, 1, q, q + 1$ then prove

$$\boxed{\mathfrak{T}_{k;L} = P_{q,L} - \log R_{\chi,L,1} + \log R_{\chi,L,q} + \log R_{\chi,L,q+1}.} \quad (\text{IK013})$$

All source and relation powers of β_K are accounted for by IK011; their four quotient contributions cancel. Equations IK012 and IK013 compute the same original finite return.

16.4 Full tails when the independent order is comparable to q

We prove that the actual f_L is retained even when $L = q/4$. Let $0 \leq L \leq q/4$ and $r \in \{1, q, q+1\}$. Keep the full forms $A_{\chi, L, r} = \text{Gram}_\sigma[f_L \chi S^j]_{j < r}$ and $A_{0, L, r} = \text{Gram}_\sigma[f_L(S)(S-c)^q S^j]_{j < r}$. On the bulk B of BSL6 their density ratio is exactly $\exp \psi_\chi$, so the finite inequalities BSL8 hold with these two forms and the same b_-, b_+ , for every such L .

Here are full tail estimates on this larger degree domain. Put $\Phi_L(y) = \prod_{j < L} (y^2 + t_j^2)$. For real z ,

$$\frac{\Phi_L(qz)}{\Phi_L(q)} \leq 1 \quad (|z| \leq 1), \quad \frac{\Phi_L(qz)}{\Phi_L(q)} \leq |z|^{2L} \quad (|z| \geq 1), \quad \Phi_L(qz) \geq \Phi_L(q) \quad (1 \leq z \leq 2). \quad (\text{IKO14})$$

Each inequality follows factor by factor from $(z^2 + (t_j/q)^2)/(1 + (t_j/q)^2)$; it does not replace this factor by z^2 in any determinant. For $Q(z) = P(c + iqz)$ with $\deg P \leq r-1$, the proved Legendre coefficient inequality CTR40 gives $|Q(z)|^2 \leq r^2 8^{2q} \int_1^2 |Q|^2$ on $|z| \leq \epsilon$, and $|Q(z)|^2 \leq r^2 ((131/32)|z|)^{2r-2} \int_1^2 |Q|^2$ on $|z| \geq 64$. The original density bounds are $C_L \exp(-\pi|y|) \leq \sigma(y) \leq C_U \exp(-|y|)$, with $C_U/C_L = (\cos 1)^{-1/2}$, proved in BRD23–24. Pairing each displacement root with its negative gives $|\chi(c + iy)|^2 \leq (y^2 + k^2(d+g))^q$. On $1 \leq z \leq 2$ the paired reverse bound gives $|\chi(c + iqz)|^2 \geq q^{2q} (3/4)^q$; on $|z| \leq \epsilon$ the upper bound is $q^{2q} (2\epsilon^2)^q$. Together with IKO14 these prove, for each of the two forms,

$$H^I \preceq \kappa_{I, r} H^B \quad (\text{IKO15})$$

with the literal $\kappa_{I, r}$ of BSL9. The factor $\Phi_L(q)$, the Jacobian q , and the factor q^{2q} occur in both numerator and bulk denominator before cancellation. The power form gives a stronger inequality because its paired bounds are exact powers of z .

For $|z| \geq 64$, the paired upper bound is $|\chi(c + iqz)|^2 \leq q^{2q} |z|^{2q} \exp(q/(4 \cdot 64^2))$, using $k\sqrt{d+g}/q \leq \epsilon/2 \leq 1/2$ and $\log(1+x) \leq x$. Relative to the bulk lower factor, its added exponent is at most $q \log(4/3) + q/(4 \cdot 64^2) < q/2$. The polynomial exponent in the far integral is $D = 2q + 2L + 2r - 2 \leq (9/2)q \leq 5q$. Using both tails and

$$\int_{64}^{\infty} z^D \exp(-qz) dz \leq \frac{64^D \exp(-64q)}{q - D/64}, \quad q - D/64 \geq 59q/64,$$

gives a coefficient at most

$$\frac{128r^2}{59q\sqrt{\cos 1}} \exp(-[64 - 2\pi - 5 \log 64 - 2 \log(131/32) - 1/2]q).$$

The bracket is greater than 20: use $\pi < 4$, $\log 2 < 1$, and $\log(131/32) < 2$, the latter following from the positive exponential series at 2. Since $r \leq q+1 \leq 2q$, this proves $H^F \preceq \kappa_F H^B$ with the same κ_F as BSL9. The power form has no added paired exponent and satisfies the same bound. Thus every original tail value obeys $0 \leq t_H \leq L_r$.

Define $\eta_{L, r} = \log \det A_{\chi, L, r} - \log \det A_{0, L, r}$. We have proved throughout the stated macroscopic range

$$rb_- - L_r \leq \eta_{L, r} \leq rb_+ + L_r. \quad (\text{IKO16})$$

Let $\mathfrak{T}_{0; q, L}$ be IKO13 with the homogeneous relation polynomial $(S-c)^q$ and with the same actual f_L . Its exact difference from the original packet return is

$$\begin{aligned} \mathfrak{T}_{k; L} - \mathfrak{T}_{0; q, L} &= (\eta_{L, q} - \eta_{0, q}) + (\eta_{L, q+1} - \eta_{0, q+1}) \\ &\quad - (\eta_{L, 1} - \eta_{0, 1}), \\ |\mathfrak{T}_{k; L} - \mathfrak{T}_{0; q, L}| &\leq (2q+2)(b_+ - b_-) + 2(L_q + L_{q+1} + L_1) = O(q/e). \end{aligned} \quad (\text{IKO17})$$

This is an exact transport and a proved error on the original packet. In particular it is $o(q^2)$ when $L = q/4$, with all root locations and the whole multiplying polynomial still in its finite formulas.

16.5 The invariant arithmetic receiver and transition law

The original arithmetic measure, quotient, and observation are fixed throughout. Their already proved comparison with σ is $\mathcal{B}_k^{\text{ar}} = \mathcal{B}_k^\sigma + \delta_k^\sigma$. Substituting IK012 proves the exact identity for every $L \geq 0$

$$\boxed{\mathcal{B}_k^{\text{ar}} = \mathcal{B}_k^{(K)} - \mathfrak{T}_{k;L} + \delta_k^\sigma.} \quad (\text{IK018})$$

For two orders $K_a = 4L_a + 1$, $K_b = 4L_b + 1$ it also proves

$$\mathcal{B}_k^{(K_b)} - \mathcal{B}_k^{(K_a)} = \mathfrak{T}_{k;L_b} - \mathfrak{T}_{k;L_a}. \quad (\text{IK019})$$

This transition satisfies the additive cocycle identity for three orders by direct subtraction. When $L_b \geq L_a$, its underlying source map is the literal multiplication by $f_{L_b}/f_{L_a} = \prod_{j=L_a}^{L_b-1} (S - c - t_j)$, with the inverse given by division on its zero-product-jet fibre, exactly as in IK07. The observation on that fibre is IK08 with this quotient germ. Applying BSL20 to each of these positive quotient metrics propagates the same identity through the original kernel and boundary factors, including the full frame determinant before its signed cancellation.

Thus IK02 realizes the proposed comparable order in the complete original construction. The baseline and return both change by the explicit quantity IK013, and their arithmetic combination in IK018 is fixed. Their actual macroscopic coefficients are calculated with the full f_L field in the accompanying independent-order energy proof; a comparison based only on matching powers does not specify those coefficients.

17 The actual macroscopic marked Gamma equilibrium

The parameter in this section is $\lambda \geq 0$, and $\kappa = 2\lambda$. The original packet centre remains $c = k/2$; the auxiliary cutoff κ is obtained by the explicit parameter maps $K \mapsto L = (K - 1)/4$, $(L, q) \mapsto L/q$, and $\lambda \mapsto \kappa = 2\lambda$. The parameter λ is the limit of the actual ratio L/q in the marked factor with $K = 4L + 1$; neither K nor L is identified with the original packet parameter k . Define, on the original limiting coordinate $x > 0$,

$$\begin{aligned} h_\lambda^{\text{mark}}(x) &= \int_0^\lambda \log(x + 4t^2) dt \\ &= \lambda \log(x + 4\lambda^2) - 2\lambda + \sqrt{x} \arctan(2\lambda/\sqrt{x}), \\ V_\lambda(x) &= \pi\sqrt{x} - 2\log x - 2h_\lambda^{\text{mark}}(x). \end{aligned} \quad (\text{KME1})$$

The equality is verified by differentiating the displayed antiderivative with respect to λ and evaluating its zero value at $\lambda = 0$. Every additive constant in this potential is retained. In terms of κ ,

$$V_\lambda(x) = 2\sqrt{x} \arctan(\sqrt{x}/\kappa) - 2\log x - \kappa \log(x + \kappa^2) + 2\kappa, \quad V'_\lambda(x) = \int_\kappa^\infty \frac{du}{x + u^2} - \frac{2}{x}. \quad (\text{KME2})$$

At $\kappa = 0$ use $\arctan(\sqrt{x}/0) = \pi/2$. For $\kappa > 0$ the derivative is $\arctan(\sqrt{x}/\kappa)/\sqrt{x} - 2/x$; integration of $1/(x + u^2)$ proves the same formula at $\kappa = 0$ by its limiting value.

For a probability measure on $(0, \infty)$ put

$$\mathcal{E}_\lambda(\mu) = \int V_\lambda d\mu - \iint \log|x - y| d\mu(x) d\mu(y), \quad F_\lambda = -\inf_\mu \mathcal{E}_\lambda(\mu). \quad (\text{KME3})$$

This is the supremum convention for F_λ used in the determinant calculation. The potential tends to $+\infty$ at both ends, and $V_\lambda(x) = \pi\sqrt{x} - (2 + \kappa)\log x + O(x^{-1})$ as $x \rightarrow \infty$. For $\kappa > 0$, the integral of $\log(x + u^2)$ over $0 < u < \kappa$ has a finite limit as $x \downarrow 0$; for $\kappa = 0$ it vanishes identically. Thus the coefficient of the divergent $-\log x$ term at zero is always exactly 2. The extended pair kernel $\frac{1}{2}V_\lambda(x) + \frac{1}{2}V_\lambda(y) - \log|x - y|$ is bounded below, since it is at least $g(x) + g(y)$ with $g(x) = V_\lambda(x)/2 - \log(1 + x)$ bounded below. Finite energy forces $\int(\sqrt{x} + |\log x|) d\mu < \infty$ by the same lower bound. This specifies the energy on all probability measures, with value $+\infty$ allowed, without subtracting divergent terms.

17.1 Constructive existence and uniqueness of the two endpoints

We retain the original positive coordinate x , the parameter $0 \leq \lambda \leq 1/4$, and the exact parameter map $\kappa = 2\lambda$. The derivative of the potential under consideration is

$$V'_\lambda(x) = \int_\kappa^\infty \frac{du}{x + u^2} - \frac{2}{x}, \quad x > 0. \quad (\text{KEP1})$$

We prove directly that there is exactly one ordered pair $0 < a < b$ solving the two endpoint equations

$$\int_\kappa^\infty \frac{du}{\sqrt{(a + u^2)(b + u^2)}} = \frac{2}{\sqrt{ab}}, \quad \int_\kappa^\infty \left(1 - \frac{u^2}{\sqrt{(a + u^2)(b + u^2)}}\right) du = 4. \quad (\text{KEP2})$$

The proof supplies a monotone construction, its exact parameter derivative, and uniform positive endpoint bounds on the full parameter interval. It uses no theorem asserting the number of support intervals of an equilibrium measure.

For positive a, b , allowing $a = b$ during the construction, set

$$R_u = \sqrt{(a + u^2)(b + u^2)}, \quad G_\kappa(a, b) = \sqrt{ab} \int_\kappa^\infty \frac{du}{R_u}, \quad B_\kappa(a, b) = \int_\kappa^\infty \left(1 - \frac{u^2}{R_u}\right) du. \quad (\text{KEP3})$$

These definitions are in the original x coordinate: in particular a diagonal value $a = b = s$ below means the endpoint $x = s$, not $x = \sqrt{s}$. The integrands are positive. Their convergence at infinity follows from the explicit inequalities, valid for every $T > 0$,

$$0 < \int_T^\infty \frac{du}{R_u} \leq \frac{1}{T}, \quad 0 < \int_T^\infty \left(1 - \frac{u^2}{R_u}\right) du \leq \frac{a+b}{2T} + \frac{ab}{6T^3}. \quad (\text{KEP4})$$

Indeed $R_u \geq u^2$, and the exact rationalization

$$1 - \frac{u^2}{R_u} = \frac{(a+b)u^2 + ab}{R_u(R_u + u^2)} \leq \frac{a+b}{2u^2} + \frac{ab}{2u^4}$$

proves the second estimate. At finite u , positive a, b keep the denominators nonzero. Both functions are smooth in $a, b > 0$ and $\kappa \geq 0$. To justify this assertion explicitly, on each compact positive set of a, b , every fixed order of a, b derivatives of the integrands is bounded at finite u and is $O(u^{-2})$ at infinity. A κ derivative evaluates an integrand or one of its derivatives at its finite lower limit. The same formulas extend smoothly to κ in a neighbourhood of zero, by allowing a short negative lower-limit interval; this is only a local parameter extension of the integrals, without changing the original domain $\kappa \geq 0$.

Subscripts a, b in the following equations denote partial derivatives with κ fixed. Direct differentiation, including the prefactor $\sqrt{ab}$ in G_κ , gives

$$\begin{aligned} (B_\kappa)_a &= \frac{1}{2} \int_\kappa^\infty \frac{u^2 du}{(a+u^2)R_u} > 0, & (B_\kappa)_b &= \frac{1}{2} \int_\kappa^\infty \frac{u^2 du}{(b+u^2)R_u} > 0, \\ (G_\kappa)_a &= \sqrt{\frac{b}{a}} (B_\kappa)_a > 0, & (G_\kappa)_b &= \sqrt{\frac{a}{b}} (B_\kappa)_b > 0. \end{aligned} \quad (\text{KEP5})$$

Strict positivity holds also at $\kappa = 0$, because the displayed integrands are strictly positive for every $u > 0$. For example

$$\partial_a \left(\frac{\sqrt{ab}}{R_u} \right) = \frac{1}{2} \sqrt{\frac{b}{a}} \frac{u^2}{(a+u^2)R_u},$$

which proves the first derivative identity for G_κ without dropping its coefficient.

Fix $\kappa \geq 0$. On the diagonal the exact value is

$$G_\kappa(s, s) = B_\kappa(s, s) = s \int_\kappa^\infty \frac{du}{s+u^2} = \sqrt{s} \left(\frac{\pi}{2} - \arctan \frac{\kappa}{\sqrt{s}} \right). \quad (\text{KEP6})$$

This is strictly increasing with $s > 0$ by (KEP5), tends to zero as $s \downarrow 0$, and tends to infinity as $s \rightarrow \infty$. The first limit also follows from its upper bound $\pi\sqrt{s}/2$; the last follows because the parenthetical expression tends to $\pi/2$. There is therefore exactly one $s = s(\kappa) > 0$ with

$$G_\kappa(s, s) = 2, \quad B_\kappa(s, s) = 2. \quad (\text{KEP7})$$

For each $0 < a < s$, there is exactly one $b = b_\kappa(a) > s$ satisfying

$$G_\kappa(a, b_\kappa(a)) = 2. \quad (\text{KEP8})$$

To prove the assertion, first $G_\kappa(a, s) < G_\kappa(s, s) = 2$ by strict monotonicity. For fixed $a > 0$, the integral representation

$$G_\kappa(a, b) = \int_\kappa^\infty \frac{\sqrt{a}}{\sqrt{a+u^2}} \frac{\sqrt{b}}{\sqrt{b+u^2}} du$$

increases as $b \rightarrow \infty$ to the divergent integral $\int_\kappa^\infty \sqrt{a}/\sqrt{a+u^2} du$. Monotone convergence thus proves $G_\kappa(a, b) \rightarrow \infty$. Continuity and strict monotonicity in b prove (KEP8) and its uniqueness. The nonzero derivative $(G_\kappa)_b$ proves that $b_\kappa(a)$ is smooth, with the exact derivative

$$b'_\kappa(a) = -\frac{(G_\kappa)_a}{(G_\kappa)_b} = -\frac{b}{a} \frac{(B_\kappa)_a}{(B_\kappa)_b} < 0, \quad b = b_\kappa(a). \quad (\text{KEP9})$$

We need both end limits of this entire curve. As $a \uparrow s$, $b_\kappa(a)$ decreases to s . Indeed it decreases to a finite limit at least s , since it is bounded above by any one value $b_\kappa(a_0)$ with $a_0 < s$. Continuity of G_κ and uniqueness of the solution $G_\kappa(s, b) = 2$ identify that limit as s . At the other end,

$$b_\kappa(a) \longrightarrow \infty \quad \text{as } a \downarrow 0. \quad (\text{KEP10})$$

Otherwise monotonicity would bound $b_\kappa(a)$ by a finite M for all sufficiently small a , whence $2 = G_\kappa(a, b_\kappa(a)) \leq G_\kappa(a, M)$. The latter quantity tends to zero: its integrand tends to zero at every $u > 0$, and for $0 < a \leq a_0$ it is dominated by the integrable integrand defining $G_\kappa(a_0, M)$. The single value $u = 0$, when present as an endpoint, has zero measure. Dominated convergence gives the contradiction.

Now define the second endpoint functional on this curve by $Q_\kappa(a) = B_\kappa(a, b_\kappa(a))$, for $0 < a < s$. Its derivative, with the full coefficient from (KEP9), is

$$Q'_\kappa(a) = (B_\kappa)_a + (B_\kappa)_b b'_\kappa(a) = \left(1 - \frac{b}{a}\right) (B_\kappa)_a < 0. \quad (\text{KEP11})$$

At the diagonal end, continuity and (KEP7) give $Q_\kappa(a) \rightarrow 2$ as $a \uparrow s$. At the other end $Q_\kappa(a) \rightarrow \infty$. For the latter assertion, once $\sqrt{b} > \kappa$ we have the concrete lower bound

$$B_\kappa(a, b) \geq \int_\kappa^{\sqrt{b}} \left(1 - \frac{u}{\sqrt{b}}\right) du = \frac{\sqrt{b}}{2} - \kappa + \frac{\kappa^2}{2\sqrt{b}}. \quad (\text{KEP12})$$

The inequality follows from $R_u \geq u\sqrt{b}$ for $u \geq 0$ and the nonnegative integrand of B_κ outside the indicated interval. At $u = 0$ it follows by continuity. Equation (KEP10) makes the last expression tend to infinity. The intermediate value theorem and the strict derivative in (KEP11) now give exactly one

$$a(\kappa) \in (0, s(\kappa)) \quad \text{with} \quad Q_\kappa(a(\kappa)) = 4, \quad b(\kappa) = b_\kappa(a(\kappa)) > s(\kappa). \quad (\text{KEP13})$$

These are the required endpoints in (KEP2).

This construction covers every possible ordered solution. In fact if $a < b$ and $G_\kappa(a, b) = 2$, then $G_\kappa(a, a) < 2 < G_\kappa(b, b)$, so necessarily $a < s(\kappa) < b$. Uniqueness of (KEP8) then places the pair on the same curve, and the strict monotonicity in (KEP11) proves that the second equation can hold only at (KEP13). Thus the endpoint uniqueness is proved directly from the actual integrals, before any use of global energy uniqueness.

The construction also provides finite monotone brackets. First bracket the diagonal solution of (KEP7) by positivity and its two end limits. For a chosen $a < s$, increase b until $G_\kappa(a, b) > 2$ and then use strict monotonicity to locate (KEP8). Take $a = s/2^j$ successively; the limit $Q_\kappa(a) \rightarrow \infty$ ensures that a finite j gives $Q_\kappa(a) > 4$, whereas the diagonal end has value 2. These values bracket the unique solution of (KEP13). The omitted positive tails in evaluating either integral have the explicit bounds (KEP4), with an additional factor $\sqrt{ab}$ for G_κ . This supplies a construction by the displayed original integrals and monotone root intervals; no numerical root is assumed in the existence proof.

We next prove parameter regularity and retain its exact derivative. The endpoint Jacobian at any ordered positive pair is

$$\mathcal{J}_\kappa(a, b) := \det \begin{pmatrix} (G_\kappa)_a & (G_\kappa)_b \\ (B_\kappa)_a & (B_\kappa)_b \end{pmatrix} = \frac{b-a}{\sqrt{ab}} (B_\kappa)_a (B_\kappa)_b > 0. \quad (\text{KEP14})$$

The integral smoothness proved after (KEP4) and this nonzero determinant give smooth dependence of the locally determined solution on κ . Uniqueness of the ordered solution makes all local solutions agree where their parameter intervals overlap. Hence $a(\kappa)$, $b(\kappa)$ are smooth for all $\kappa \geq 0$, including one-sided derivatives of every order at zero and a smooth local extension through zero. The lower-limit derivatives used here are exactly

$$\partial_\kappa G_\kappa(a, b) = -\frac{\sqrt{ab}}{R_\kappa}, \quad \partial_\kappa B_\kappa(a, b) = -\left(1 - \frac{\kappa^2}{R_\kappa}\right), \quad R_\kappa = \sqrt{(a + \kappa^2)(b + \kappa^2)}. \quad (\text{KEP15})$$

Differentiating both equations at the endpoints gives the original two-coordinate linear system

$$\begin{pmatrix} (G_\kappa)_a & (G_\kappa)_b \\ (B_\kappa)_a & (B_\kappa)_b \end{pmatrix} \begin{pmatrix} a'(\kappa) \\ b'(\kappa) \end{pmatrix} = \begin{pmatrix} \sqrt{ab}/R_\kappa \\ 1 - \kappa^2/R_\kappa \end{pmatrix}, \quad (a, b) = (a(\kappa), b(\kappa)). \quad (\text{KEP16})$$

Cramer's rule, followed by the exact identities in (KEP5), yields

$$\begin{aligned} a'(\kappa) &= \frac{\sqrt{a/b}(B_\kappa)_b}{\mathcal{J}_\kappa(a, b)} \left(\frac{b + \kappa^2}{R_\kappa} - 1 \right) > 0, \\ b'(\kappa) &= \frac{\sqrt{b/a}(B_\kappa)_a}{\mathcal{J}_\kappa(a, b)} \left(1 - \frac{a + \kappa^2}{R_\kappa} \right) > 0. \end{aligned} \quad (\text{KEP17})$$

Both strict signs follow from $a + \kappa^2 < R_\kappa < b + \kappa^2$, which holds because $a < b$. The map back to the stipulated parameter is

$$a_\lambda = a(2\lambda), \quad b_\lambda = b(2\lambda), \quad \frac{da_\lambda}{d\lambda} = 2a'(2\lambda), \quad \frac{db_\lambda}{d\lambda} = 2b'(2\lambda). \quad (\text{KEP18})$$

For every fixed $C_* > 0$, continuity and strict endpoint ordering give the quantitative existence of positive finite constants

$$a(0) \leq a(\kappa) < b(\kappa) \leq b(C_*) \quad (0 \leq \kappa \leq C_*), \quad \Delta_{C_*} := \min_{0 \leq \kappa \leq C_*} (b(\kappa) - a(\kappa)) > 0. \quad (\text{KEP19})$$

The first bounds also use the proved monotonicity. The minimum in the second statement is attained by continuity on the compact interval; its attaining pair still has $a < b$ by (KEP13), so its value is strictly positive. The same compactness argument applied to (KEP14) gives $\min_{0 \leq \kappa \leq C_*} \mathcal{J}_\kappa(a(\kappa), b(\kappa)) > 0$. All fixed orders of endpoint derivatives are bounded on this compact interval by smoothness and the local extension at zero. In particular these statements hold uniformly for the entire original range $C_* = 1/2$, equivalently $0 \leq \lambda \leq 1/4$.

Finally the endpoint at $\kappa = 0$ is connected exactly to the original elliptic equations. Put $u_0 = \sqrt{a(0)}$, $v_0 = \sqrt{b(0)}$, and $k_0 = \sqrt{1 - u_0^2/v_0^2}$. For arbitrary positive $a < b$, substitution $u = \sqrt{a} \tan \theta$ proves

$$\int_0^\infty \frac{du}{\sqrt{(a+u^2)(b+u^2)}} = \frac{1}{\sqrt{b}} K\left(\sqrt{1-a/b}\right), \quad K(k) = \int_0^{\pi/2} \frac{d\theta}{\sqrt{1-k^2 \sin^2 \theta}}.$$

To check the second integral including its factor, use $\sqrt{x} = (2/\pi) \int_0^\infty x/(x+u^2) du$. Integrating this positive identity against $dx/[\pi \sqrt{(b-x)(x-a)}]$ on $[a, b]$ and using

$$\frac{1}{\pi} \int_a^b \frac{dx}{(x+u^2)\sqrt{(b-x)(x-a)}} = \frac{1}{R_u}$$

gives

$$\frac{1}{\pi} \int_a^b \frac{\sqrt{x} dx}{\sqrt{(b-x)(x-a)}} = \frac{2}{\pi} B_0(a, b).$$

The substitution $x = b \cos^2 \theta + a \sin^2 \theta$ evaluates the left side as $2\sqrt{b} E(\sqrt{1-a/b})/\pi$, with $E(k) = \int_0^{\pi/2} \sqrt{1-k^2 \sin^2 \theta} d\theta$. All exchanges are Tonelli exchanges of positive integrands. The arcsine integral used above follows itself by this same trigonometric substitution and then the tangent substitution in $\int_0^{\pi/2} d\theta/(b \cos^2 \theta + a \sin^2 \theta + u^2)$. Therefore the endpoint equations at $\kappa = 0$ are precisely

$$u_0 K(k_0) = 2, \quad v_0 E(k_0) = 4, \quad k_0^2 = 1 - u_0^2/v_0^2. \quad (\text{KEP20})$$

Thus the entire parameter family has been constructed from the actual endpoint system and returns to the previously calculated elliptic endpoint at zero, with every coordinate map and coefficient explicit.

17.2 The positive density and the full variational inequality

Use the unique endpoints $a = a(\kappa)$ and $b = b(\kappa)$ established in (KEP1)–(KEP20), so $0 < a < b$. Write

$$m_{\text{end}} = \frac{a+b}{2}, \quad d_{\text{end}} = \frac{b-a}{2}, \quad D_s = \sqrt{(a+s)(b+s)}, \quad R_u = D_{u^2}, \quad D_0 = \sqrt{ab}. \quad (\text{KME4})$$

The two actual endpoint equations are

$$\int_{\kappa}^{\infty} \frac{du}{R_u} = \frac{2}{D_0}, \quad \int_{\kappa}^{\infty} \left(1 - \frac{u^2}{R_u}\right) du = 4. \quad (\text{KME5})$$

Define, for every $x > 0$,

$$\begin{aligned} H_{\kappa}(x) &= \frac{2}{xD_0} - \int_{\kappa}^{\infty} \frac{du}{(x+u^2)R_u} \\ &= \frac{1}{x} \int_{\kappa}^{\infty} \frac{u^2 du}{(x+u^2)R_u} > 0, \\ \rho_{\kappa}(x) &= \frac{\sqrt{(b-x)(x-a)}}{2\pi} H_{\kappa}(x) \quad (a < x < b), \end{aligned} \quad (\text{KME6})$$

with density zero outside $[a, b]$. The second identity is precisely the first endpoint equation in (KME5); it proves strict positivity rather than assuming it. All integrals converge absolutely. On a compact positive x interval their derivatives in x converge locally uniformly: the last integrand in (KME6) is $O(u^2)$ at zero and $O(u^{-2})$ at infinity. Thus H_{κ} is smooth and strictly positive on $[a, b]$.

Let $\mathcal{R}(z) = \sqrt{(z-a)(z-b)}$ on $\mathbb{C} \setminus [a, b]$, with $\mathcal{R}(z)/z \rightarrow 1$ at infinity. In particular $\mathcal{R}(x)$ is negative for $x < a$ and positive for $x > b$, and its upper boundary value in the interval is $i\sqrt{(b-x)(x-a)}$. The elementary arcsine and semicircle integrals give

$$\begin{aligned} \frac{1}{\pi} \int_a^b \frac{dt}{(z-t)\sqrt{(b-t)(t-a)}} &= \frac{1}{\mathcal{R}(z)}, \\ \frac{1}{\pi} \int_a^b \frac{\sqrt{(b-t)(t-a)}}{z-t} dt &= z - m_{\text{end}} - \mathcal{R}(z), \\ \frac{1}{\pi} \int_a^b \frac{\sqrt{(b-t)(t-a)}}{t+s} dt &= m_{\text{end}} + s - D_s. \end{aligned} \quad (\text{KME7})$$

To verify the first equation directly for $z > b$, put $t = m_{\text{end}} + d_{\text{end}} \cos \theta$ and $w = \tan(\theta/2)$; its integral becomes $\pi^{-1} \int_0^{\infty} 2 dw / [(z-b) + (z-a)w^2]$. Analytic continuation proves the stated domain. Polynomial division of $(b-t)(t-a)$ then proves the other two identities. Partial fractions yield

$$A_s(z) := \frac{1}{\pi} \int_a^b \frac{\sqrt{(b-t)(t-a)}}{(t+s)(z-t)} dt = 1 - \frac{D_s + \mathcal{R}(z)}{z+s}. \quad (\text{KME8})$$

At $z = -s$ the right side has its removable value.

The mass of the density is exactly

$$\begin{aligned} \int_a^b \rho_{\kappa}(t) dt &= \frac{m_{\text{end}} - D_0}{D_0} - \frac{1}{2} \int_{\kappa}^{\infty} \left(\frac{m_{\text{end}} + u^2}{R_u} - 1 \right) du \\ &= -1 + \frac{1}{2} \int_{\kappa}^{\infty} \left(1 - \frac{u^2}{R_u} \right) du = 1. \end{aligned} \quad (\text{KME9})$$

The cancellation of the terms containing m_{end} uses the first equation in (KME5), and the last equality uses its second equation. If $G_{\kappa}(z) = \int_a^b \rho_{\kappa}(t)/(z-t) dt$, the same substitutions give, initially off the negative real axis and off $[a, b]$,

$$G_{\kappa}(z) = \frac{1}{D_0} A_0(z) - \frac{1}{2} \int_{\kappa}^{\infty} \frac{A_{u^2}(z)}{R_u} du \quad (\text{KME10})$$

$$= \frac{1}{2} \left\{ \int_{\kappa}^{\infty} \frac{du}{z+u^2} - \frac{2}{z} - \mathcal{R}(z)H_{\kappa}(z) \right\} = \frac{V'_{\lambda}(z) - \mathcal{R}(z)H_{\kappa}(z)}{2}.$$

All the exchanges converge absolutely on compact subsets of this domain; $A_{u^2}(z) = O(u^{-2})$ at infinity. The constant terms cancel by (KME5). The integral definition on the first line provides the continuation of G_{κ} to all of $\mathbb{C} \setminus [a, b]$.

Set

$$U_{\kappa}(x) = V_{\lambda}(x) - 2 \int_a^b \log |x - t| \rho_{\kappa}(t) dt. \quad (\text{KME11})$$

It is continuous on $(0, \infty)$: the density is bounded and compactly supported, and the integral of $|\log |x - t||$ over a small interval about x tends uniformly to zero with its length. In the interior, the derivative of its logarithmic potential is the principal value integral. Subtracting $\rho_{\kappa}(x)$ on a symmetric neighbourhood of x proves this directly, because the remaining numerator is $O(|x - t|)$. Taking the real boundary value of (KME10) therefore gives $U'_{\kappa} = 0$ there. The same subtraction proves that this real boundary value is the indicated principal value. There is a finite constant ℓ_{κ} with

$$U_{\kappa} = \ell_{\kappa} \text{ on } [a, b], \quad U'_{\kappa}(x) = \mathcal{R}(x)H_{\kappa}(x) \begin{cases} < 0, & 0 < x < a, \\ > 0, & x > b. \end{cases} \quad (\text{KME12})$$

It follows that $U_{\kappa}(x) > \ell_{\kappa}$ at every positive point outside $[a, b]$. This proves the exterior inequality on the entire original domain, including the interval adjacent to zero.

We give the energy positivity argument needed for global uniqueness. For a finite signed real measure η of total mass zero with $|\log |x - y||$ integrable against $|\eta| \otimes |\eta|$, the identity

$$-\log r = \frac{1}{2} \int_0^{\infty} \frac{e^{-tr^2} - e^{-t}}{t} dt \quad (r > 0) \quad (\text{KME13})$$

follows by differentiation in r and evaluation at $r = 1$. Each truncated half-integral has absolute value at most $|\log r|$ and the same sign as $-\log r$. Dominated convergence and the zero mass consequently give

$$\begin{aligned} - \iint \log |x - y| d\eta(x) d\eta(y) &= \frac{1}{2} \int_0^{\infty} \frac{1}{t} \iint e^{-t(x-y)^2} d\eta(x) d\eta(y) dt \geq 0, \\ \iint e^{-t(x-y)^2} d\eta(x) d\eta(y) &= \frac{1}{\sqrt{4\pi t}} \int_{\mathbb{R}} e^{-\xi^2/(4t)} |\widehat{\eta}(\xi)|^2 d\xi. \end{aligned} \quad (\text{KME14})$$

The second identity follows by completing the square in the scalar Gaussian Fourier integral and integrating it against the finite measure. If $\eta \neq 0$, its Fourier transform is nonzero somewhere and, being continuous, makes the last integral strictly positive for every $t > 0$. For completeness, a measure with identically zero Fourier transform vanishes: its convolution with every centred Gaussian is zero by that same formula, and Gaussian approximate identities converge uniformly on each compactly supported continuous test function. Thus (KME14) has equality only for $\eta = 0$.

Let $\mu_{\kappa}(dx) = \rho_{\kappa}(x)dx$ and let μ have finite energy. For $\eta = \mu - \mu_{\kappa}$ the absolute logarithmic integrability just used holds. The self term for μ follows from its finite energy and logarithmic moment; that for μ_{κ} follows from bounded compact support. For the cross term, the negative logarithmic singularity integrated against the bounded density is uniformly bounded, and its positive part is bounded by $\log(1+x) + \log(1+y)$. Expanding the two energies now gives

$$\mathcal{E}_{\lambda}(\mu) - \mathcal{E}_{\lambda}(\mu_{\kappa}) = \int (U_{\kappa} - \ell_{\kappa}) d\mu - \iint \log |x - y| d\eta(x) d\eta(y) \geq 0. \quad (\text{KME15})$$

The inequality is strict unless $\mu = \mu_{\kappa}$. Infinite energy competitors cannot lower the finite energy of μ_{κ} . Therefore the explicit density above is the unique global minimizing probability measure, and $F_{\lambda} = -\mathcal{E}_{\lambda}(\mu_{\kappa})$ is its actual energy value.

17.3 Entropy and continuity on the actual parameter interval

The endpoint proof gives smooth endpoints and $0 < a(\kappa) < b(\kappa) < \infty$ on every compact κ interval. On such an interval their distance from zero and their separation have positive lower bounds. Uniform differentiation in (KME6) proves that $H_\kappa(x)$ and its parameter derivatives are continuous and bounded on all the resulting supports. The positive integral in (KME6) also gives a positive lower bound there. In the fixed coordinate $x = a + (b - a)y$, the probability measure is

$$d\mu_\kappa(x) = \frac{(b - a)^2}{2\pi} \sqrt{y(1 - y)} H_\kappa(a + (b - a)y) dy, \quad 0 < y < 1. \quad (\text{KME16})$$

Its smooth coefficients are uniformly bounded on compact parameter intervals, with each parameter derivative dominated by a constant times $\sqrt{y(1 - y)}$. This proves continuity, and smoothness for smooth observables, of all the integrals used below.

The probability reference in the original Gamma ensemble is $d\omega(x) = \frac{1}{2}x^{-1/2}e^{-\sqrt{x}}dx$. The exact density ratio on the support is

$$\frac{d\mu_\kappa}{d\omega}(x) = 2\sqrt{x} e^{\sqrt{x}} \rho_\kappa(x). \quad (\text{KME17})$$

The factors other than the square-root endpoint factor are bounded above and below by positive numbers uniformly for κ in a compact interval. It follows from (KME16) that

$$\int \left| \log \frac{d\mu_\kappa}{d\omega} \right| d\mu_\kappa < \infty, \quad \iint |\log |x - y|| d\mu_\kappa(x) d\mu_\kappa(y) < \infty.$$

Indeed the first absolute integral is bounded by a constant times $\int_0^1 \sqrt{y(1 - y)}(1 + |\log y| + |\log(1 - y)|)dy$, which is finite; for example $\int_0^1 y^{1/2} |\log y| dy = 4/9$ by differentiation of $\int_0^1 y^p dy = 1/(p + 1)$. The second integral follows from bounded compact support as above. These facts hold in particular uniformly on the actual range $0 \leq \lambda \leq 1/4$.

17.4 Explicit elementary endpoint integrals for the energy

Define $d_{\text{end}}, m_{\text{end}}, D_s$ as in (KME4) and, for $z, s \geq 0$, put

$$C_z = \frac{m_{\text{end}} + z + D_s}{2}, \quad \theta_s = \frac{d_{\text{end}}}{m_{\text{end}} + s + D_s}, \quad (\text{KME18})$$

$$Q_s(z) = (m_{\text{end}} + s - D_s) \log C_z - d_{\text{end}} \theta_s - 2D_s \log(1 - \theta_s \theta_s).$$

All logarithm arguments are positive: $0 < \theta_s \leq \theta_0 < 1$. The exact identity behind these formulas is

$$Q_s(z) = \frac{1}{\pi} \int_a^b \frac{\sqrt{(b - x)(x - a)}}{x + s} \log(x + z) dx. \quad (\text{KME19})$$

To prove it, substitute $x = m_{\text{end}} + d_{\text{end}} \cos \phi$, and use the convergent series

$$\log(x + z) = \log C_z + 2 \sum_{j \geq 1} \frac{(-1)^{j+1} \theta_z^j}{j} \cos(j\phi), \quad \frac{1}{x + s} = \frac{1}{D_s} \left(1 + 2 \sum_{j \geq 1} (-\theta_s)^j \cos(j\phi) \right).$$

Their averages are respectively $\log C_z$ and $[\log C_z + 2 \log(1 - \theta_s \theta_s)]/D_s$ after multiplication. The average of $x \log(x + z)$ is $m_{\text{end}} \log C_z + d_{\text{end}} \theta_s$. Finally divide $(b - x)(x - a)/(x + s) = -x + 2m_{\text{end}} + s - D_s^2/(x + s)$ and combine these three averages. This proves (KME19) with every constant in (KME18).

The logarithmic expectation, including the added mark parameter z , is

$$M_\kappa(z) := \int \log(x + z) d\mu_\kappa(x) = \frac{1}{2} \int_\kappa^\infty \frac{Q_0(z) - Q_{u^2}(z)}{R_u} du. \quad (\text{KME20})$$

For proof, use (KME6) in the first expression, giving $Q_0(z)/D_0 - \frac{1}{2} \int_{\kappa}^{\infty} Q_{u^2}(z)/R_u du$; the first endpoint equation gives the stated form. The exchanges are absolute, since $\log(x+z)$ is bounded on the support, $Q_0(z) - Q_{u^2}(z) = O(u^2)$ near zero, and $Q_{u^2}(z) = O(u^{-2})$ at infinity by (KME19).

For the logarithmic potential at the right endpoint define

$$\begin{aligned} C_{\text{edge}} &= \frac{d_{\text{end}}}{2} = \frac{b-a}{4}, \\ Q_s^{\text{edge}} &= (m_{\text{end}} + s - D_s) \log C_{\text{edge}} + d_{\text{end}} - 2D_s \log(1 + \theta_s), \\ J_{\kappa} &= \frac{1}{2} \int_{\kappa}^{\infty} \frac{Q_0^{\text{edge}} - Q_{u^2}^{\text{edge}}}{R_u} du = \int \log(b-x) d\mu_{\kappa}(x). \end{aligned} \quad (\text{KME21})$$

These identities follow by the same calculation with $\log(b-x) = \log(d_{\text{end}}/2) - 2 \sum_{j \geq 1} \cos(j\phi)/j$. One may justify this endpoint series by first replacing its coefficient 1 by $r < 1$ and then taking $r \uparrow 1$; the possible singularity is bounded by a constant plus $|\log \phi|$, integrable on $[0, \pi]$. The resulting averages are $\log(d_{\text{end}}/2)$, $m_{\text{end}} \log(d_{\text{end}}/2) - d_{\text{end}}$, and $[\log(d_{\text{end}}/2) + 2 \log(1 + \theta_s)]/D_s$, respectively. This proves the displayed Q_s^{edge} and then J_{κ} by the same endpoint cancellation as (KME20). All the final integrals converge absolutely, including the logarithmic singularity at $x = b$.

There is now an explicit energy formula involving only the uniquely specified endpoints and real one-dimensional integrals:

$$F_{\lambda} = J_{\kappa} - \frac{1}{2} V_{\lambda}(b) - 3 - \kappa + M_{\kappa}(0) + \frac{\kappa}{2} M_{\kappa}(\kappa^2). \quad (\text{KME22a})$$

Here is a direct derivation of the exact constant $3 + \kappa$. Push the minimizing probability measure forward by the explicit coordinate dilation $x \mapsto rx$ for $r > 0$. Differentiating its energy at its minimizing value $r = 1$ is legitimate on its compact positive support and gives $\int x V'_{\lambda}(x) d\mu_{\kappa}(x) = 1$: the logarithmic double integral contributes exactly $-\log r$ to the energy. By (KME2) this is $\int \sqrt{x} \arctan(\sqrt{x}/\kappa) d\mu_{\kappa}(x) = 3$. Taking the expectation of the first formula in (KME2) now gives $\int V_{\lambda} d\mu_{\kappa} = 6 + 2\kappa - 2M_{\kappa}(0) - \kappa M_{\kappa}(\kappa^2)$. Finally $\ell_{\kappa} = V_{\lambda}(b) - 2J_{\kappa}$, and the integrated Euler equality gives $F_{\lambda} = -\frac{1}{2}(\int V_{\lambda} d\mu_{\kappa} + \ell_{\kappa})$. These identities prove (KME22a).

An equivalent representation, useful for checking all the additive constants directly against the Stieltjes integral, is

$$F_{\lambda} = J_{\kappa} + M_{\kappa}(0) - \frac{1}{2} V_{\lambda}(b) - \frac{1}{2} \int_{\kappa}^{\infty} (M_{\kappa}(u^2) - 2 \log u) du + \kappa \log \kappa - \kappa. \quad (\text{KME22})$$

At $\kappa = 0$ set $\kappa \log \kappa = 0$. To prove the formula without an unspecified energy constant, first observe the scalar identity

$$V_{\lambda}(x) = \int_{\kappa}^{\infty} \log(1 + x/u^2) du - 2 \log x - 2\kappa \log \kappa + 2\kappa. \quad (\text{KME23})$$

The identity $\int_0^{\infty} \log(1 + x/u^2) du = \pi \sqrt{x}$ follows by differentiation in $x > 0$ and then its zero limit as $x \downarrow 0$ (or by $u = \sqrt{x}v$ and one integration by parts). Subtracting the integral over $[0, \kappa]$ proves (KME23), including the integral $\int_0^{\kappa} 2 \log u du = 2\kappa \log \kappa - 2\kappa$. Its expectation is absolutely finite: near zero $|\log u|$ is integrable, and at infinity $0 \leq M_{\kappa}(u^2) - 2 \log u = \int \log(1 + x/u^2) d\mu_{\kappa} \leq b/u^2$. Since $\ell_{\kappa} = V_{\lambda}(b) - 2J_{\kappa}$ by (KME11)–(KME12), integration of the Euler equality gives $F_{\lambda} = -\frac{1}{2}(\int V_{\lambda} d\mu_{\kappa} + \ell_{\kappa})$. Substituting (KME23) proves exactly (KME22).

The difference of the actual energies also has the following exact form:

$$\frac{dF_{\lambda}}{d\lambda} = 2M_{2\lambda}(4\lambda^2), \quad F_{\lambda} - F_0 = 2 \int_0^{\lambda} M_{2t}(4t^2) dt. \quad (\text{KME24})$$

Here the subscript of M_{κ} is the lower Stieltjes endpoint κ , so the parameter relation in the formula is explicit. For a direct proof, the difference $V_{\lambda+h}(x) - V_{\lambda}(x) = -2 \int_{\lambda}^{\lambda+h} \log(x + 4t^2) dt$ is exact. Testing

the two unique minimizers in each other's variational problem bounds the difference quotient of F between the integrals of $2h^{-1} \int_{\lambda}^{\lambda+h} \log(x+4t^2)dt$ against these two measures. Formula (KME16) and compact positive support bounds show that both bounds tend to $2M_{2\lambda}(4\lambda^2)$. This includes the right derivative at zero. Continuity of this derivative and the fundamental theorem of calculus prove the second equation in (KME24).

The formulas above apply, in particular, to the actual macroscopic choice $K = q+1, L = q/4$ when q is a positive multiple of four. Its limiting mark parameter is exactly $\lambda = 1/4$ and the lower Stieltjes endpoint is exactly $\kappa = 1/2$. The energy in (KME22) or (KME24) is computed for the entire changed potential (KME1), with its additive constants and original $-2 \log x$ term preserved.

Finally the actual energy is convex as a function of λ . For each fixed compactly supported probability measure of finite logarithmic energy on $(0, \infty)$, its objective in (KME3) has second parameter derivative $16\lambda \int (x+4\lambda^2)^{-1} d\mu(x) \geq 0$. The supremum of these convex functions is F_{λ} : its proved minimizer belongs to this class for every parameter. Therefore the defining convexity inequality passes to that supremum, and (KME24) gives the unconditional tangent estimate

$$F_{\lambda} - F_0 \geq 2\lambda M_0(0) \quad (\lambda \geq 0). \quad (\text{KME25})$$

All the quantities on the right are given by (KME18)–(KME20) at the uniquely constructed zero-parameter endpoints.

This note records an exploratory comparison family of Gamma measures. The independent order K is a comparison parameter: the identities below do not derive this order from the original tensor, exterior, or spectral-sum operation. The original packet parameter k and its canonical source remain as specified by those original constructions.

18 The independent order and the unchanged original coordinate

Keep the original packet integers and centre

$$k = 4l + 1, \quad e = 1 + k(m - 1), \quad q = e(k + 1)^2 = 2n, \quad c = k/2,$$

and choose a separate integer L with $1 \leq L \leq q/4$. Its Gamma order is $K = 4L + 1$. The original packet order k has not been replaced by K , and the source coordinate remains $S = c + iy$. Since q is divisible by four, the exact choice $L = q/4$ gives $K = q + 1$. The original Gamma measure and the actual multiplying polynomial are

$$d\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy, \quad \sigma(\mathbb{R}) = \sqrt{2\pi}, \quad f_L(S) = \prod_{j=0}^{L-1} (S - c - 2j - 1/2), \quad (\text{MFG1})$$

$$\Phi_L(y) = |f_L(c + iy)|^2 = \prod_{j=0}^{L-1} (y^2 + (2j + 1/2)^2). \quad (\text{MFG2})$$

We first calculate the exact homogeneous relation source $\chi_0(S) = (S - c)^q$. The original displaced packet returns by its separate full-source comparison, with the same f_L retained.

Put $\mu_a = \int y^{2a} d\sigma(y)$ and let $\nu = (y \mapsto y^2)_* \sigma$ be the full square pushforward, including both half-axes. For integers $a, s \geq 0$ define

$$H_s^{(a;L)} = \det \left[\int_0^\infty t^{a+i+j} \prod_{r=0}^{L-1} (t + (2r + 1/2)^2) d\nu(t) \right]_{i,j < s}, \quad H_0^{(a;L)} = 1. \quad (\text{MFG3})$$

At $L = 0$ these are exactly the old $H_s^{(a)}$. The full real-line relation matrices have entries

$$\int y^{2q+i+j} \Phi_L(y) d\sigma(y).$$

Their even/odd congruence gives the literal paired determinant

$$Z_{q,L} = H_n^{(q;L)} H_{n+1}^{(q;L)} (H_n^{(q+1;L)})^2, \quad Z_{q,0} = Z_q. \quad (\text{MFG4})$$

The same row and column permutation is used, so its determinant sign is squared. The low relation ratio and the signed source product are

$$r_{q,L} = \frac{\int y^{2q} \Phi_L(y) d\sigma(y)}{\mu_q},$$

$$P_{q,L} = - \sum_{j=0}^{2L-1} \left[\sum_{a=q}^{2q-1} \log(a + j + 1/2) + \sum_{a=q+1}^{2q} \log(a + j + 1/2) \right]. \quad (\text{MFG5})$$

Here the complete source determinant ratio for this same f_L is

$$\frac{\det H_N^{\Phi_L \sigma}}{\det H_N^{\sigma}} = \prod_{a=0}^N \prod_{j=0}^{2L-1} (a + j + 1/2).$$

For clarity its monic norms follow with the original mass as follows. Gamma recurrence gives $\Phi_L(y)d\sigma(y) = 4^L |\Gamma(L + 1/4 + iy/2)|^2 dy / (2\pi)$. Write $\alpha = L + 1/4$. The Euler integral and its Fourier autocorrelation give the exact tilted mass

$$\int e^{ty} \Phi_L(y) d\sigma(y) = \sqrt{2} \Gamma(2\alpha) (\cos t)^{-2\alpha} \quad (|t| < \pi/2).$$

Indeed, for $g_\alpha(x) = e^{\alpha x - e^x}$, its Fourier transform is $\Gamma(\alpha + i\xi)$; the autocorrelation integral at displacement u is $\Gamma(2\alpha) / (2 \cosh(u/2))^{2\alpha}$ after $v = e^x$. Fourier inversion, $y = 2\xi$, and analytic continuation through the strip justified by the Euler contour bound give the displayed formula, including its factor $4^L 2^{1-2\alpha} = \sqrt{2}$. Define the monic polynomials by

$$\sum_{j \geq 0} p_j^{(\alpha)}(y) \frac{z^j}{j!} = (1 + z^2)^{-\alpha} e^{y \arctan z}.$$

Integrating the product of two such series and using

$$\cos(\arctan z + \arctan w) = (1 - zw) / \sqrt{(1 + z^2)(1 + w^2)}$$

gives $\sqrt{2} \Gamma(2\alpha) (1 - zw)^{-2\alpha}$. For sufficiently small real z, w , differentiated integration is justified by the same exponential bound. Coefficient comparison proves the squared norm $\sqrt{2} j! \Gamma(j + 2L + 1/2)$ and all off-diagonal inner products zero. At $L = 0$ the same norm is $\sqrt{2} j! \Gamma(j + 1/2)$. Their ratio is exactly the displayed finite product. Multiplication by i^j and $y = (S - c)/i$ makes these polynomials monic in S without changing their norms; all monic coefficient transformations have determinant one. This proves the source determinant identity, and cancellation gives precisely the two source product ranges in MFG5. Thus the actual homogeneous four-endpoint return for this independent Gamma order is

$$\boxed{T_{q,L}^{\text{hom}} = P_{q,L} - \log r_{q,L} + \log(Z_{q,L}/Z_{q,0})}. \quad (\text{MFG6})$$

This finite identity uses Φ_L throughout; it does not substitute the weight y^{2L} for MFG2.

19 The actual macroscopic potential from the full polynomial

Write $\lambda_q = L/q$ and suppose $\lambda_q \rightarrow \lambda \in (0, 1/4]$. For the literal scale $t = q^2 x$ define

$$\phi_{q,L}(x) = \prod_{j=0}^{L-1} \left(x + \left(\frac{2j + 1/2}{q} \right)^2 \right), \quad \Phi_L(q\sqrt{x}) = q^{2L} \phi_{q,L}(x). \quad (\text{MFG7})$$

Define the explicit integral

$$h_\lambda(x) = \int_0^\lambda \log(x + 4u^2) du = \lambda \log(x + 4\lambda^2) - 2\lambda + \sqrt{x} \arctan \frac{2\lambda}{\sqrt{x}}. \quad (\text{MFG8})$$

The equality follows by differentiating the displayed antiderivative in its upper limit, with value zero at that limit zero.

For fixed $x > 0$, the integrand in MFG8 is increasing with $u \geq 0$. The point $(j + 1/4)/q$ lies in the j th cell of width $1/q$. Both the sum of its values and q times the integral lie between the same left and right sums. Their difference is therefore bounded by the endpoint difference. This proves the global finite estimate

$$|\log \phi_{q,L}(x) - q h_{\lambda_q}(x)| \leq \log \left(1 + \frac{4\lambda_q^2}{x} \right). \quad (\text{MFG9})$$

In particular the convergence to h_λ after division by q is uniform on every compact subinterval of $(0, \infty)$.

The exact macroscopic field is consequently

$$\boxed{V_\lambda(x) = \pi\sqrt{x} - 2\log x - 2h_\lambda(x).} \quad (\text{MFG10})$$

For $\lambda > 0$ it has the equivalent derivative

$$V'_\lambda(x) = \frac{\arctan(\sqrt{x}/(2\lambda))}{\sqrt{x}} - \frac{2}{x} = \int_{2\lambda}^{\infty} \frac{du}{x+u^2} - \frac{2}{x}. \quad (\text{MFG11})$$

Indeed $\partial_x h_\lambda(x) = \arctan(2\lambda/\sqrt{x})/(2\sqrt{x})$, and the two positive arctangents sum to $\pi/2$. At $\lambda = 0$, MFG10 returns to $V_0(x) = \pi\sqrt{x} - 2\log x$. The extra term in MFG10 has order one in this macroscopic field; removing it would change the variational quantity being calculated.

For precision define

$$F_\lambda = \sup_{\eta \in \mathcal{P}((0,\infty))} \iint \left[\log|x-y| - \frac{V_\lambda(x) + V_\lambda(y)}{2} \right] d\eta(x)d\eta(y). \quad (\text{MFG12})$$

The definition is the single bounded-above kernel integral. The explicit macroscopic equilibrium provider KME1-24 in `../equilibrium/MACROSCOPIC_MARKED_EQUILIBRIUM.tex`, with its complete `endpoint_existence.tex` KEP1-20, constructs its unique positive density ρ_λ on $[a_\lambda, b_\lambda] \subset (0, \infty)$, proves the full exterior variational inequality, and proves finite entropy relative to the measure ω below. Its endpoint system is

$$\begin{aligned} \int_{2\lambda}^{\infty} \frac{du}{\sqrt{(a_\lambda + u^2)(b_\lambda + u^2)}} &= \frac{2}{\sqrt{a_\lambda b_\lambda}}, \\ \int_{2\lambda}^{\infty} \left[1 - \frac{u^2}{\sqrt{(a_\lambda + u^2)(b_\lambda + u^2)}} \right] du &= 4, \\ \rho_\lambda(x) &= \frac{\sqrt{(b_\lambda - x)(x - a_\lambda)}}{2\pi x} \int_{2\lambda}^{\infty} \frac{u^2 du}{(x + u^2)\sqrt{(a_\lambda + u^2)(b_\lambda + u^2)}}. \end{aligned} \quad (\text{MFG13})$$

The symbol ρ_λ here denotes exactly that provider's ρ_κ under $\kappa = 2\lambda$; the original centre remains $c = k/2$. KME15 proves its full variational inequality and KME16-17 prove its compact-support and entropy properties. The endpoint existence calculation supplies the unique ordered solution, including $\lambda = 0$. At zero this is exactly the original equilibrium with $uK(\kappa) = 2$, $vE(\kappa) = 4$. The following argument proves the new ensemble passage from the literal polynomial to this field.

20 The full original ensemble, including the discrete factors

Retain the exact original density map used in GEL3:

$$d\omega(x) = \frac{1}{2}x^{-1/2}e^{-\sqrt{x}}dx, \quad B_q(x) = 2qe^{\sqrt{x}}\sigma(q\sqrt{x}), \quad d\nu(q^2x) = B_q(x)d\omega(x).$$

The measure ω has mass one. Expansion of the two Vandermonde determinants and the substitution $t_i = q^2x_i$ give

$$\boxed{H_s^{(a;L)} = q^{2as+2s(s-1)+2Ls} J_{s,q,L}(a),} \quad J_{s,q,L}(a) = \frac{1}{s!} \int \prod_{i < j} (x_i - x_j)^2 \prod_{i=1}^s x_i^a \phi_{q,L}(x_i) B_q(x_i) d\omega(x_i). \quad (\text{MFG14})$$

The exact powers come from the original t^a factors, squared Vandermonde differences, and the L original quadratic factors per variable, respectively. Every original mass and Jacobian remains in $B_q d\omega$.

For sequences with $s \rightarrow \infty$, $q/s \rightarrow 2$, $a/s \rightarrow 2$ and $L/q \rightarrow \lambda \in (0, 1/4]$, we prove

$$\boxed{\frac{\log J_{s,q,L}(a)}{s^2} \rightarrow F_\lambda.} \quad (\text{MFG15})$$

We use the previously proved original Gamma inequalities, valid at every real y , with $C_\Gamma = \Gamma(1/4)^2/(2\pi)$:

$$C_\Gamma(1 + 4y^2)^{-3/4}e^{-\pi|y|/2} \leq \sigma(y) \leq C_\Gamma e^{1/2}(1 + 4y^2)^{1/4}e^{-\pi|y|/2}. \quad (\text{MFG16})$$

In particular, on every fixed compact interval $[a, b] \subset (0, \infty)$, $\log B_q(x) = -\pi q\sqrt{x}/2 + O(\log q)$ uniformly. Globally the same bounds give

$$B_q(x) \leq C_q \exp[-(\pi q/2 - 3/2)\sqrt{x}], \quad C_q = \frac{\Gamma(1/4)^2}{\pi} q e^{1/2}(1 + 2q)^{1/2}. \quad (\text{MFG17})$$

Upper bound with the singular endpoints retained. Put $R(x) = 1 + \sqrt{x} + |\log x|$. MFG9 gives a sequence $\varepsilon_q \rightarrow 0$ such that, globally on $x > 0$,

$$|q^{-1} \log \phi_{q,L}(x) - h_\lambda(x)| \leq \varepsilon_q R(x). \quad (\text{MFG18})$$

To check this global assertion, the endpoint error in MFG9 is at most $C + |\log x|$ for $0 < x \leq 1$ and at most a constant for $x \geq 1$. For all sufficiently large q , λ_q lies in a fixed positive compact interval around λ . The upper-limit derivative $\log(x + 4u^2)$ in that interval is bounded in absolute value by $C + \log(1 + x) \leq C + 2\sqrt{x}$. The difference $h_{\lambda_q} - h_\lambda$ therefore satisfies MFG18 as well. Also $|h_\lambda(x)| \leq CR(x)$, directly from MFG8.

After removing $C_q^s/s!$ by MFG17, the exponent of the remaining integral is

$$2 \sum_{i < j} \log |x_i - x_j| - s \sum_i V_s(x_i),$$

$$V_s(x) = \left(\frac{\pi q}{2s} - \frac{3}{2s} \right) \sqrt{x} - \frac{a}{s} \log x - \frac{1}{s} \log \phi_{q,L}(x).$$

The parameter limits and MFG18 show $|V_s - V_\lambda| \leq o(1)R$. For $s > 1$, this exponent equals $\sum_{i \neq j} K_{Q_s}(x_i, x_j)$, where $Q_s = sV_s/(s-1)$ and $K_Q(x, y) = \log |x - y| - (Q(x) + Q(y))/2$. Therefore for each sufficiently small fixed $\varepsilon > 0$, it is bounded above eventually by $\sum_{i \neq j} K_{V_\lambda - \varepsilon R}(x_i, x_j)$.

Here is the complete truncation step for this field. For $0 \leq \varepsilon \leq \varepsilon_0 < \min(1, \pi/2)$, put $Q_\varepsilon = V_\lambda - \varepsilon R$ and $r_\varepsilon(x) = \log(1 + x) - Q_\varepsilon(x)/2$. The value $h_\lambda(x)$ has a finite limit at zero and grows only logarithmically at infinity. Thus r_ε tends to $-\infty$ at both endpoints, uniformly for this interval of ε , and is uniformly bounded above. The inequality $K_{Q_\varepsilon}(x, y) \leq r_\varepsilon(x) + r_\varepsilon(y)$ proves that

$$K_{\varepsilon,M}(x, y) = \max(K_{Q_\varepsilon}(x, y), -M)$$

extends continuously to the compact square $[0, \infty]^2$; its value on the diagonal and wherever a coordinate is an endpoint is $-M$. Let $\Phi_{\varepsilon,M}$ be its largest integral over probability measures on that compact interval. For the empirical measure $\eta_s = s^{-1} \sum_i \delta_{x_i}$,

$$\sum_{i \neq j} K_{Q_\varepsilon}(x_i, x_j) \leq s^2 \iint K_{\varepsilon,M} d\eta_s^2 + sM \leq s^2 \Phi_{\varepsilon,M} + sM.$$

Because ω has mass one and $\log(C_q^s/s!)/s^2 \rightarrow 0$, this proves $\limsup s^{-2} \log J_{s,q,L}(a) \leq \Phi_{\varepsilon,M}$.

For completeness, $\Phi_{\varepsilon,M} \downarrow F(Q_\varepsilon)$ as $M \rightarrow \infty$. Choose maximizing probabilities for increasing M and pass to a weak subsequence on the compact interval. For every fixed truncation level, continuity of its kernel and its domination of subsequent truncations show that the limiting probability has untruncated kernel integral at least $\lim_M \Phi_{\varepsilon,M}$. Monotone convergence applies to a common upper bound minus the kernels. This integral is finite from below, because a compact interval probability

with finite logarithmic energy is a competitor of finite value at every truncation. It follows that the limiting probability has no mass at the two endpoints and gives a legitimate competitor for $F(Q_\varepsilon)$. The reverse inequality follows by testing every such competitor before truncation.

Finally $F(Q_\varepsilon) \rightarrow F_\lambda$ as $\varepsilon \downarrow 0$. The inequality $F(Q_\varepsilon) \geq F_\lambda$ follows from $R \geq 0$. For each fixed M , the truncated kernels converge uniformly to $K_{0,M}$: uniform coercivity puts all pairs with a coordinate outside a fixed compact interval at value $-M$, and on the remaining compact square the boundedness of R gives uniform convergence. Thus $\limsup_{\varepsilon \downarrow 0} F(Q_\varepsilon) \leq \Phi_{0,M}$ for every M . Letting $M \rightarrow \infty$ gives the claim. These steps prove the required upper bound in MFG15.

Lower bound on the proved actual equilibrium. Use the probability $\rho_\lambda(x)dx$ from MFG13. Its support is compact inside $(0, \infty)$, its logarithmic energy is finite, and its entropy relative to ω is finite, as proved by the explicit equilibrium construction. Restrict the ensemble integral to that support and apply Jensen's inequality after changing each $d\omega$ to $\rho_\lambda dx$. Writing $f = d(\rho_\lambda dx)/d\omega$, the result is

$$\begin{aligned} \log J_{s,q,L}(a) \geq & -\log(s!) + s(s-1) \iint \log|x-y| \rho_\lambda(x) \rho_\lambda(y) dx dy \\ & + as \int \log x \rho_\lambda(x) dx + s \int \log \phi_{q,L}(x) \rho_\lambda(x) dx \\ & + s \int \log B_q(x) \rho_\lambda(x) dx - s \int \log f(x) \rho_\lambda(x) dx. \end{aligned}$$

All terms are integrable under these proved density properties. Divide by s^2 , apply the compact-uniform limits MFG16 and MFG18, and use $q/s \rightarrow 2$, $a/s \rightarrow 2$. The right side tends to the energy of ρ_λ for V_λ , which equals F_λ by its full variational inequality. This proves the lower bound and completes MFG15 for all four moving sizes and exponents needed in MFG4.

21 Both high endpoints, the source products, and the actual low factor

Apply MFG14–15 to $(s, a) = (n, q), (n+1, q), (n, q+1), (n+1, q+1)$. The same proof covers their $L=0$ counterparts with the moving exponents $a/s \rightarrow 2$: set $\phi_{q,0} = 1$ and $h_0 = 0$ identically. Then the upper field still satisfies $|V_s - V_0| \leq o(1)R$, and the lower Jensen bound uses the already proved ρ_0 . This gives the same value $F_0 = \mathcal{F}(2, \pi)$ as GEL10 while explicitly retaining these moving denominator exponents. Subtract these four denominator limits. The exact new scale power is $2L[n + (n+1) + n + n] = 2L(2q+1)$. The four squared sizes divided by q^2 sum to a quantity tending to one. Therefore

$$\boxed{\log(Z_{q,L}/Z_{q,0}) = 2L(2q+1) \log q + q^2(F_\lambda - F_0) + o(q^2).} \quad (\text{MFG19})$$

Define the explicit source integral

$$\begin{aligned} Q(\lambda) &= 2 \int_1^2 \int_0^{2\lambda} \log(s+t) dt ds = 2[A(2+2\lambda) - A(1+2\lambda) - A(2) + A(1)], \\ A(v) &= \frac{v^2}{2} \log v - \frac{3v^2}{4}, \quad A''(v) = \log v. \end{aligned} \quad (\text{MFG20})$$

The closed form follows by integrating the two literal endpoint ranges. There are exactly $4Lq$ original factors in MFG5. After extracting q from each factor, use cells of side $1/q$ in $[1, 2] \times [0, 2L/q]$. The first range samples the left s endpoint, the second the right s endpoint, and both sample the midpoint in t . Both partial derivatives of $\log(s+t)$ have absolute value at most one there. Each individual cell error, after multiplication by q^2 , is at most $3/(2q)$. Summing over the $2Lq$ cells in each range gives the finite bound

$$\boxed{|P_{q,L} + 4Lq \log q + q^2 Q(\lambda_q)| \leq 6L.} \quad (\text{MFG21})$$

This preserves the half-integer shifts and both unequal endpoint ranges.

The low factor has an explicit finite control which also preserves the actual polynomial. The original full-source bounds are

$$C_L e^{-\pi|y|} \leq \sigma(y) \leq C_U e^{-|y|}, \quad C_L = \frac{\pi}{\Gamma(3/4)^2}, \quad C_U = \frac{\Gamma(1/4)^2}{2\pi\sqrt{\cos 1}}.$$

Define

$$A_{q,L} = \int_0^\infty z^{2q} \phi_{q,L}(z^2) \sigma(qz) dz, \quad B_q^{\text{low}} = \int_0^\infty z^{2q} \sigma(qz) dz.$$

Evenness and the exact substitution $y = qz$ give $r_{q,L} = q^{2L} A_{q,L} / B_q^{\text{low}}$; both half-axis factors and Jacobians cancel only in this particular ratio. On $1 \leq z \leq 2$, all factors of $\phi_{q,L}(z^2)$ are at least one, so both $A_{q,L}$ and B_q^{low} are at least $C_L e^{-2\pi q}$. Since $(2j + 1/2)/q \leq 1/2$ for every stipulated $j < L \leq q/4$, $\phi_{q,L}(z^2) \leq (1 + z)^{2L}$ on the whole positive axis. For $D = 2q + 2L \leq 5q/2$, the elementary bounds

$$(1 + z)^D \leq 2^D (1 + z^D), \quad D! \leq D^D$$

give

$$A_{q,L} \leq \frac{2C_U}{q} 5^{5q/2}, \quad B_q^{\text{low}} \leq \frac{C_U}{q} 4^q.$$

These are obtained by integrating 1 and z^D against e^{-qz} ; their exact integrals are $1/q$ and $D!/q^{D+1}$. Thus

$$\begin{aligned} \log \frac{C_L}{C_U} + \log q - (2\pi + \log 4)q &\leq \log r_{q,L} - 2L \log q \\ &\leq \log \frac{2C_U}{C_L} - \log q + (2\pi + \frac{5}{2} \log 5)q. \end{aligned} \tag{MFG22}$$

The remaining low logarithm is therefore $O(q)$ uniformly on the stated macroscopic order range. No power approximation to Φ_L was used.

22 The evaluated macroscopic coefficient on the actual polynomial source

Substitute MFG19, MFG21 and MFG22 into the exact MFG6. The scale terms cancel with their full integer coefficients:

$$-4Lq \log q + 2L(2q + 1) \log q - 2L \log q = 0.$$

Continuity of the explicit Q and $\lambda_q \rightarrow \lambda$ give

$$\frac{T_{q,L}^{\text{hom}}}{q^2} \rightarrow \Psi(\lambda) := F_\lambda - F_0 - Q(\lambda), \quad 0 < \lambda \leq 1/4. \tag{MFG23}$$

In particular $L = q/4$, $K = q + 1$ gives the exact coefficient $\Psi(1/4)$, defined by the proved field MFG10 and the elementary closed expression MFG20. This is the q^2 term of the actual Gamma polynomial source, rather than the coefficient obtained by changing that polynomial to a power.

For comparison, the exact identity connecting the two source weights is

$$\Phi_L(y) = |y|^{2L} \prod_{j=0}^{L-1} \left(1 + \frac{(2j + 1/2)^2}{y^2} \right) \quad (y \neq 0). \tag{MFG24}$$

On $y = q\sqrt{x}$ the logarithm of the second factor divided by q tends to $h_\lambda(x) - \lambda \log x$. Hence the homogeneous power field $\pi\sqrt{x} - (2 + 2\lambda) \log x$ and the actual field MFG10 are related by the exact limiting density correction $-2[h_\lambda(x) - \lambda \log x]$. The latter correction remains in MFG23 at fixed positive λ .

The derivative proved in KME24 by testing the two actual minimizers in each other's variational problems is $F'_\lambda = 2 \int \log(x + 4\lambda^2) \rho_\lambda(x) dx$. Since $Q'(\lambda) = 4 \int_1^2 \log(s + 2\lambda) ds$, it gives the exact integral form

$$\Psi(\lambda) = \int_0^\lambda \left[2 \int \log(x + 4t^2) \rho_t(x) dx - 4 \int_1^2 \log(s + 2t) ds \right] dt. \quad (\text{MFG25})$$

At zero the endpoint family returns to the original equilibrium. Consequently $\Psi'(0) = 2\mathcal{L} - 4(2\log 2 - 1) = \mathcal{C}_\Gamma$, the previously proved small-order coefficient, which is strictly positive. This establishes the exact local bridge to the earlier order- lq calculation while retaining the finite positive- λ field needed for $K/q \rightarrow 1$.

Dependencies and the original receiving map. The Gamma bounds and the unmodified $L = 0$ ensemble limit are GEL1–10. The constructive actual equilibrium and its full variational identities are the companion macroscopic equilibrium provider KME1–24, with complete endpoint existence KEP1–20. Its KME22a evaluates each F_λ by real one-dimensional endpoint integrals, retaining all additive constants in MFG10. The argument above proves the discrete polynomial limit, the complete ensemble passage, all scale cancellations, and the finite low and source controls. Transfer from χ_0 to the original displaced χ uses its original source comparison with f_L kept intact. The baseline at the same independent order K , its arithmetic deficit, and the mixed kernel/boundary maps must then enter their exact receiving equality at that same K .

23 A strictly positive macroscopic return at the exact order $K = q + 1$

Exploratory comparison status. This proof evaluates the auxiliary family of IKO, retaining its exact relation to the original source. Its tested choice $K = q + 1$ is an externally selected comparison parameter. The original tensor source and its canonical mixed-cohomology maps remain those of order k ; this calculation supplies no rule selecting a new canonical order. The simultaneous baseline and return identities KPR15–16 are kept in full, as required to use the comparison without an ad hoc insertion.

Retain the proved equilibrium $\rho_0(x)dx$ for the exact field $V_0(x) = \pi\sqrt{x} - 2\log x$, with support $[a_*, b_*]$ inside $(0, \infty)$. Here a_*, b_* denote these equilibrium endpoints only; the earlier scalar error allowances retain their earlier decorated notation. Write

$$\begin{aligned} M_1 &= \int \sqrt{x} \rho_0(x) dx = 6/\pi, & M_2 &= \int x \rho_0(x) dx, \\ M_3 &= \int x^{3/2} \rho_0(x) dx, & \mathcal{L} &= \int \log x \rho_0(x) dx. \end{aligned} \tag{KPR1}$$

The first identity is EIQ31 with its original parameters. All moments here are finite because the equilibrium support is compact and bounded away from zero. We shall prove the sign of the actual independent-order coefficient using these integrals and explicit finite inequalities.

23.1 A further exact moment identity from a polynomial transport

For sufficiently small real t of either sign, the map $T_t(x) = x + tx^2$ is positive and strictly increasing on $[a_*, b_*]$. For example $|t| < 1/(4b_*)$ ensures both assertions. Its probability pushforward is therefore an admissible competitor in the exact variational definition of F_0 . For $x \neq y$ its pair difference is

$$T_t(x) - T_t(y) = (x - y)(1 + t(x + y)), \quad 1 + t(x + y) > 0.$$

The change in pair energy is exactly the integral of $\log(1 + t(x + y))$, and its derivative at zero is $2M_2$. Differentiation is justified by the uniform positive lower bound on the last factor and the bounded support. The derivative of the potential integral is

$$\int x^2 V'_0(x) \rho_0(x) dx = \frac{\pi}{2} M_3 - 2M_2.$$

The original equilibrium maximizes the functional, and zero is an interior value of this explicit admissible parameter. Hence its derivative is zero. We have proved

$$\boxed{M_3 = \frac{8}{\pi} M_2.} \tag{KPR2}$$

Applying Cauchy's inequality to the two functions $x^{1/4}$ and $x^{3/4}$ in this original probability space gives $M_2^2 \leq M_1 M_3$. Since $M_2 > 0$, KPR1–2 imply the finite bound

$$\boxed{0 < M_2 \leq \frac{48}{\pi^2} < \frac{16}{3}.} \tag{KPR3}$$

The last inequality uses $\pi > 3$. A direct rational verification is

$$\pi = 4 \int_0^1 \frac{dx}{1+x^2} > 4 \sum_{j=0}^7 \frac{(-1)^j}{2j+1} = \frac{135904}{45045} > 3.$$

The strict integral inequality follows by retaining the positive remainder $x^{16}/(1+x^2)$ in the finite geometric identity.

23.2 A specified dilated competitor for the full marked field

For $\lambda \geq 0$ retain exactly

$$h_\lambda(x) = \int_0^\lambda \log(x + 4u^2) du, \quad V_\lambda(x) = \pi\sqrt{x} - 2\log x - 2h_\lambda(x), \quad F_\lambda = \sup_\eta \mathcal{E}_{V_\lambda}(\eta), \quad (\text{KPR4})$$

where $\mathcal{E}_V(\eta) = \iint \log|x - y| d\eta^2 - \int V d\eta$ has the precise single-kernel interpretation of MFG12. For any $s > 0$ take the explicit pushforward $x \mapsto s^2x$ of $\rho_0 dx$. Its value relative to F_0 is exactly

$$\begin{aligned} \mathcal{E}_{V_\lambda}((s^2 \cdot)_* \rho_0) - F_0 &= (6 + 4\lambda) \log s - 6(s - 1) + 2\lambda \mathcal{L} \\ &\quad + 2 \int_0^\lambda \int \log \left(1 + \frac{4u^2}{s^2 x} \right) \rho_0(x) dx du. \end{aligned} \quad (\text{KPR5})$$

To verify every term, the pair energy gains $2 \log s$; the original $-\pi\sqrt{x}$ term gains $-\pi(s - 1)M_1 = -6(s - 1)$; the original $2 \log x$ gains $4 \log s$. Finally split the full marked logarithm only by its exact identity $\log(s^2 x + 4u^2) = 2 \log s + \log x + \log(1 + 4u^2/(s^2 x))$ and integrate it over its original interval. All terms are finite on the specified support.

For $A \geq 0$, the function $x \mapsto \log(1 + A/x)$ is convex on $(0, \infty)$: for $A > 0$ its second derivative is $A(2x + A)/(x^2(x + A)^2) > 0$, and for $A = 0$ it is zero. Jensen's inequality followed by the elementary integral inequality $\log(1 + v) \geq v/(1 + v)$ for $v \geq 0$ therefore gives

$$\begin{aligned} 2 \int_0^\lambda \int \log \left(1 + \frac{4u^2}{s^2 x} \right) \rho_0(x) dx du &\geq 2 \int_0^\lambda \log \left(1 + \frac{4u^2}{s^2 M_2} \right) du \\ &\geq \frac{8\lambda^3}{3(s^2 M_2 + 4\lambda^2)}. \end{aligned} \quad (\text{KPR6})$$

The last line retains the numerator $4u^2$, and bounds its denominator $s^2 M_2 + 4u^2$ from above by $s^2 M_2 + 4\lambda^2$ before integration. The logarithmic inequality follows, for example, by differentiating $\log(1 + v) - v/(1 + v)$, whose derivative is $v/(1 + v)^2 \geq 0$.

Use precisely $\lambda = 1/4$ and $s = 7/6$. Equations KPR3 and KPR6 give

$$2 \int_0^{1/4} \int \log \left(1 + \frac{4u^2}{(7/6)^2 x} \right) \rho_0(x) dx du \geq \frac{9}{1622}, \quad (\text{KPR7})$$

since $(7/6)^2(16/3) + 1/4 = 811/108$ and $8(1/4)^3/[3(811/108)] = 9/1622$. The supremum $F_{1/4}$ is at least the value of this stated competitor. Consequently

$$F_{1/4} - F_0 \geq \frac{1}{2} \mathcal{L} + 7 \log(7/6) - 1 + \frac{9}{1622}. \quad (\text{KPR8})$$

23.3 A positive rational lower bound, with exact logarithmic errors

The proved original recurrence bound and its exact limit GEL18a are

$$2\mathcal{L} - 4(2 \log 2 - 1) = \mathcal{C}_\Gamma \geq 4 \log \frac{27}{8\pi}.$$

Thus $\mathcal{L}/2 \geq \log(27/(2\pi)) - 1$ after combining the displayed logarithms. The exact polynomial division identity $22/7 - \pi = \int_0^1 x^4(1 - x)^4/(1 + x^2) dx > 0$ implies

$$\frac{1}{2} \mathcal{L} > \log(189/44) - 1. \quad (\text{KPR9})$$

The literal source integral MFG20 at this same parameter is

$$Q(1/4) = \frac{25}{4} \log(5/2) - \frac{9}{4} \log(3/2) - 4 \log 2 - \frac{3}{2}. \quad (\text{KPR10})$$

Combining KPR8–10 with the proved definition $\Psi(1/4) = F_{1/4} - F_0 - Q(1/4)$ yields

$$\Psi(1/4) > \log(189/44) + 7 \log(7/6) - \frac{25}{4} \log(5/2) + \frac{9}{4} \log(3/2) + 4 \log 2 - \frac{1}{2} + \frac{9}{1622}. \quad (\text{KPR11})$$

Here are finite rational bounds for every logarithm in this expression. For $x > 1$ set $r = (x - 1)/(x + 1)$. Integration of $2/(1 - t^2)$ from 0 to r gives

$$S_{20}(r) = 2 \sum_{j=0}^{19} \frac{r^{2j+1}}{2j+1} < \log x < S_{20}(r) + \frac{2r^{41}}{41(1-r^2)}. \quad (\text{KPR12})$$

The inequalities retain the positive remaining series and bound each of its denominators below by 41. For the five rational arguments used here, substituting the displayed r gives the following exact rational enclosures. Every entry in the last two columns is an integer, to be divided by 10^8 .

x	r	lower numerator	upper numerator
189/44	145/233	145755738	145755739
7/6	1/13	15415067	15415068
5/2	3/7	91629073	91629074
3/2	1/5	40546510	40546511
2	1/3	69314718	69314719

(KPR13)

For completeness these rational comparisons require only $n_-/10^8 \leq S_{20}(r)$ and $S_{20}(r) + 2r^{41}/[41(1-r^2)] \leq n_+/10^8$ in each row. Their denominators are positive; inserting its displayed rational r and multiplying the denominators verifies each finite inequality. The accompanying exact-arithmetic script performs exactly these ten stated integer comparisons and reproduces the following final fraction; its proof inputs are the explicit sum and remainder KPR12. Taking lower entries at positive coefficients and the upper entry at the negative coefficient in KPR11 gives

$$\Psi(1/4) > \frac{9279677}{40550000000} > \frac{1}{5000} > 0. \quad (\text{KPR14})$$

Indeed the difference between the middle two fractions is $1169677/40550000000 > 0$. This sign proof uses no numerical quadrature of the equilibrium or finite matrix experiment.

23.4 The original packet at the requested comparable order

At $L = q/4$, $K = q + 1$, the actual homogeneous-polynomial theorem MFG23 and the exact original-packet transport IKO17 prove

$$\frac{\mathfrak{T}_{k;q/4}}{q^2} \longrightarrow \Psi(1/4) > \frac{1}{5000}. \quad (\text{KPR15})$$

This is the positive return of the original packet with the new independent Gamma order. Both original high ranks, the low relation factor, and the entire $f_{q/4}$ occur in the finite formula IKO13.

The exact same-order identities IKO18–19, the proved original baseline BSL16, and $\delta_k^\sigma/(kq) \rightarrow 0$ now give

$$\begin{aligned} \frac{\mathcal{B}_k^{(q+1)}}{q^2} &\longrightarrow C_B + \Psi(1/4), \\ \frac{\mathcal{B}_k^{\text{ar}} - \mathcal{B}_k^{(q+1)}}{q^2} &\longrightarrow -\Psi(1/4), \\ \frac{\mathcal{B}_k^{\text{ar}}}{q^2} &\longrightarrow C_B > 0. \end{aligned} \quad (\text{KPR16})$$

The scalar cancellation in the last line has the exact original finite identity $\mathcal{B}_k^{\text{ar}} = \mathcal{B}_k^{(q+1)} - \mathfrak{T}_{k;q/4} + \delta_k^\sigma$ underneath it. The mixed kernel and boundary receive the same equality through IKO8 and BSL20,

with every original observation direction and off-diagonal Gram block retained. Thus the comparable-order construction has produced and evaluated a positive q^2 return, and its simultaneous baseline change is explicitly accounted for.